GENERAL INFORMATION

DESCRIPTION AND OPERATION

ENGINE CONTROL MODULE (ECM) - 3.0L TDV6 - HYBRID ELECTRIC VEHICLE

WARNINGS:

- **A risk assessment must be performed before any work is undertaken.**
- **All hybrid electric vehicle work must be performed by a suitably qualified person.**
- **Appropriate Personal Protection Equipment (PPE) must be worn when working on or near a hybrid electric vehicle high voltage system.**
- **The hybrid electric vehicle battery pack casing must not be opened due to the risk of exposure to hazardous voltage, which may cause serious injury or death.**
- **The electric power inverter converter casing must not be disassembled due to the risk of exposure to hazardous voltage, which may cause serious injury or death.**
- **If the hybrid electric vehicle battery pack is damaged or overcharged, there is a risk of exposure to hazardous voltage and/or highly corrosive electrolyte mist. If liquid or vapour is observed leaking from the hybrid electric vehicle battery pack,**

take the following action:

- Evacuate the area
- Notify manager
- Do not breathe smoke/vapour
- Contain spillage using spill kit
- Wash any spillage off body and clothing (remove contaminated clothing)

- If the hybrid electric vehicle warning indicator is illuminated, the Battery Energy Control Module (BECM) cannot isolate the high voltage cables. Live working will be necessary to rectify the fault.
- The motor generator can generate a voltage if the wheels are rotated, even if the hybrid electric vehicle system has been made safe.

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance, always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Engine Control Module (ECM) - Hybrid Electric Vehicle. For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1087-93	LIN Bus "A" - No operation	▪ The engine control module has detected that the component is not operating ▪ Generator LIN bus communication circuit failure	▪ Refer to the electrical circuit diagrams and check the generator LIN bus circuit, for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
B10A2-32	Crash Input - Signal low time < minimum	**NOTE:** Circuit reference INI ▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Restraints control module failure ▪ Auxiliary junction box failure ▪ Harness failure	▪ This DTC is set when the 'airba deployed' signal supplied by th restraints control module is outside the specification expected by the engine contro module ▪ Check the restraints control module for DTCs and refer to t relevant DTC index ▪ Check auxiliary junction box fo DTCs and refer to the relevant DTC index ▪ Refer to electrical circuit diagrams and check the supplementary restraints syster input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junctior box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
B10A2-35	Crash Input - Signal high time > maximum	**NOTE:** Circuit reference INI ▪ This DTC is set when the	▪ This DTC is set when the 'airba deployed' signal supplied by th restraints control module is outside the specification expected by the engine contro module ▪ Check the restraints control module for DTCs and refer to t

			engine control module detects that the signal high time is greater than the maximum value ▪ Restraints control module failure ▪ Auxiliary junction box failure ▪ Harness failure	relevant DTC index ▪ Check auxiliary junction box for DTCs and refer to the relevant DTC index ▪ Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis men tab and retest
	B10A2-36	Crash Input - Signal frequency too low	**NOTE:** Circuit reference INI ▪ The engine control module detected excessive duration for one cycle of the output across a specified sample size ▪ Restraints control module failure ▪ Auxiliary junction box failure ▪ Harness failure	▪ This DTC is set when the 'airbag deployed' signal supplied by the restraints control module is outside the specification expected by the engine control module ▪ Check the restraints control module for DTCs and refer to the relevant DTC index ▪ Check auxiliary junction box for DTCs and refer to the relevant DTC index ▪ Refer to electrical circuit diagrams and check the supplementary restraints system input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis men tab and retest

B10A2-37	Crash Input - Signal frequency too high	**NOTE:** Circuit reference INI - The engine control module detected insufficient duration for one cycle of the output across a specified sample size - Restraints control module failure - Auxiliary junction box failure - Harness failure	- This DTC is set when the 'airba deployed' signal supplied by th restraints control module is outside the specification expected by the engine contro module - Check the restraints control module for DTCs and refer to t relevant DTC index - Check auxiliary junction box for DTCs and refer to the relevant DTC index - Refer to electrical circuit diagrams and check the supplementary restraints syster input circuit for failures. This circuit is a single wire which connects the restraints control module to the auxiliary junction box and the engine control module. Check this circuit for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
B11D9-00	Vehicle Battery - No sub type information	- Harness failure - Battery failure - Battery monitoring system module failure	- This DTC is set when the batter monitoring system fails a diagnostic check - Refer to the battery care requirements section and verify that the vehicle battery is fully charged and serviceable before continuing with further diagnostic tests - Refer to the electrical circuit diagrams and check the connections between the batte and the battery monitoring module are clean and secure - Ensure that full battery voltage present on the monitor line pin the battery monitoring system module connector. Ensure the battery ground connection is clean and secure

				▪ Check the LIN bus connections to the battery monitoring syste module. Check LIN bus integrit Check and install a new battery monitoring system module as required
	B11DB-87	Battery Monitoring Module - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module communication with battery monitoring system module) ▪ Harness failure ▪ Battery monitoring system module failure	▪ This DTC is set when the engin control module has lost communication with the battery monitoring system module ▪ Refer to the battery care requirements section and verify that the vehicle battery is fully charged and serviceable before continuing with further diagnostic tests ▪ Refer to the electrical circuit diagrams and check the connections between the batte and the battery monitoring module are clean and secure ▪ Ensure that full battery voltage present on the monitor line pin the battery monitoring system module connector. Ensure the battery ground connection is clean and secure ▪ Check the LIN bus connections to the battery monitoring syste module. Check LIN bus integrit Check and install a new battery monitoring system module as required
	B1206-68	Crash Occurred - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Event information - The engine control module	▪ This DTC is set if the restraints control module has deployed tl restraints systems following activation of the crash sensors. Check the restraints control module for DTCs and refer to t relevant DTC index

			has received a crash signal from the restraints control module	
	P0030-11	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit short to ground	**NOTE:** Circuit reference LPPH_A ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Heated oxygen sensor heater control circuit short circuit to ground ▪ Heated oxygen sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycl Bank 1 Sensor 1 (0x03A1). Refe to the electrical circuit diagram and check the heated oxygen sensor heater control (heater ground) circuit for short circuit ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P0030-12	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit short to battery	**NOTE:** Circuit reference LPPH_A ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure - Heated oxygen sensor heater control circuit short circuit to power	▪ Using the manufacturer approv diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycl Bank 1 Sensor 1 (0x03A1). Refe to the electrical circuit diagram and check the heated oxygen sensor heater control (heater ground) circuit for short circuit power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required

			■ Heated oxygen sensor failure	
	P0030-13	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit open	**NOTE:** Circuit reference LPPH_A ■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ■ Harness failure - Heated oxygen sensor heater control circuit open circuit ■ Heated oxygen sensor failure	■ Using the manufacturer approv diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycl Bank 1 Sensor 1 (0x03A1). Refe to the electrical circuit diagram and check the heated oxygen sensor heater control (heater ground) circuit for open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ■ Check and install a new heated oxygen sensor as required
	P0030-4B	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Over temperature	■ The engine control module detected an internal temperature above the expected range ■ Harness failure - Heated oxygen sensor heater control circuit short circuit to power ■ Heated oxygen sensor failure	■ Using the manufacturer approv diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycl Bank 1 Sensor 1 (0x03A1). Refe to the electrical circuit diagram and check the heated oxygen sensor heater control (heater ground) circuit for short circuit power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ■ Check and install a new heated oxygen sensor as required
	P0033-00	Turbocharger/Supercharger Bypass Valve Control Circuit / Open - No sub type information	**NOTE:** Circuit reference CRV	■ Refer to the electrical circuit diagrams and check the charge air recirculation solenoid contro circuit between the engine control module and the control valve for open circuit. Check th power supply to the control

			• Harness failure - Charge air recirculation solenoid control circuit open circuit • Charge air recirculation solenoid failure	valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest • Check and install a new charge air recirculation solenoid as required
	P0034-00	Turbocharger/Supercharger Bypass Valve Control Circuit Low - No sub type information	**NOTE:** Circuit reference CRV • Harness failure - Charge air recirculation solenoid circuit short circuit to ground • Charge air recirculation solenoid failure	• Refer to the electrical circuit diagrams and check the charge air recirculation solenoid circuit between the engine control module and the control valve fo a short circuit to ground. Check the power supply to the contro valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest • Check and install a new charge air recirculation solenoid as required
	P0035-00	Turbocharger/Supercharger Bypass Valve Control Circuit High - No sub type information	**NOTE:** Circuit reference CRV • Harness failure - Charge air recirculation solenoid circuit short circuit to power • Charge air recirculation solenoid failure	• Refer to the electrical circuit diagrams and check the charge air recirculation solenoid circuit between the engine control module and the control valve fo a short circuit to power. Check the power supply to the contro valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis Men tab and retest • Check and install a new charge air recirculation solenoid as required
	P0039-4B	Turbocharger/Supercharger Bypass Valve Control Circuit Range/Performance - Over temperature	• The engine control module detected an internal temperature above the expected range • Harness failure - Charge air recirculation solenoid	• Refer to the electrical circuit diagrams and check the charge air recirculation solenoid circuit between the engine control module and the control valve fo a short circuit to ground, short circuit to power, high resistance Check the power supply to the

			▪ circuit short to ground, short circuit to power, high resistance ▪ Charge air recirculation solenoid failure	control valve. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new charge air recirculation solenoid as required
	P004A-00	Turbocharger/Supercharger Boost Control B Circuit / Open - No sub type information	▪ Harness failure - Variable geometry turbine vane actuator control circuit open circuit ▪ Variable geometry turbine vane actuator control unit failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533), Boost Pressure Actuator Bank 2 - Controller output (0x03DE). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for open circuit. Check both of the circuits for open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new variabl geometry turbine vane actuato control unit as required
	P004A-71	Turbocharger/Supercharger Boost Control B Circuit / Open - Actuator stuck	▪ The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Harness failure - Variable geometry turbine vane actuator control circuit open circuit ▪ Variable geometry turbine vane actuator control unit failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533), Boost Pressure Actuator Bank 2 - Controller output (0x03DE). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for open circuit. Check both of the circuits for open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new variabl geometry turbine vane actuato control unit as required

P004B-16	Turbocharger/Supercharger Boost Control B Circuit Range/Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Variable geometry turbine vane actuator control circuit ▪ Variable geometry turbine vane actuator control unit failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for high resistance or short circuit to another circuit. Check both of the circuits for failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new variabl geometry turbine vane actuato control unit as required
P004B-19	Turbocharger/Supercharger Boost Control B Circuit Range/Performance - Circuit current above threshold	▪ The engine control module has measured current flow above a specified range ▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to ground, short circuit to power, high resistance ▪ Variable geometry turbine vane actuator control unit failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to ground, short circuit t power, high resistance. Check both of the circuits for failures. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new variabl geometry turbine vane actuato control unit as required
P004B-1D	Turbocharger/Supercharger Boost Control B Circuit Range/Performance - Circuit current out of range	▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ▪ Harness failure - Variable geometry turbine vane actuator control circuit	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to another circuit. Check

			▪ Mechanical failure on actuator mechanism ▪ Variable geometry turbine vane actuator control unit failure	both of the circuits for failures. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check for mechanical failure on actuator mechanism ▪ Check and install a new variabl geometry turbine vane actuato control unit as required
	P004B-4B	Turbocharger/Supercharger Boost Control B Circuit Range/Performance - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to ground, short circuit to power, high resistance. ▪ Mechanical failure on actuator mechanism ▪ Variable geometry turbine vane actuator control unit failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to ground, short circuit t power, high resistance. Check both of the circuits for failures. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check for mechanical failure on actuator mechanism ▪ Check and install a new variabl geometry turbine vane actuato control unit as required
	P004C-00	Turbocharger/Supercharger Boost Control B Circuit Low - No sub type information	▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to ground	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to ground. Check both o the circuits for failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

P004C-11	Turbocharger/Supercharger Boost Control B Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to ground	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to ground. Check both o the circuits for failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P004C-12	Turbocharger/Supercharger Boost Control B Circuit Low circuit - Short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to power	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to power. Check both circuits for failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P004C-77	Turbocharger/Supercharger Boost Control B Circuit Low - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to ground, high resistance, open circuit ▪ Harness failure - Variable geometry turbocharger	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to ground, high resistanc open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the variabl geometry turbocharger sensor circuit for short circuit to groun

		▪ sensor circuit, short circuit to ground, high resistance, open circuit ▪ Mechanical failure on actuator mechanism ▪ Variable geometry turbine vane actuator control unit failure	high resistance, open circuit ▪ Check for mechanical failure on actuator mechanism ▪ Check and install a new variable geometry turbine vane actuator control unit as required
P004D-00	Turbocharger/Supercharger Boost Control B Circuit High - No sub type information	▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to power	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the boost control circuit for short circuit to power. Check both circuits for failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P004D-77	Turbocharger/Supercharger Boost Control B Circuit High - Commanded Position Not Reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Harness failure - Variable geometry turbine vane actuator control circuit short circuit to power ▪ Harness failure - Variable geometry turbocharger sensor circuit, short circuit to power ▪ Mechanical failure on actuator mechanism ▪ Variable geometry turbine vane actuator control unit failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Boost Actuator Control Bank 2 (0x0533). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator control circuit for shor circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check the variabl geometry turbocharger sensor circuit for short circuit to power ▪ Check for mechanical failure on actuator mechanism ▪ Check and install a new variable geometry turbine vane actuato control unit as required

	P006A-00	MAP - Mass or Volume Air Flow Correlation - No sub type information	▪ Intake air system, high pressure boost leak bank 1 ▪ Intake air system, high pressure boost leak bi-turbo mode bank 2 ▪ Intake air system, low pressure boost leak bank 1 ▪ Intake air system, low pressure boost leak bank 2	▪ If this DTC is logged with P124 00, P0402-00 & P00BF-07, suspect intake air system, high pressure boost leak bank 1 ▪ If this DTC is logged with P124 00, suspect intake air system, high pressure boost leak bi-turb mode bank 2 ▪ If this DTC is logged with P00B 07 & P0401-00, suspect intake system, low pressure boost leak bank 1 ▪ If this DTC is logged with P00B 00, suspect intake air system, lo pressure boost leak bank 2 ▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine
	P006A-22	MAP - Mass or Volume Air Flow Correlation - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Error path indicating whether over boost monitoring is active or not ▪ Failure in induction air circuit ▪ Failure in charge air circuit ▪ Mass air flow sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Manifold absolute pressure sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Mass air flow sensor	▪ Check for other DTCs associate with intake and charge air flow control actuators, refer to the relevant DTC index and act on those DTCs first ▪ Refer to workshop manual and check intake air circuit, charge circuit and associated control valves, intercooler and air filter for leaks, blockages, restriction and control valve actuator malfunctions ▪ Refer to the electrical circuit diagrams and check both mass air flow sensor circuits for short circuit to ground, short circuit t power, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check manifold absolute pressure sensor wiring circuits f short circuit to ground, short circuit to power, high resistance open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle

			failure ▪ Manifold absolute pressure sensor failure	all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor or manifold absolut pressure sensor as required
	P0070-16	Ambient Air Temperature Sensor Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference AAT ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Ambient air temperature sensor circuit ▪ Ambient air temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Ambient Ai Temperature Sensor Voltage (0x03BA), Ambient Air Temperature (0xF466). This DT(is set when the ambient air temperature sensor signal line voltage at the engine control module is less than the thresho value. Refer to the electrical circuit diagrams and check both sides of the circuit for high resistance or short circuits. Rep harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new ambien air temperature sensor as required
	P0070-17	Ambient Air Temperature Sensor Circuit - Circuit voltage above threshold	**NOTE:** Circuit reference AAT ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Ambient air temperature sensor circuit ▪ Ambient air temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Ambient Ai Temperature Sensor Voltage (0x03BA), Ambient Air Temperature (0xF466). This DT(is set when the ambient air temperature sensor signal line voltage at the engine control module is greater than the threshold value. Refer to the electrical circuit diagrams and check both sides of the circuit f high resistance or short circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new ambien air temperature sensor as

			required
P007A-16	Charge Air Cooler Temperature Sensor Circuit (Bank 1) - Circuit voltage below threshold	**NOTE:** Circuit reference ACT ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Air charge temperature sensor circuit ▪ Air charge temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Charge Air Temperature Voltage (0x03EE). This DTC is set when the air charge temperature sensor sigr line voltage at the engine contr module is less than the thresho value. Refer to the electrical circuit diagrams and check both sides of the circuit for high resistance or short circuits. Rep harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new charge air temperature sensor as required
P007A-17	Charge Air Cooler Temperature Sensor Circuit (Bank 1) - Circuit voltage above threshold	**NOTE:** Circuit reference ACT ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Air charge temperature sensor circuit ▪ Air charge temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Air Charge Temperature (0x051C). This DT is set when the air charge temperature sensor signal line voltage at the engine control module is greater than the threshold value. Refer to the electrical circuit diagrams and check both sides of the circuit f high resistance or short circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new charge air temperature sensor as required
P007A-62	Charge Air Cooler Temperature Sensor Circuit (Bank 1) - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility	▪ Using the manufacturer approv diagnostic system check datalogger signals, Charge Air Temperature Voltage (0x03EE). Refer to the electrical circuit diagrams and check air charge

			- Harness failure - Air charge temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit - Air charge temperature sensor failure	temperature sensor circuit for short circuit to ground, short circuit to power, high resistance open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new charge air temperature sensor as required
	P0087-00	Fuel Rail/System Pressure - Too Low - No sub type information	- Low level fuel condition - Fuel gauge sender unit sticking - Fuel lines restricted - Fuel lines leaking - Fuel pump failure - Fuel rail pressure sensor signal circuit short circuit to ground, short circuit to power, open circuit, high resistance - Fuel rail pressure sensor failure	- Check the fuel volume available in the fuel tank is sufficient, add fuel as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check the fuel gauge sender units for correct operation - Check fuel lines for restriction - Check fuel lines for leakage - Check for fuel pump related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, short circuit t power, open circuit, high resistance. Repair harness as required - Check and install a new fuel rai pressure sensor as required
	P0087-16	Fuel Rail/System Pressure - Too Low - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Fuel rail pressure sensor circuit, short circuit to ground, open circuit, high resistance - Fuel rail pressure sensor	- Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair harness required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new fuel rai pressure sensor as required

			failure	
	P0087-21	Fuel Rail/System Pressure - Too Low - Signal amplitude < minimum	- The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low - Fuel lines restricted - Fuel lines leaking - Fuel injector's stuck open, leaking - Fuel pump failure - Fuel rail pressure sensor circuit short circuit to ground, open circuit, high resistance - Fuel pressure control valve circuit short circuit to ground, open circuit, high resistance - Fuel volume control valve circuit short circuit to ground, open circuit, high resistance - Fuel rail pressure sensor failure - Fuel pressure control valve failure - Fuel volume control valve failure	- Check fuel lines for restriction - Check fuel lines for leakage - Check fuel injector's for stuck open, leakage - Check for fuel pump related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to ground, open circuit, high resistance. Repair harness required - Refer to the electrical circuit diagrams and check fuel pressu control valve circuit for short circuit to ground, open circuit, high resistance. Repair harness required - Refer to the electrical circuit diagrams and check fuel volum control valve circuit for short circuit to ground, open circuit, high resistance. Repair harness required - Check and install a new fuel rai pressure sensor as required - Check and install a new fuel pressure control valve as requir - Check and install a new fuel volume control valve as require
	P0088-00	Fuel Rail/System Pressure - Too High - No sub type information	- Fuel gauge sender unit sticking - Fuel rail pressure sensor signal circuit short circuit to power - Fuel rail pressure sensor ground circuit open circuit, high resistance - Fuel pressure control valve circuit short circuit to power, ground, open	- Check the fuel gauge sender units for correct operation - Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power. Repair harness as required - Refer to the electrical circuit diagrams and check fuel rail pressure sensor ground circuit f open circuit, high resistance. Repair harness as required

			circuit, high resistance ▪ Fuel volume control valve circuit short circuit to power, ground, open circuit, high resistance ▪ Fuel rail pressure sensor failure ▪ Fuel pressure control valve failure ▪ Fuel volume control valve failure	▪ Refer to the electrical circuit diagrams and check fuel pressu control valve circuit for short circuit to power, ground, open circuit, high resistance. Repair harness as required ▪ Refer to the electrical circuit diagrams and check fuel volum control valve circuit for short circuit to power, ground, open circuit, high resistance. Repair harness as required ▪ Check and install a new fuel rai pressure sensor as required ▪ Check and install a new fuel pressure control valve as requir ▪ Check and install a new fuel volume control valve as require
	P0088-17	Fuel Rail/System Pressure - Too High - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Fuel rail pressure sensor signal circuit short circuit to power	▪ Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power
	P0088-22	Fuel Rail/System Pressure - Too High - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Fuel gauge sender unit sticking ▪ Fuel lines restricted ▪ Fuel rail pressure sensor circuit short circuit to power, ground, open circuit, high resistance ▪ Fuel pressure control valve circuit short circuit to power, ground, open circuit, high resistance	▪ Check the fuel gauge sender units for correct operation ▪ Check fuel lines for restriction ▪ Refer to the electrical circuit diagrams and check fuel rail pressure sensor circuit for short circuit to power, ground, open circuit, high resistance. Repair harness as required ▪ Refer to the electrical circuit diagrams and check fuel pressu control valve circuit for short circuit to power, ground, open circuit, high resistance. Repair harness as required ▪ Refer to the electrical circuit diagrams and check fuel volum control valve circuit for short

			▪ Fuel volume control valve circuit short circuit to power, ground, open circuit, high resistance ▪ Fuel rail pressure sensor failure ▪ Fuel pressure control valve failure ▪ Fuel volume control valve failure ▪ Fuel pump failure	circuit to power, ground, open circuit, high resistance. Repair harness as required ▪ Check and install a new fuel rai pressure sensor as required ▪ Check and install a new fuel pressure control valve as requir ▪ Check and install a new fuel volume control valve as require ▪ Check for fuel pump related DTCs and refer to the relevant DTC index
	P0089-21	Fuel Pressure Regulator Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Fuel pressure regulator signal circuit short circuit to ground, open circuit ▪ Fuel pressure regulator reference voltage circuit high resistance ▪ Fuel pressure regulator failure	▪ Refer to the electrical circuit diagrams and check fuel pressu regulator circuit for short circuit to ground, open circuit. Repair harness as required ▪ Refer to the electrical circuit diagrams and check fuel pressu regulator reference voltage circuit for high resistance. Repa harness as required ▪ Check and install a new fuel pressure regulator as required
	P0089-22	Fuel Pressure Regulator Performance - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Fuel pressure regulator signal circuit short circuit to power ▪ Fuel pressure regulator failure	▪ Refer to the electrical circuit diagrams and check fuel pressu regulator circuit for short circuit to power. Repair harness as required ▪ Check and install a new fuel pressure regulator as required
	P0090-13	Fuel Pressure Regulator 1 Control Circuit/Open - Circuit open		▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Pressu Relief Control Valve Duty Cycle

			NOTE: Circuit reference PCV ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Fuel pressure control valve control circuit open circuit	(0x03C3). Refer to the electrica circuit diagrams and check the fuel pressure control valve control circuit for open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0090-4B	Fuel Pressure Regulator 1 Control Circuit/Open - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Fuel pressure control valve signal circuit short circuit to power, open circuit ▪ Fuel pressure regulator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Pressu Relief Control Valve Duty Cycle (0x03C3). Refer to the electrica circuit diagrams and check fuel pressure control valve circuit fo short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new fuel pressure regulator as required
	P0091-11	Fuel Pressure Regulator 1 Control Circuit Low - Circuit short to ground	**NOTE:** Circuit reference PCV ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Pressu Relief Control Valve Duty Cycle (0x03C3). Refer to the electrica circuit diagrams and check the fuel pressure control valve circu for short circuit to ground. Che power supply circuit to control valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			another value was expected ▪ Harness failure - Fuel pressure control valve circuit short circuit to ground ▪ Fuel pressure control valve failure	▪ Check and install a new fuel pressure control valve as requir
	P0091-16	Fuel Pressure Regulator 1 Control Circuit Low - Circuit voltage below threshold	**NOTE:** Circuit reference PCV ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Fuel pressure control valve circuit ▪ Fuel pressure control valve failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Pressu Relief Control Valve Duty Cycle (0x03C3). This DTC is set when the voltage on the fuel pressure control valve circuit is less than the threshold expected by the engine control module. Refer to the electrical circuit diagrams a check the fuel pressure control valve circuit for high resistance, short circuit to ground. Check power supply circuit to the fuel pressure control valve. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel pressure control valve as requir
	P0092-12	Fuel Pressure Regulator 1 Control Circuit High - Circuit short to battery	**NOTE:** Circuit reference PCV ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Pressu Relief Control Valve Duty Cycle (0x03C3). Refer to the electrica circuit diagrams and check the fuel pressure control valve circu for short circuit to power. Chec power supply circuit to control valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			▪ Harness failure - Fuel pressure control valve circuit short circuit to power	
	P0092-17	Fuel Pressure Regulator 1 Control Circuit High - Circuit voltage above threshold	**NOTE:** Circuit reference PCV ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Fuel pressure control valve circuit ▪ Fuel pressure control valve failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Pressu Relief Control Valve Duty Cycle (0x03C3). This DTC is set when the voltage on the fuel pressure control valve circuit is more tha the threshold expected by the engine control module. Refer to the electrical circuit diagrams a check the fuel pressure control valve circuit for high resistance, short circuit to power. Check power supply circuit to the fuel pressure control valve. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel pressure control valve as requir
	P00AA-16	Intake Air Temperature Sensor 1 Circuit (Bank 2) - Circuit voltage below threshold	**NOTE:** Circuit reference IAT ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Intake air temperature sensor circuit ▪ Intake air temperature sensor failure (part of mass air flow sensor)	▪ Using the manufacturer approv diagnostic system check datalogger signals, Intake Air Temperature Sensor (0x1279), Intake Air Temperature (0xF40F This DTC is set when the voltag on the intake air temperature sensor circuit is less than the threshold expected by the engine control module. The intake air temperature sensor is integrated into the electronics package of the mass air flow sensor and shares the same ground circuit as the mass air flow element of the sensor. Ref to the electrical circuit diagram and check all the circuits connected to the mass air flow sensor for short circuit to grour short circuit to power, open circuit, high resistance. Check mass air flow sensor 2 circuits a

				both sensors share the ground connection. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new intake a temperature sensor as required
	P00AA-17	Intake Air Temperature Sensor 1 Circuit (Bank 2) - Circuit voltage above threshold	**NOTE:** Circuit reference IAT ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Intake air temperature sensor circuit ▪ Intake air temperature sensor failure (part of mass air flow sensor)	▪ Using the manufacturer approv diagnostic system check datalogger signals, Intake Air Temperature Sensor (0x1279), Intake Air Temperature (0xF40F This DTC is set when the voltag on the intake air temperature sensor circuit is more than the threshold expected by the engine control module. The intake air temperature sensor is integrated into the electronics package of the mass air flow sensor and shares the same ground circuit as the mass air flow element of the sensor. Ref to the electrical circuit diagram and check all the circuits connected to the mass air flow sensor for short circuit to grour short circuit to power, open circuit, high resistance. Check mass air flow sensor 2 circuits a both sensors share the ground connection. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new intake a temperature sensor as required
	P00AA-62	Intake Air Temperature Sensor 1 Circuit (Bank 2) - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Harness failure - Intake air temperature sensor circuit ▪ Intake air temperature	▪ Refer to the electrical circuit diagrams and check all the circuits connected to the mass a flow sensor for short circuit to ground, short circuit to power, open circuit, high resistance. Check mass air flow sensor 2 circuits as both sensors share th ground connection. Repair harness as required. Using the

		sensor failure (part of mass air flow sensor)	manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new intake temperature sensor as required
P00BC-00	Mass or Volume Air Flow A Circuit Range/Performance - Air Flow Too Low - No sub type information	▪ Charge air solenoid stuck shut bi-turbo mode ▪ Turbine intake solenoid stuck shut ▪ Intake system low pressure boost leak bank 2	▪ If this DTC is logged with P124 00, suspect charge air solenoid stuck shut bi-turbo mode ▪ If this DTC is logged with P22D 77 & P22CF-71, suspect turbine intake solenoid stuck shut ▪ If this DTC is logged with P006 00, suspect intake system low pressure boost leak bank 2
P00BC-07	Mass or Volume Air Flow A Circuit Range/Performance - Air Flow Too Low - Mechanical failures	▪ Air flow disruption at sensing element of mass air flow sensor ▪ Mass air flow sensor failure	▪ This DTC is set if the mass air flow signal from sensor 1 is less than the value expected by the engine control module. Refer t the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow sensor as required
P00BC-64	Mass or Volume Air Flow A Circuit Range/Performance - Air Flow Too Low - Signal Plausibility Failure	▪ The engine control module detected plausibility failures ▪ Charge air solenoid stuck shut bi-turbo mode ▪ Turbine intake solenoid stuck shut ▪ Intake system low pressure boost leak bank 2	▪ If this DTC is logged with P124 00, suspect charge air solenoid stuck shut bi-turbo mode ▪ If this DTC is logged with P22D 77 & P22CF-71, suspect turbine intake solenoid stuck shut ▪ If this DTC is logged with P006 00, suspect intake system low pressure boost leak bank 2 ▪ Refer to the relevant sections o the workshop manual and chec

				the induction system for air leal and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P00BD-00	Mass or Volume Air Flow A Circuit Range/Performance - Air Flow Too High - No sub type information	▪ Air leak at air intake system ▪ Harness failure - Mass air flow sensor circuit short circuit to power ▪ Mass air flow sensor failure	▪ Refer to the relevant sections o the workshop manual and chec the induction system for air leal ▪ Refer to the electrical circuit diagrams and check mass air flo sensor for short circuit to powe Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow sensor as required
	P00BD-07	Mass or Volume Air Flow A Circuit Range/Performance - Air Flow Too High - Mechanical failures	▪ Charge air solenoid stuck open mono turbo mode ▪ Turbine intake solenoid leakage when closed ▪ Turbine intake solenoid stuck open ▪ Intake air system, blocked low pressure air intake. This failure mode can be caused by snow packing in the intake system. Symptoms often disappear after the vehicle has been warmed and heat soaked. Similar symptoms to seized primary turbo ▪ Primary turbo charger seized	▪ If this DTC is logged with P023 94 suspect, charge air solenoid stuck open mono turbo mode ▪ If this DTC is logged with P023 94 suspect, turbine intake solenoid leakage when closed ▪ If this DTC is logged with P023 94, P22D2-77, P1247-00 & P22CF-71 suspect, turbine intal solenoid stuck open ▪ If this DTC is logged with P023 94, P22D2-77, P1247-00 & P22CF-71 suspect, intake air system, blocked low pressure a intake ▪ If this DTC is logged with P00B 07 & P1247-00 suspect, primary turbocharger seized ▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine

P00BD-64	Mass or Volume Air Flow A Circuit Range/Performance - Air Flow Too High - Signal Plausibility Failure	▪ The engine control module detected plausibility failures ▪ Charge air solenoid stuck open mono turbo mode ▪ Turbine intake solenoid leakage when closed ▪ Turbine intake solenoid stuck open mono turbo mode ▪ Intake system high pressure bank 2 in mono turbo mode	▪ If this DTC is logged with P023 94 suspect, charge air solenoid stuck open mono turbo mode ▪ If this DTC is logged with P023 94 suspect, turbine intake solenoid leakage when closed ▪ If this DTC is logged with P023 94, P22D2-77, P1247-00 & P22CF-71 suspect, turbine intal solenoid stuck open ▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine
P00BE-00	Mass or Volume Air Flow B Circuit Range/Performance - Air Flow Too Low - No sub type information	▪ Charge air recirculation solenoid stuck open bi-turbo mode ▪ Air flow disruption at sensing element of mass air flow sensor 2 ▪ Harness failure - Mass air flow sensor circuit short circuit to ground, high resistance, open circuit ▪ Mass air flow sensor 2 failure	▪ If this DTC is logged with P124 00, suspect charge air recirculation solenoid stuck ope Refer to the relevant sections o the workshop manual and chec the charge air recirculation solenoid circuit for short circuit power, short circuit to ground, open circuit. Refer to the releva sections of the workshop manu and check the induction system for air leaks, and obstructions t flow. Check the condition of th air filter and examine the induction pipes for debris whicl could disrupt air flow at the sensing element ▪ Check and install a new charge air recirculation solenoid as required ▪ Refer to the electrical circuit diagrams and check mass air flo sensor 2 for short circuit to ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor 2 as required

	P00BE-07	Mass or Volume Air Flow B Circuit Range/Performance - Air Flow Too Low - Mechanical failures	▪ Air flow disruption at sensing element of mass air flow ▪ Mass air flow failure	▪ This DTC is set if the mass air flow signal from sensor 2 is less than the value expected by the engine control module. Refer t the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow as required
	P00BF-00	Mass or Volume Air Flow B Circuit Range/Performance - Air Flow Too High - No sub type information	▪ Air leak at air intake system ▪ Harness failure - Mass air flow sensor circuit short circuit to power ▪ Mass air flow sensor 2 failure	▪ If this DTC is logged with P00B 00, suspect charge air recirculation solenoid stuck ope Refer to the relevant sections o the workshop manual and chec the charge air recirculation solenoid circuit for short circuit power, short circuit to ground, open circuit. Refer to the releva sections of the workshop manu and check the induction system for air leaks ▪ Check and install a new charge air recirculation solenoid as required. ▪ Refer to the electrical circuit diagrams and check mass air flo sensor 2 for short circuit to power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor 2 as required
	P00BF-07	Mass or Volume Air Flow B Circuit Range/Performance - Air Flow Too High - Mechanical failures	▪ Intake air system, high pressure boost leak	▪ If this DTC is logged with P124 00, P006A-00 & P00BF-07, suspect intake air system, high pressure boost leak

P00CF-62	Secondary Compressor Outlet Pressure Sensor - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Charge air circuit leak or blockage ▪ Charge air recirculation solenoid failure ▪ Charge air solenoid failure ▪ Mechanical failure - Sensor hose blocked or leaking ▪ Harness failure - Charge air pressure sensor circuit ▪ Charge air pressure sensor failure	▪ This DTC is set when the engin control module detects a signa compare failure on the charge a pressure sensor signal. Check t sensor hose for blockages or leaks. Check the sensor and harness for mechanical damage caused by heat or chaffing ▪ Refer to the electrical circuit diagrams and check the secondary compressor outlet pressure sensor power and ground supplies for short circui to ground, open circuit. Check the signal line for short circuit t ground, short circuit to power, open circuit. Repair harness as required ▪ Check and install a new secondary compressor outlet pressure sensor as required
P0101-21	Mass or Volume Air Flow A Circuit Range/Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Air flow disruption at sensing element of mass air flow sensor ▪ Harness failure - Mass air flow sensor circuit short circuit to ground, high resistance, open circuit ▪ Mass air flow sensor failure	▪ This DTC is set if the mass air flow signal from sensor 1 is less than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element ▪ Refer to the electrical circuit diagrams and check mass air flo sensor circuit for short circuit to ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor as required
P0101-22	Mass or Volume Air Flow A Circuit Range/Performance	▪ The engine control module measured a	▪ This DTC is set if the mass air flow signal from sensor 1 is mo

		- Signal amplitude > maximum	signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Air flow disruption at sensing element of mass air flow sensor ▪ Harness failure - Mass air flow sensor circuit short circuit to power ▪ Mass air flow sensor failure	than the value expected by the engine control module. Refer t the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element ▪ Refer to the electrical circuit diagrams and check mass air fl sensor circuit for short circuit t power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor as required
	P0101-92	Mass or Volume Air Flow Sensor "A" Circuit Range/Performance - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Air flow disruption at sensing element of mass air flow sensor ▪ Harness failure - Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Mass air flow sensor failure	▪ This DTC is set if the mass air flow signal from sensor 1 is less than the value expected by the engine control module. Refer t the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element ▪ Refer to the electrical circuit diagrams and check mass air fl sensor circuit for short circuit t ground, short circuit to power, open circuit, high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow sensor as required
	P0102-00	Mass or Volume Air Flow A Circuit Low - No sub type information		▪ Using the manufacturer approv diagnostic system check datalogger signals, Air Flow Ra From mass air flow Sensor Bank

			NOTE: Circuit reference MAF_B ▪ Air flow disruption at sensing element of mass air flow sensor ▪ Harness failure - Mass air flow sensor circuits ▪ Mass air flow sensor failure	(0x0504). This DTC is set if the mass air flow signal from senso is less than the value expected the engine control module. Ref to the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element ▪ Refer to the electrical circuit diagrams and check the mass a flow sensor power and ground circuits for high resistance, shor circuit to power, short circuit to ground. Check the mass air flov signal line for high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow sensor as required
	P0103-00	Mass or Volume Air Flow A Circuit High - No sub type information	**NOTE:** Circuit reference MAF_B ▪ Air flow disruption at sensing element of mass air flow sensor ▪ Harness failure - Mass air flow sensor circuits ▪ Mass air flow sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Air Flow Ra From Mass Air Flow Sensor Bar 1 (0x0504). This DTC is set if th mass air flow signal from senso is greater than the value expected by the engine contro module. Refer to the relevant sections of the workshop manu and check the induction system for air leaks, and obstructions t flow. Check the condition of the air filter and examine the induction pipes for debris whicl could disrupt air flow at the sensing element. ▪ Refer to the electrical circuit diagrams and check the mass a flow sensor power and ground circuits for high resistance, shor circuit to power, short circuit to ground. Check the mass air flov signal line for high resistance.

				Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow sensor as required
	P0105-16	Manifold Absolute Pressure/BARO circuit - Circuit voltage below threshold	**NOTE:** Circuit reference MAP_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold absolute pressure sensor	▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for open circuit, high resistance, short circuit to ground. Check the sensor grou for open circuit, short circuit to power. Check the signal circuit for short circuit to ground, high resistance. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new manifo absolute pressure sensor as required ▪ Using the manufacturer approv diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0322), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). This DTC is set when the voltage on the manifold absolute pressure sensor signal circuit is less than the threshold expected by the engine contro module. Refer to the workshop manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are n blockages or air leaks preventir communication of pressure changes between the manifold and the sensor
	P0105-17	Manifold Absolute Pressure/BARO circuit - Circuit voltage above		▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for short circuit tc

		threshold	**NOTE:** Circuit reference MAP_A ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold pressure sensor	power. Check the sensor groun for open circuit, short circuit to power. Check the signal circuit for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new manifo absolute pressure sensor as required ▪ Using the manufacturer approv diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). This DTC is set when the voltage on the manifold absolute pressure sensor signal circuit is greater than the threshold expected by the engine control module. Refer t the workshop manual and chec the manifold absolute pressure sensor is correctly connected t the air intake manifold and the are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor
	P0105-65	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit - Signal has too few transitions / events	**NOTE:** Circuit reference MAP_A ▪ The engine control module monitored a parameter over time within specified limits and detected fewer than the expected number of transitions ▪ Harness failure - Manifold absolute	▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for short circuit t power. Check the sensor groun for open circuit, short circuit to power. Check the signal circuit for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new manifo absolute pressure sensor as required ▪ Using the manufacturer approv

		▪ pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold pressure sensor	diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). Refer to the worksho manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are n blockages or air leaks preventir communication of pressure changes between the manifold and the sensor
P0107-00	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit Low - No sub type information	**NOTE:** Circuit reference MAP_A ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold pressure sensor	▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for short circuit tc power. Check the sensor groun for open circuit, short circuit to power. Check the signal circuit for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new manifo absolute pressure sensor as required ▪ Using the manufacturer approv diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). Refer to the worksho manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are n blockages or air leaks preventir communication of pressure changes between the manifold and the sensor
P0107-	Manifold Absolute		▪ Refer to the electrical circuit

	16	Pressure/Barometric Pressure Sensor Circuit Low - Circuit voltage below threshold	**NOTE:** Circuit reference MAP_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure ▪ Blockage or leak between the intake manifold and the manifold pressure sensor	diagrams and check the 5 volt supply circuit for open circuit, short circuit to power, short circuit to ground. Check the sensor ground for open circuit, short circuit to power. Check th signal circuit for open circuit, short circuit to power, short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new manifo absolute pressure sensor as required ▪ Using the manufacturer approv diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). This DTC is set when the manifold absolute pressure sensor signal voltage is lower than expected by the engine control module. Refer to the workshop manual and check th manifold absolute pressure sensor is correctly connected to the air intake manifold and ther are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor
	P0108-00	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High - No sub type information	**NOTE:** Circuit reference MAP_A ▪ Harness failure - Manifold absolute pressure sensor circuits ▪ Manifold absolute pressure sensor failure	▪ Refer to the electrical circuit diagrams and check the 5 volt supply circuit for open circuit, short circuit to power, short circuit to ground. Check the sensor ground for open circuit, short circuit to power. Check th signal circuit for open circuit, short circuit to power, short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store

			Blockage or leak between the intake manifold and the manifold pressure sensor	DTCs using the 'Diagnosis men tab and retest - Check and install a new manifo absolute pressure sensor as required - Using the manufacturer approv diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). Refer to the worksho manual and check the manifold absolute pressure sensor is correctly connected to the air intake manifold and there are n blockages or air leaks preventir communication of pressure changes between the manifold and the sensor
	P0108-17	Manifold Absolute Pressure/Barometric Pressure Sensor Circuit High - Circuit voltage above threshold	**NOTE:** Circuit reference MAP_A - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Manifold absolute pressure sensor circuits - Manifold absolute pressure sensor failure - Blockage or leak between the intake manifold and the manifold pressure sensor	- Refer to the electrical circuit diagrams and check the 5 volt supply circuit for open circuit, short circuit to power, short circuit to ground. Check the sensor ground for open circuit, short circuit to power. Check th signal circuit for open circuit, short circuit to power, short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new manifo absolute pressure sensor as required - Using the manufacturer approv diagnostic system check datalogger signals, Corrected Intake Manifold Absolute Pressure (0x0504), Manifold Absolute Pressure Sensor Voltage (0x0301), Manifold Absolute Pressure Bank 1 (0x052C). This DTC is set when the manifold absolute pressure

				sensor signal voltage is greater than expected by the engine control module. Refer to the workshop manual and check th manifold absolute pressure sensor is correctly connected to the air intake manifold and ther are no blockages or air leaks preventing communication of pressure changes between the manifold and the sensor
	P010A-92	Mass or Volume Air Flow Sensor "B" Circuit - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Air flow disruption at sensing element of mass air flow ▪ Harness failure - Mass air flow sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Mass air flow failure	▪ This DTC is set if the mass air flow signal from sensor 2 is greater than the value expecte by the engine control module. Refer to the relevant sections o the workshop manual and chec the induction system for air leal and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element ▪ Refer to the electrical circuit diagrams and check mass air flo sensor 2 for short circuit to ground, short circuit to power, open circuit, high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new mass ai flow sensor 2 as required
	P010B-21	Mass or Volume Air Flow B Circuit Range/Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Air flow disruption at sensing element of mass air flow ▪ Harness failure - Mass air flow sensor circuit short circuit to ground,	▪ This DTC is set if the mass air flow signal from sensor 2 is less than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element ▪ Refer to the electrical circuit

			- high resistance, open circuit - Mass air flow failure	diagrams and check mass air flo sensor 2 for short circuit to ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new mass ai flow sensor 2 as required
	P010B-22	Mass or Volume Air Flow B Circuit Range/Performance - Signal amplitude > maximum	- The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high - Air flow disruption at sensing element of mass air flow sensor 2 - Harness failure - Mass air flow sensor circuit short circuit to power - Mass air flow failure	- This DTC is set if the mass air flow signal from sensor 2 is mor than the value expected by the engine control module. Refer to the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element - Refer to the electrical circuit diagrams and check mass air flo sensor 2 circuit for short circuit power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new mass ai flow sensor 2 as required
	P010C-00	Mass or Volume Air Flow B Circuit Low - No sub type information	**NOTE:** Circuit reference MAF_A - Air flow disruption at sensing element of mass air flow - Harness failure - Mass air flow sensor circuits - Mass air flow failure	- Using the manufacturer approv diagnostic system check datalogger signals, Air Flow Ra From Mass Air Flow Sensor Bar 2 (0x0505). This DTC is set if th mass air flow signal from senso is less than the value expected the engine control module. Ref to the relevant sections of the workshop manual and check th induction system for air leaks, and obstructions to flow. Check the condition of the air filter an examine the induction pipes fo debris which could disrupt air flow at the sensing element

			▪ Refer to the electrical circuit diagrams and check the power and ground circuits for the mas air flow sensors for high resistance, short circuit to powe short circuit to ground. Check t mass air flow signal line betwee the engine control module and the mass air flow sensor for hig resistance. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor as required
P010D-00	Mass or Volume Air Flow B Circuit High - No sub type information	**NOTE:** Circuit reference MAF_A ▪ Air flow disruption at sensing element of mass air flow ▪ Harness failure - Mass air flow sensor circuits ▪ Mass air flow failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Air Flow Ra From Mass Air Flow Sensor Bar 2 (0x0505). This DTC is set if th mass air flow signal from senso is greater than the value expected by the engine contro module. Refer to the relevant sections of the workshop manu and check the induction system for air leaks, and obstructions t flow. Check the condition of th air filter and examine the induction pipes for debris whicl could disrupt air flow at the sensing element ▪ Refer to the electrical circuit diagrams and check the power and ground circuits for the mas air flow sensors for high resistance, short circuit to powe short circuit to ground. Check t mass air flow signal line betwee the engine control module and the mass air flow sensor for hig resistance. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new mass ai flow sensor as required

P0116-00	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - No sub type information	**NOTE:** Circuit reference ECT_A ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuits ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DT is set if the engine coolant temperature value fails plausibility checks by the engin control module. Refer to the workshop manual and check th engine cooling system to ensur the coolant condition and level correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuits for high resistance, shor circuit to other circuits. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine coolant temperature sensor as required
P0116-16	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Circuit voltage below threshold	**NOTE:** Circuit reference ECT_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuits ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DT is set if the engine coolant temperature sensor signal voltage is less than the value expected by the engine contro module. Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuits for high resistance, shor circuit to other circuits. Repair harness as required. Using the manufacturer approved diagnostic system clear all store

				DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine coolant temperature sensor as required
	P0116-17	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference ECT_A ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuits ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DT is set if the engine coolant temperature sensor signal voltage is greater than the valu expected by the control modul Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuits for high resistance, shor circuit to other circuits. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine coolant temperature sensor as required
	P0116-21	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuit short circuit to ground, high resistance, open circuit	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DT is set if the engine coolant temperature sensor signal amplitude is lower than the valu expected by the control modul Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine

			▪ Engine coolant temperature sensor failure	coolant temperature sensor circuit for short circuit to groun high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine coolant temperature sensor as required
	P0116-22	Engine Coolant Temperature Sensor 1 Circuit Range/Performance - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Engine coolant level or flow failure ▪ Harness failure - Engine coolant temperature sensor circuit short circuit to power ▪ Engine coolant temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Coolant Temperature Sensor (0x0357), Engine Coolant Temperature (0xF405). This DT is set if the engine coolant temperature sensor signal amplitude is greater than the value expected by the control module. Refer to the workshop manual and check the engine cooling system to ensure the coolant condition and level is correct ▪ Refer to the electrical circuit diagrams and check the engine coolant temperature sensor circuit for short circuit to powe Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine coolant temperature sensor as required
	P0130-00	O2 Sensor Circuit (Bank 1 Sensor 1) - No sub type information	**NOTE:** Circuit reference LPV_A LPPC_A ▪ Harness failure - Heated	▪ Refer to the electrical circuit diagrams and check the heatec oxygen sensor circuits for open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated

			oxygen sensor circuit open circuit ▪ Heated oxygen sensor component failure	oxygen sensor as required
	P0130-11	O2 Circuit (Bank 1, Sensor 1) - Circuit short to ground	**NOTE:** Circuit reference LPV_A ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Heated oxygen sensor circuit short circuit to ground ▪ Heated oxygen sensor component failure	▪ Refer to the electrical circuit diagrams and check the heatec oxygen sensor circuits for short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new heated oxygen sensor as required
	P0130-12	O2 Circuit (Bank 1, Sensor 1) - Circuit short to battery	**NOTE:** Circuit reference LPV_A ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure - Heated oxygen sensor circuit short circuit to power ▪ Heated oxygen sensor	▪ Refer to the electrical circuit diagrams and check the heatec oxygen sensor circuits for short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new heated oxygen sensor as required

		component failure	
P0130-13	O2 Circuit (Bank 1, Sensor 1) - Circuit open	**NOTE:** Circuit reference LPPC_A LPV_A ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Heated oxygen sensor circuit open circuit ▪ Heated oxygen sensor component failure	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
P0130-1A	O2 Circuit (Bank 1, Sensor 1) - Circuit resistance below threshold	**NOTE:** Circuit reference LPTR_A ▪ Harness failure - Heated oxygen sensor circuit failure ▪ Heated oxygen sensor component failure	▪ This DTC is set when the heate oxygen sensor internal trim resistance value is less than tha expected by the engine contro module. Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits f short circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
P0130-1B	O2 Circuit (Bank 1, Sensor 1) - Circuit resistance above threshold	**NOTE:** Circuit reference LPTR_A ▪ The engine control module has infered a circuit resistance above a specified range	▪ This DTC is set when the oxyge sensor internal trim resistance value is greater than that expected by the engine contro module. Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits f short circuits, open circuits. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store

			- Harness failure - Heated oxygen sensor circuit failure - Heated oxygen sensor component failure	DTCs using the 'Diagnosis men tab and retest - Check and install a new heated oxygen sensor as required
	P0130-26	O2 Circuit (Bank 1, Sensor 1) - Signal rate of change below threshold	- The signal transitions more slowly than is reasonably allowed - Exhaust system leak - Fuel control system failure - Heated oxygen sensor to engine control module circuit short circuit to ground, short circuit to power, high resistance - Heated oxygen sensor failure	- Check for and rectify any exhau leak between cylinder head anc catalytic converter. Check heate oxygen sensor is correctly installed in exhaust manifold - Check fuel control system for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check heated oxygen sensor to engine contro module circuit for short circuit t ground, short circuit to power, high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new heated oxygen sensor as required
	P0133-00	O2 Circuit Slow Response (Bank 1, Sensor 1) - No sub type information	- Exhaust system leak - Fuel control system failure - Heated oxygen sensor to engine control module wiring shield high resistance - Heated oxygen sensor failure	- Check for and rectify any exhau leak between cylinder head anc catalytic converter. Check heate oxygen sensor is correctly installed in exhaust manifold - Check fuel control system for related DTCs and refer to the relevant DTC index - Refer to the electrical circuit diagrams and check heated oxygen sensor to engine contro module wiring shield for high resistance. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new heated oxygen sensor as required

P0135-16	O2 Sensor Heater Circuit (Bank 1 Sensor 1) - Circuit voltage below threshold	**NOTE:** Circuit reference LPPH_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Heated oxygen sensor to engine control module circuit short circuit to ground, open circuit ▪ Heated oxygen sensor failure	▪ Check fuel control system for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor to engine contro module circuit for short circuit t ground, open circuit ▪ Check and install a new heated oxygen sensor as required
P0148-00	Fuel Delivery Error - No sub type information	▪ This DTC is set after the engine control module internal monitoring function has evaluated high pressure fuel pump fatigue and wear against fuel rail pressure	▪ Check high pressure fuel pump for mechanical damage and excessive wear ▪ Check and install a new high pressure fuel pump as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P0148-91	Fuel Delivery Error - Parametric	▪ The engine control module has detected that the component parameter e.g. capacitance or inductance is outside its expected range ▪ This DTC is set as part of the high pressure fuel pump protection strategy. The DTC has set because the high pressure fuel pump is reaching a calibrated number of engine re-starts	▪ Check high pressure fuel pump for mechanical damage and excessive wear ▪ Check and install a new high pressure fuel pump as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

		▪ First level of high press fuel pump re-starts has been reached	
P0148-93	Fuel Delivery Error - No operation	▪ The engine control module has detected that the component is not operating ▪ This DTC is set as part of the high pressure fuel pump protection strategy. The DTC has set because the high pressure fuel pump is reaching a calibrated number of engine re-starts ▪ Second level of high press fuel pump re-starts has been reached	▪ Check high pressure fuel pump for mechanical damage and excessive wear ▪ Check and install a new high pressure fuel pump as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P0181-16	Fuel Temperature Sensor A Circuit Range/Performance - Circuit voltage below threshold	**NOTE:** Circuit reference IFTS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Fuel temperature sensor circuit failure ▪ Sensor component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fue Rail Temperature Sensor Voltag (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is les than that expected by the engi control module. Refer to the electrical circuit diagrams and check the fuel temperature sensor signal circuit for short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel temperature sensor as required
P0181-17	Fuel Temperature Sensor A Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference IFTS	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fue Rail Temperature Sensor Voltag (0x033F). This DTC is set when

			- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Fuel temperature sensor circuit failure - Sensor component failure	the voltage on the fuel temperature sensor circuit is greater than that expected by the engine control module. Ref to the electrical circuit diagram and check the fuel temperature sensor signal and ground circui for short circuit to power, open circuit or high resistance. Repai harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new fuel temperature sensor as required
	P0182-00	Fuel Temperature Sensor A Circuit Low - No sub type information	**NOTE:** Circuit reference IFTS - Harness failure - Fuel temperature sensor circuit failure - Sensor component failure	- Using the manufacturer approv diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fue Rail Temperature Sensor Voltag (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is les than that expected by the engi control module. Refer to the electrical circuit diagrams and check the fuel temperature sensor signal circuit for short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new fuel temperature sensor as required
	P0183-00	Fuel Temperature Sensor A Circuit High - No sub type information	**NOTE:** Circuit reference IFTS - Harness failure - Fuel temperature sensor	- Using the manufacturer approv diagnostic system check datalogger signals, Fuel Temperature "A" (0x0522), Fue Rail Temperature Sensor Voltag (0x033F). This DTC is set when the voltage on the fuel temperature sensor circuit is greater than that expected by

			circuit failure ▪ Sensor component failure	the engine control module. Ref to the electrical circuit diagram and check the fuel temperature sensor signal and ground circui for short circuit to power, open circuit or high resistance. Repai harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel temperature sensor as required
	P0191-16	Fuel Rail Pressure Sensor A Circuit Range/Performance - Circuit voltage below threshold	**NOTE:** Circuit reference RPS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Fuel supply system failure ▪ Harness failure - Fuel rail pressure sensor circuit failure ▪ Sensor component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressur transducer mounted on the fue rail. This DTC is set when the voltage on the signal line is less than that expected by the engi control module. Refer to the workshop manual and ensure that the fuel supply system is working correctly ▪ Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for intermittent failures, open circuits, short circuits, high resistance. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel rai pressure sensor as required
	P0191-17	Fuel Rail Pressure Sensor A Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference RPS	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressur transducer mounted on the fue rail. This DTC is set when the

			▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Fuel supply system failure ▪ Harness failure - Fuel rail pressure sensor circuit failure ▪ Sensor component failure	voltage on the signal line is greater than that expected by the engine control module. Ref to the workshop manual and ensure that the fuel supply system is working correctly ▪ Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for intermittent failures, open circuits, short circuits, high resistance. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel rai pressure sensor as required
	P0192-16	Fuel Rail Pressure Sensor A Circuit Low - Circuit voltage below threshold	**NOTE:** Circuit reference RPS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Fuel supply system failure ▪ Harness failure - Fuel rail pressure sensor circuit failure ▪ Sensor component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressur transducer mounted on the fue rail. This DTC is set when the voltage on the signal line is less than that expected by the engi control module. Refer to the workshop manual and ensure that the fuel supply system is working correctly ▪ Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for short circuit to power, short circuit to ground, open circuit, high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel rai pressure sensor as required
	P0193-	Fuel Rail Pressure Sensor A		▪ Using the manufacturer approv

	17	Circuit High - Circuit voltage above threshold	**NOTE:** Circuit reference RPS - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Fuel supply system failure - Harness failure - Fuel rail pressure sensor circuit failure - Sensor component failure	diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressur transducer mounted on the fue rail. This DTC is set when the voltage on the signal line is greater than that expected by the engine control module. Ref to the workshop manual and ensure that the fuel supply system is working correctly - Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for short circuit to power, short circuit to ground, open circuit, high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new fuel rai pressure sensor as required
	P0194-00	Fuel Rail Pressure Sensor A Circuit Intermittent/Erratic - No sub type information	- Fuel supply system failure - Harness failure - Fuel rail pressure sensor circuit failure - Sensor component failure	- Using the manufacturer approv diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). The fuel rail pressure sensor is a pressur transducer mounted on the fue rail. Refer to the workshop manual and ensure that the fue supply system is working correctly - Refer to the electrical circuit diagrams and check the sensor supply, signal and ground connections for intermittent failures, open circuits, short circuits, high resistance. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new fuel rai

				pressure sensor as required
	P0195-00	Engine Oil Temperature Sensor Circuit - No sub type information	**NOTE:** Circuit reference OTL ▪ Oil contaminated or level incorrect ▪ Harness failure - Oil level and temperature sensor circuits ▪ Oil level and temperature sensor failure	▪ Check the oil level is correct an the oil does not appear contaminated. Renew or top up oil as required ▪ Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line fo open circuit, short circuit to power, short circuit to ground, high resistance and intermitten connections. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new engine oil level and temperature senso as required
	P0195-23	Engine Oil Temperature Sensor Circuit - Signal stuck low	**NOTE:** Circuit reference OTL ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Oil contaminated or level incorrect ▪ Harness failure - Oil level and temperature sensor circuits ▪ Oil level and temperature sensor failure	▪ Check the oil level is correct an the oil does not appear contaminated. Renew or top up oil as required ▪ Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line fo open circuit, short circuit to power, short circuit to ground, high resistance and intermitten connections. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new engine oil level and temperature senso as required
	P0195-62	Engine Oil Temperature Sensor Circuit - Signal compare failure	▪ The engine control module detected failure when comparing two or	▪ Check the oil does not appear contaminated and the level is correct. Renew or top up oil as

		▪ more input parameters for plausibility ▪ Oil contaminated or level incorrect ▪ Harness failure - Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Oil level and temperature sensor failure	required ▪ Refer to the electrical circuit diagrams and check oil level an temperature sensor signal circu for short circuit to ground, shor circuit to power, high resistance open circuit, disconnected ▪ Start the engine from cold and allow to idle, check and record Sump Oil Temperature - Measured (0x03F3) datalogger signal. Continue to warm up at idle, after approximately 10 minutes check and record Sump Oil Temperature - Measured (0x03F3) signal. If value of signa has not increased by 5°C Check and install a new oil level and temperature sensor as required
P0201-00	Cylinder 1 Injector Circuit/Open - No sub type information	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ Harness failure - Cylinder 1 injector circuit open circuit ▪ Cylinder 1 injector failure	▪ This DTC is set when the engin control module detects an oper circuit on the cylinder 1 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector fo high resistance, open circuits or intermittent connections. Repai harness as required. ▪ Check and install a new fuel injector as required
P0202-00	Cylinder 2 Injector Circuit/Open - No sub type information	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 ▪ Harness failure - Cylinder 2 injector circuit open circuit ▪ Cylinder 2 injector failure	▪ This DTC is set when the engin control module detects an oper circuit on the cylinder 2 injector control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector fo high resistance, open circuits or intermittent connections. Repai harness as required ▪ Check and install a new fuel injector as required

	P0203-00	Cylinder 3 Injector Circuit/Open - No sub type information	**NOTE:** Circuit reference O_P_PVL13 O_P_PVH13 ▪ Harness failure - Cylinder 3 injector circuit open circuit ▪ Cylinder 3 injector failure	▪ This DTC is set when the engin control module detects an ope circuit on the cylinder 3 injecto control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector fo high resistance, open circuits o intermittent connections. Repa harness as required ▪ Check and install a new fuel injector as required
	P0204-00	Cylinder 4 Injector Circuit/Open - No sub type information	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ Harness failure - Cylinder 4 injector circuit open circuit ▪ Cylinder 4 injector failure	▪ This DTC is set when the engin control module detects an ope circuit on the cylinder 4 injecto control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector fo high resistance, open circuits o intermittent connections. Repa harness as required ▪ Check and install a new fuel injector as required
	P0205-00	Cylinder 5 Injector Circuit/Open - No sub type information	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ Harness failure - Cylinder 5 injector circuit open circuit ▪ Cylinder 5 injector failure	▪ This DTC is set when the engin control module detects an ope circuit on the cylinder 5 injecto control circuit. Refer to the electrical circuit diagrams and check the two control circuits between the engine control module and the fuel injector fo high resistance, open circuits o intermittent connections. Repa harness as required ▪ Check and install a new fuel injector as required
	P0206-00	Cylinder 6 Injector Circuit/Open - No sub type information		▪ This DTC is set when the engin control module detects an ope circuit on the cylinder 6 injecto control circuit. Refer to the electrical circuit diagrams and

		NOTE: Circuit reference O_P_PVL23 O_P_PVH23 ▪ Harness failure - Cylinder 6 injector circuit open circuit ▪ Cylinder 6 injector failure	check the two control circuits between the engine control module and the fuel injector fo high resistance, open circuits o intermittent connections. Repai harness as required ▪ Check and install a new fuel injector as required
P0216-00	Injector/Injection timing Control Circuit - No sub type information	▪ Internal engine control module power supply is unable to supply the fuel injectors with the maximum number of injections required ▪ Internal engine control module monitoring has detected a fuel pressure not able to meet the demand for the number of fuel injections ▪ At high engine speeds the internal engine control module monitoring has detected that the computing time available for the desired number of injections is not sufficient	▪ Refer to the battery care requirements section and verify that the vehicle battery is fully charged and serviceable before continuing with further diagnostic tests ▪ Check the vehicle charging system performance to ensure the voltage regulation is correc ▪ Refer to the electrical circuit diagrams and check engine control module power and ground circuits ▪ Check fuel control system for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P0219-00	Engine Overspeed Condition - No sub type information	▪ Camshaft or crankshaft position sensor circuit short circuit to ground, short circuit to power, open circuit ▪ Camshaft or crankshaft position sensor failure	▪ Refer to the electrical circuit diagrams and check camshaft and crankshaft position sensor circuits for short circuit to ground, short circuit to power, open circuit ▪ Check for engine oil ingestion t sensors. Check and install a nev camshaft and crankshaft positic sensor as required
P0235-	Turbocharger/Supercharger		▪ Using the manufacturer approv

	16	Boost Sensor A Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference SCOP ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Mechanical failure - Sensor hose blocked or leaking ▪ Failure affecting intake air circuit or charge air circuit control valves or actuators ▪ Harness failure - Charge air pressure sensor circuit ▪ Charge air pressure sensor failure	diagnostic system check datalogger signals, Turbocharger/Supercharger Boost Sensor A Circuit (0x033C Boost Absolute Pressure - Raw Value (0x033E). This DTC is set when the engine control modul detects a signal voltage from th charge air pressure sensor signa line which is less than the threshold value. Check the sensor hose for blockages or leaks ▪ Check the intake air and charge air circuits for failures including leaks, blockages, control valve actuator malfunctions ▪ Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line fo open circuit, short circuit to ground, high resistance and intermittent connections. Repai harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new charge air pressure sensor as required
	P0235-17	Turbocharger/Supercharger Boost Sensor A Circuit - Circuit voltage above threshold	**NOTE:** Circuit reference SCOP ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Mechanical failure - Sensor hose blocked or leaking	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbocharger/Supercharger Boost Sensor A Circuit (0x033C Boost Absolute Pressure - Raw Value (0x033E). This DTC is set when the engine control modul detects a signal voltage from th charge air pressure sensor signa line which is greater than the threshold value. Check the sensor hose for blockages or leaks ▪ Check the intake air and charge air circuits for failures including leaks, blockages, control valve actuator malfunctions

			▪ Failure affecting intake air circuit or charge air circuit control valves or actuators ▪ Harness failure - Charge air pressure sensor circuit ▪ Charge air pressure sensor failure	▪ Refer to the electrical circuit diagrams and check the sensor power and ground supplies for open circuit or short circuit to ground. Check the signal line fo open circuit, short circuit to power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new charge air pressure sensor as required
	P0235-94	Turbocharger/Supercharger Boost Sensor A Circuit - Unexpected operation	**NOTE:** Circuit reference SCOP ▪ The engine control module has detected that the component is operating in a way or at a time that it has not been commanded to operate ▪ Charge air solenoid stuck open mono turbo mode ▪ Turbine intake solenoid leakage when closed ▪ Turbine intake solenoid stuck open ▪ Intake air system, blocked low pressure air intake. This failure mode can be caused by snow packing in the intake system. Symptoms often disappear after the vehicle has been warmed and heat soaked. Similar symptoms to seized primary turbo	▪ If this DTC is logged with P00B 07 suspect, charge air solenoid stuck open mono turbo mode ▪ If this DTC is logged with P00B 07 suspect, turbine intake solenoid leakage when closed ▪ If this DTC is logged with P00B 07, P22D2-77, P1247-00 & P22CF-71 suspect, turbine intal solenoid stuck open ▪ If this DTC is logged with P00B 07, P22D2-77, P1247-00 & P22CF-71 suspect, intake air system, blocked low pressure a intake ▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine

P023D-00	Manifold Absolute Pressure-Turbocharger/Supercharger Boost Sensor A Correlation - No sub type information	▪ Induction system air leak or blockage ▪ Charge air system leak or blockage ▪ Manifold absolute pressure sensor A failure ▪ Variable geometry turbocharger actuator A sticking, failure ▪ Turbocharger A failure	▪ Check induction system for leal blockages ▪ Check charge air system for leaks, blockages. Check for related DTCs and refer to the relevant DTC index ▪ Check and install a new manifo absolute pressure sensor as required ▪ Check and install a new variabl geometry turbocharger actuato as required ▪ Check turbocharger rod connection and oil seals
P0251-13	Injection Pump Fuel Metering Control A - Circuit open	**NOTE:** Circuit reference MEU ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Fuel volume control valve circuit open circuit ▪ Fuel volume control valve failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volum Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). Refer to the electrica circuit diagrams and check the fuel volume control valve circui between the engine control module and the fuel volume control valve for open circuit. Check power supply to fuel volume control valve. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel volume control valve as require
P0251-4B	Injection Pump Fuel Metering Control A - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Fuel volume control valve circuit short circuit to ground, short circuit to power, high resistance	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volum Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). Refer to the electrica circuit diagrams and check the fuel volume control valve circui between the engine control module and the fuel volume

			▪ Fuel volume control valve failure	control valve for short circuit to ground, short circuit to power, high resistance. Check power supply to fuel volume control valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel volume control valve as require
	P0252-16	Injection Pump Fuel Metering Control A Range/Performance - Circuit voltage below threshold	**NOTE:** Circuit reference MEU ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Fuel volume control valve circuit ▪ Fuel volume control valve failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volum Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is less than that expected by th engine control module. Refer to the electrical circuit diagrams a check the fuel volume control valve circuit between the engin control module and the fuel volume control valve for an intermittent open circuit, high resistance, short circuit to ground. Check the power supp to fuel volume control valve. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel volume control valve as require
	P0252-17	Injection Pump Fuel Metering Control A Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference MEU ▪ The engine control module measured a voltage above a	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volum Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is greater than that expected b the engine control module. Ref

			specified range but not necessarily a short circuit to power ▪ Harness failure - Fuel volume control valve circuit ▪ Fuel volume control valve failure	to the electrical circuit diagram and check the fuel volume control valve circuit between th engine control module and the volume control valve for a short circuit to power. Check the power supply to fuel volume control valve. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new fuel volume control valve as require
	P0253-00	Injection Pump Fuel Metering Control A Low - No sub type information	**NOTE:** Circuit reference MEU ▪ Harness failure - Fuel volume control valve circuit short circuit to ground, open circuit ▪ Fuel volume control valve failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is less than that expected by th engine control module. Refer to the electrical circuit diagrams a check the fuel volume control valve circuit between the engin control module and the fuel volume control valve for a short circuit to ground, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel volume control valve as require
	P0254-00	Injection Pump Fuel Metering Control A High - No sub type information	**NOTE:** Circuit reference MEU ▪ Harness failure - Fuel volume control valve	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). This DTC is set when the voltage on the signal circuit to the fuel volume control valve is greater than that expected b

			circuit short circuit to power ▪ Fuel volume control valve failure	the engine control module. Ref to the electrical circuit diagram and check the fuel volume control valve circuit between th engine control module and the fuel volume control valve for a short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel volume control valve as require
	P0255-00	Injection Pump Fuel Metering Control A Intermittent - No sub type information	▪ Harness failure - Fuel volume control valve circuit intermittent short circuit to ground, short circuit to power, high resistance ▪ Fuel volume control valve intermittent failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Volume Control Valve Duty Cycle (0x03C2), Fuel Volume Control Valve Current - Measured (0x03EA). Refer to the electrica circuit diagrams and check the fuel volume control valve circui between the engine control module and the fuel volume control valve for an intermitten short circuit to ground, short circuit to power, high resistance Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel volume control valve as require
	P0261-00	Cylinder 1 Injector Circuit Low - No sub type information	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ Harness failure - Injector control circuit short circuit ▪ Injector failure	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 1 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

				▪ Check and install a new fuel injector as required
	P0261-11	Cylinder 1 Injector Circuit Low - Circuit short to ground	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Injector control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 1 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to ground Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0261-23	Cylinder 1 Injector Circuit Low - Signal stuck low	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Harness failure - Short circuit between injector control circuits on different cylinders	▪ Refer to the electrical circuit diagrams and check the injecto control circuits between the engine control module and the cylinder 1 injector for short circ to other injector control circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0264-00	Cylinder 2 Injector Circuit Low - No sub type information		▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 2 injector for short circ to ground or short between the

			NOTE: Circuit reference O_P_PVL12 O_P_PVH12 ▪ Harness failure - Injector control circuit short circuit ▪ Injector failure	two wires. This circuit is a twiste pair, check both high and low sides for short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0264-11	Cylinder 2 Injector Circuit Low - Circuit short to ground	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Injector control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 2 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to ground Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0264-23	Cylinder 2 Injector Circuit Low - Signal stuck low	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Harness failure - Short	▪ Refer to the electrical circuit diagrams and check the injecto control circuits between the engine control module and the cylinder 2 injector for short circ to other injector control circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

		circuit between injector control circuits on different cylinders	
P0267-00	Cylinder 3 Injector Circuit Low - No sub type information	**NOTE:** Circuit reference O_P_PVL13 O_P_PVH13 ▪ Harness failure - Injector control circuit short circuit ▪ Injector failure	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 3 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
P0267-11	Cylinder 3 Injector Circuit Low - Circuit short to ground	**NOTE:** Circuit reference O_P_PVL13 O_P_PVH13 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Injector control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 3 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to ground Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
P0267-23	Cylinder 3 Injector Circuit Low - Signal stuck low		▪ Refer to the electrical circuit diagrams and check the injecto control circuits between the engine control module and the cylinder 3 injector for short circ to other injector control circuits Repair harness as required. Usi

			NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Harness failure - Short circuit between injector control circuits on different cylinders	the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0270-00	Cylinder 4 Injector Circuit Low - No sub type information	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ Harness failure - Injector control circuit short circuit ▪ Injector failure	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 4 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0270-11	Cylinder 4 Injector Circuit Low - Circuit short to ground	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 4 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to ground Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			measurement when another value was expected ▪ Harness failure - Injector control circuit short circuit to ground	▪ Check and install a new fuel injector as required
	P0270-23	Cylinder 4 Injector Circuit Low - Signal stuck low	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Harness failure - Short circuit between injector control circuits on different cylinders	▪ Refer to the electrical circuit diagrams and check the injecto control circuits between the engine control module and the cylinder 4 injector for short circ to other injector control circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0273-00	Cylinder 5 Injector Circuit Low - No sub type information	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ Harness failure - Injector control circuit short circuit ▪ Injector failure	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 5 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0273-11	Cylinder 5 Injector Circuit Low - Circuit short to ground		▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 5 injector for short circ to ground or short between the

			NOTE: Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Injector control circuit short circuit to ground	two wires. This circuit is a twiste pair, check both high and low sides for short circuit to ground Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0273-23	Cylinder 5 Injector Circuit Low - Signal stuck low	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Harness failure - Short circuit between injector control circuits on different cylinders	▪ Refer to the electrical circuit diagrams and check the injecto control circuits between the engine control module and the cylinder 5 injector for short circ to other injector control circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0276-00	Cylinder 6 Injector Circuit Low - No sub type information	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ Harness failure - Injector control circuit short	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 6 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to power. Repair harness as required. Usi

			circuit ▪ Injector failure	the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0276-11	Cylinder 6 Injector Circuit Low - Circuit short to ground	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Injector control circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the injecto control circuit between the engine control module and the cylinder 6 injector for short circ to ground or short between the two wires. This circuit is a twiste pair, check both high and low sides for short circuit to ground Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel injector as required
	P0276-23	Cylinder 6 Injector Circuit Low - Signal Stuck Low	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ The engine control module measures a signal that remains low when transitions are expected ▪ Harness failure - Short circuit between injector control circuits on different cylinders	▪ Refer to the electrical circuit diagrams and check the injecto control circuits between the engine control module and the cylinder 6 injector for short circ to other injector control circuits Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P02CD-00	Cylinder 1 Fuel Injector Offset Learning at Max	▪ Corrected set point voltage of the piezo	▪ Check for other related DTCs and refer to the relevant DTC

		Limit - No sub type information	actuator violates the on board diagnostic limit ▪ Fuel injector internal components out of tolerance	index. Using the manufacturer approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new a fuel injector as required
	P02CF-00	Cylinder 2 Fuel Injector Offset Learning at Max Limit - No sub type information	▪ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit ▪ Fuel injector internal components out of tolerance	▪ Check for other related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new a fuel injector as required
	P02D1-00	Cylinder 3 Fuel Injector Offset Learning at Max Limit - No sub type information	▪ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit ▪ Fuel injector internal components out of tolerance	▪ Check for other related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new a fuel injector as required
	P02D3-00	Cylinder 4 Fuel Injector Offset Learning at Max Limit - No sub type information	▪ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit ▪ Fuel injector internal components out of tolerance	▪ Check for other related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new a fuel injector as required
	P02D5-00	Cylinder 5 Fuel Injector Offset Learning at Max Limit - No sub type information	▪ Corrected set point voltage of the piezo actuator violates the on board diagnostic limit ▪ Fuel injector internal components out of tolerance	▪ Check for other related DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new a fuel injector as required
	P02D7-00	Cylinder 6 Fuel Injector Offset Learning at Max Limit - No sub type	▪ Corrected set point voltage of the piezo actuator violates the on	▪ Check for other related DTCs and refer to the relevant DTC index. Using the manufacturer

	information	board diagnostic limit ▪ Fuel injector internal components out of tolerance	approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new a fuel injector as required
P02EE-17	Cylinder 1 Injector Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 1 inject control circuit that is above the diagnostic threshold. The inject control circuit consists of a twisted pair of wires between t engine control module and the Piezo actuator within the inject Refer to the electrical circuit diagrams and check both the control circuits (high and low) fo open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits Repair wiring harness as require ▪ Check and install a new a fuel injector as required
P02EE-1C	Cylinder 1 Injector Circuit Range/Performance - Circuit voltage out of range	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ The engine control module measured a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 1 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
P02EE-	Cylinder 1 Injector Circuit		▪ This DTC is set when the engin

	68	Range/Performance - Event information	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 - The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module - Harness failure - Injector control circuit - Fuel injector failure	control module monitors a voltage on the cylinder 1 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required - Check and install a new a fuel injector as required
	P02EF-17	Cylinder 2 Injector Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engin control module monitors a voltage on the cylinder 2 inject control circuit that is above the diagnostic threshold. The inject control circuit consists of a twisted pair of wires between t engine control module and the Piezo actuator within the inject Refer to the electrical circuit diagrams and check both the control circuits (high and low) fo open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits Repair wiring harness as require - Check and install a new a fuel injector as required
	P02EF-1C	Cylinder 2 Injector Circuit Range/Performance - Circuit voltage out of range		- This DTC is set when the engin control module monitors a voltage on the cylinder 2 inject control circuit that is out of

			NOTE: Circuit reference O_P_PVL12 O_P_PVH12 - The engine control module measured a voltage outside of the expected range, but not identified as too high or too low - Harness failure - Injector control circuit - Fuel injector failure	range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required - Check and install a new a fuel injector as required
	P02EF-68	Cylinder 2 Injector Circuit Range/Performance - Event information	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 - The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engin control module monitors a voltage on the cylinder 2 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required - Check and install a new a fuel injector as required
	P02F0-17	Cylinder 3 Injector Circuit Range/Performance - Circuit voltage above threshold		- This DTC is set when the engin control module monitors a voltage on the cylinder 3 inject control circuit that is above the diagnostic threshold. The inject control circuit consists of a

			NOTE: Circuit reference O_P_PVL13 O_P_PVH13 - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Harness failure - Injector control circuit - Fuel injector failure	twisted pair of wires between t engine control module and the Piezo actuator within the inject Refer to the electrical circuit diagrams and check both the control circuits (high and low) fo open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits Repair wiring harness as require - Check and install a new a fuel injector as required
	P02F0-1C	Cylinder 3 Injector Circuit Range/Performance - Circuit voltage out of range	**NOTE:** Circuit reference O_P_PVL13 O_P_PVH13 - The engine control module measured a voltage outside of the expected range, but not identified as too high or too low - Harness failure - Injector control circuit - Fuel injector failure	- This DTC is set when the engin control module monitors a voltage on the cylinder 3 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required - Check and install a new a fuel injector as required
	P02F0-68	Cylinder 3 Injector Circuit Range/Performance - Event information	**NOTE:** Circuit reference O_P_PVL13 O_P_PVH13 - The engine control module indicated the	- This DTC is set when the engin control module monitors a voltage on the cylinder 3 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and

			detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
	P02F1-17	Cylinder 4 Injector Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 4 inject control circuit that is above the diagnostic threshold. The inject control circuit consists of a twisted pair of wires between t engine control module and the Piezo actuator within the inject Refer to the electrical circuit diagrams and check both the control circuits (high and low) f open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits Repair wiring harness as require ▪ Check and install a new a fuel injector as required
	P02F1-1C	Cylinder 4 Injector Circuit Range/Performance - Circuit voltage out of range	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module measured a voltage outside of the expected range, but not identified as too high or too low	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 4 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance,

			▪ Harness failure - Injector control circuit ▪ Fuel injector failure	short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
	P02F1-68	Cylinder 4 Injector Circuit Range/Performance - Event information	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 4 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
	P02F2-17	Cylinder 5 Injector Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 5 inject control circuit that is above the diagnostic threshold. The inject control circuit consists of a twisted pair of wires between t engine control module and the Piezo actuator within the inject Refer to the electrical circuit diagrams and check both the control circuits (high and low) f open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits Repair wiring harness as require ▪ Check and install a new a fuel

			injector as required
P02F2-1C	Cylinder 5 Injector Circuit Range/Performance - Circuit voltage out of range	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module measured a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 5 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
P02F2-68	Cylinder 5 Injector Circuit Range/Performance - Event information	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 5 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
P02F3-	Cylinder 6 Injector Circuit		▪ This DTC is set when the engin

	17	Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	control module monitors a voltage on the cylinder 6 inject control circuit that is above the diagnostic threshold. The inject control circuit consists of a twisted pair of wires between t engine control module and the Piezo actuator within the inject Refer to the electrical circuit diagrams and check both the control circuits (high and low) fo open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits Repair wiring harness as require ▪ Check and install a new a fuel injector as required
	P02F3-1C	Cylinder 6 Injector Circuit Range/Performance - Circuit voltage out of range	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ The engine control module measured a voltage outside of the expected range, but not identified as too high or too low ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 6 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
	P02F3-68	Cylinder 6 Injector Circuit Range/Performance - Event information	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ The engine control	▪ This DTC is set when the engin control module monitors a voltage on the cylinder 6 inject control circuit that is out of range. The injector control circu consists of a twisted pair of wir between the engine control module and the Piezo actuator within the injector. Refer to the electrical circuit diagrams and

			module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Harness failure - Injector control circuit ▪ Fuel injector failure	check both the control circuits (high and low) for open circuit, short circuit to ground, short circuit to power, intermittent connections, high resistance, short to or interference from other circuits. Repair wiring harness as required ▪ Check and install a new a fuel injector as required
	P0300-00	Random Misfire Detected - No sub type information	▪ Fuel injector circuit failure(s) (injector DTCs also flagged) ▪ Fuel system failure	▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index. Refe to the electrical circuit diagram and check injector circuits for short circuit to ground, short circuit to power, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check for fuel system failure. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0301-00	Cylinder 1 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to groun short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index ▪ Check for fuel injector failure o blockage. Carry out cylinder compression tests

	P0302-00	Cylinder 2 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to groun short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
	P0303-00	Cylinder 3 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to groun short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
	P0304-00	Cylinder 4 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to groun short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder

				compression tests
	P0305-00	Cylinder 5 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to groun short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
	P0306-00	Cylinder 6 Misfire Detected - No sub type information	▪ Fuel injector electrical circuit failure(s) (injector DTCs also flagged) ▪ Fuel injector failure ▪ Cylinder compression low	▪ Refer to the electrical circuit diagrams and check injector circuit for short circuit to groun short circuit to power, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check for cylinder mis-fire, glov plug and injector DTCs and refe to the relevant DTC index ▪ Check for fuel injector failure or blockage. Carry out cylinder compression tests
	P0336-29	Crankshaft Position Sensor A Circuit Range/Performance - Signal Invalid	**NOTE:** Circuit reference CPS ▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Crankshaft position	▪ Refer to the electrical circuit diagrams and check crankshaft position sensor circuit for short circuit to ground, short circuit t power, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new crankshaft position sensor circu shielding as required

			▪ sensor circuit for short circuit to ground, short circuit to power, open circuit, disconnected ▪ Crankshaft position sensor circuit shielding failure ▪ Crankshaft Position sensor failure ▪ Crankshaft position sensor foreign matter on sensor face, gap incorrect ▪ Target wheel failure	▪ Check crankshaft position senso for foreign matter on crankshaf position sensor face. Check crankshaft position sensor air g ▪ Check and install a new crankshaft position sensor as required. Check and install a ne target wheel as required
	P0336-31	Crankshaft Position Sensor A Circuit Range/Performance - No Signal	**NOTE:** Circuit reference CPS ▪ The engine control module does not detect a signal which ought to be present ▪ Harness failure - Crankshaft position sensor circuits ▪ Crankshaft position sensor failure ▪ Crankshaft position sensor or reference target positioning incorrect	▪ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to th sensor, check the signal circuit f open circuits, short circuit to power, and short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the relevant section of the workshop manual. Check th sensor and crankshaft target fo damage, contamination, and correct mounting ▪ Check and install a new crankshaft position sensor as required. Check and install a ne target wheel as required
	P0336-64	Crankshaft Position Sensor A Circuit Range/Performance - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Crankshaft position sensor circuit for short circuit to ground, short circuit to power, open circuit, disconnected ▪ Crankshaft position	▪ Refer to the electrical circuit diagrams and check crankshaft position sensor circuit for short circuit to ground, short circuit t power, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes

			sensor circuit shielding failure ▪ Crankshaft Position sensor failure ▪ Crankshaft position sensor foreign matter on sensor face, gap incorrect ▪ Target wheel failure	▪ Check and install a new crankshaft position sensor circu shielding as required ▪ Check crankshaft position sens for foreign matter on crankshaf position sensor face. Check crankshaft position sensor air g ▪ Check and install a new crankshaft position sensor as required. Check and install a ne target wheel as required
	P0341-31	Camshaft Position Sensor A Circuit Range/Performance (Bank 1 or single sensor) - No Signal	**NOTE:** Circuit reference CID ▪ The engine control module does not detect a signal which ought to be present ▪ Harness failure - Camshaft position sensor circuits ▪ Camshaft position sensor failure ▪ Camshaft position sensor or reference target positioning incorrect	▪ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to th sensor, check the signal circuit f open circuit, short circuit to power, and short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the relevant section of the workshop manual. Check th sensor and camshaft target for damage, contamination, and correct mounting ▪ Check and install a new camsha position sensor as required
	P0341-3A	Camshaft Position Sensor A Circuit Range/Performance (Bank 1 or single sensor) - Incorrect has too many pulses	**NOTE:** Circuit reference CID ▪ Harness failure - Camshaft position sensor circuits ▪ Camshaft position sensor failure ▪ Camshaft position sensor or reference	▪ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to th sensor, check the signal circuit f open circuit, short circuit to power, and short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the relevant section of the workshop manual. Check th sensor and camshaft target for damage, contamination, and

		target positioning incorrect	correct mounting. Check camshaft timing is to specification ▪ Check and install a new camsha position sensor as required
P0342-64	Camshaft Position Sensor A Circuit Low (Bank 1 or single sensor) - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Harness failure - Camshaft position sensor circuit ▪ Camshaft position sensor failure ▪ Camshaft position sensor or reference target positioning incorrect	▪ Refer to the electrical circuit diagrams. Check the power supply and ground circuits to th sensor, check the signal circuit f open circuit, short circuit to power, and short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check the sensor and camshaft target for damage, contamination, and correct mounting. Check camshaft timi is to specification ▪ Check and install a new camsha position sensor as required
P0380-11	Glow Plug/Heater Circuit A - Circuit short to ground	**NOTE:** Circuit reference GPC ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure between engine control module and glow plug control module - Short circuit to ground ▪ Component failure - Glow plug control	▪ Using the manufacturer approv diagnostic system check datalogger signal, Glow Plug C Duty Cycle (0x9A04). Refer to t electrical circuit diagrams and check the control circuit from th engine control module to the glow plug control module for short circuit to ground. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module as require

		module failure	
P0380-12	Glow Plug/Heater Circuit A - Circuit short to battery	**NOTE:** Circuit reference GPC ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure between engine control module and glow plug control module - Short circuit to power ▪ Component failure - Glow plug control module failure	▪ Using the manufacturer approv diagnostic system check datalogger signal, Glow Plug C Duty Cycle (0x9A04). Refer to t electrical circuit diagrams and check the control circuit from th engine control module to the glow plug control module for short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module as require
P0380-13	Glow Plug/Heater Circuit A - Circuit open	**NOTE:** Circuit reference GPC ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure between engine control module and glow plug control module - Open circuit ▪ Component failure - Glow plug control	▪ Using the manufacturer approv diagnostic system check datalogger signal, Glow Plug C Duty Cycle (0x9A04). Refer to t electrical circuit diagrams and check the control circuit from th engine control module to the glow plug control module for short circuit to ground. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module as require

		module failure	
P0380-4B	Glow Plug/Heater Circuit A - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Glow plug heater circuit A short circuit to ground, short circuit to power, high resistance	▪ Using the manufacturer approv diagnostic system check datalogger signal, Glow Plug C Duty Cycle (0x9A04). Refer to t electrical circuit diagrams and check the glow plug heater circuit A for short circuit to ground, short circuit to power, high resistance. Repair harness required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes
P0401-00	Exhaust Gas Recirculation A Flow Insufficient Detected - No sub type information	▪ Intake air system, low pressure boost leak ▪ Exhaust gas recirculation actuator circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ If this DTC is logged with P00B 07 & P006A-00, suspect intake air system, low pressure boost leak ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required
P0402-00	Exhaust Gas Recirculation A Flow Excessive Detected - No sub type information	▪ Intake air system, high pressure boost leak ▪ Exhaust gas recirculation actuator circuit short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ If this DTC is logged with P124 00, P006A-00 & P00BF-07, suspect intake air system, high pressure boost leak ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to power, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes

				▪ Check and install a new exhaus gas recirculation actuator as required
	P0403-13	Exhaust Gas Recirculation A Control Circuit - Circuit open	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Exhaust gas recirculation actuator circuit high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation contro circuit for open circuit. This circ consists of two wires connectec between the engine control module and the exhaust gas recirculation valve motor. Checl both wires for open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation control actuat as required
	P0403-16	Exhaust Gas Recirculation A Control Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas recirculation actuator circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required

	P0403-19	Exhaust Gas Recirculation A Control Circuit - Circuit current above threshold	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module has measured current flow above a specified range ▪ Exhaust gas recirculation actuator circuit short circuit to power ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB). Refer tc the electrical circuit diagrams a check exhaust gas recirculation actuator circuit for short circuit power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required
	P0403-1D	Exhaust Gas Recirculation A Control Circuit - Circuit current out of range	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ▪ Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required
	P0403-4B	Exhaust Gas Recirculation A Control Circuit - Over temperature	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust ga

			▪ The engine control module detected an internal temperature above the expected range ▪ Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required
	P0403-71	Exhaust Gas Recirculation "A" Control Circuit / Open - Actuator stuck	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS ▪ The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Exhaust gas recirculation actuator circuit open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation actuator as required
	P0404-19	Exhaust Gas Recirculation A Control Circuit Range/Performance - Circuit current above threshold	**NOTE:** Circuit reference EGR_A_NEG EGR_A_POS	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Bank 1 Commanded (0x03FB) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to ground, short

			▪ The engine control module has measured current flow above a specified range ▪ Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance ▪ Exhaust gas recirculation actuator failure	circuit to power, high resistance Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation actuator as required
	P0405-00	Exhaust Gas Recirculation Sensor A Circuit Low - No sub type information	**NOTE:** Circuit reference EVP_A ▪ Exhaust gas recirculation actuator sensor circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 1 (0x052E) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator sensor circuit for short circuit to groun high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required
	P0405-11	Exhaust Gas Recirculation Sensor A Circuit Low - Circuit short to ground	**NOTE:** Circuit reference EVP_A ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Exhaust gas recirculation actuator position sensor	▪ Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation senso signal circuit for short circuit to ground. Check the power and ground circuits which supply th sensor. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation sensor as required

			circuit short circuit to ground ▪ Exhaust gas recirculation actuator failure	
	P0405-77	Exhaust Gas Recirculation Sensor A Circuit Low - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation actuator position sensor circuit short circuit to ground ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 1 (0x052E) ▪ Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation senso signal circuit for short circuit to ground. Check the power and ground circuits which supply th sensor. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation sensor as required
	P0406-00	Exhaust Gas Recirculation Sensor A Circuit High - No sub type information	**NOTE:** Circuit reference EVP_A ▪ Exhaust gas recirculation actuator circuit short circuit to power ▪ Exhaust gas recirculation actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 1 (0x052E) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 1 circui for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation sensor as required
	P0406-12	Exhaust Gas Recirculation Sensor A Circuit High - Circuit short to battery	**NOTE:** Circuit reference EVP_A	▪ Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation senso signal circuit for short circuit to power. Repair harness as required. Using the manufactur approved diagnostic system cle

			- The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Exhaust gas recirculation actuator position sensor circuit short circuit to power - Exhaust gas recirculation actuator failure	all stored DTCs using the 'Diagnosis menu' tab and retes Check and install a new exhaus gas recirculation sensor as required
	P0406-77	Exhaust Gas Recirculation Sensor A Circuit High - Commanded position not reachable	- The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - Exhaust gas recirculation actuator circuit short circuit to power - Exhaust gas recirculation actuator failure	- Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation senso signal circuit for short circuit to power. Check the power and ground circuits which supply th sensor. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new exhaus gas recirculation sensor as required
	P0407-00	Exhaust Gas Recirculation Sensor B Circuit Low - No sub type information	**NOTE:** Circuit reference TPS - Harness failure - Throttle position sensor circuit - Throttle position sensor failure	- Refer to the electrical circuit diagrams and check the throttl position sensor circuits for shor circuit to ground, open circuit. Repair harness as required, clea the DTC and retest system - Check and install a new throttl position sensor as required
	P0408-	Exhaust Gas Recirculation		Refer to the electrical circuit

	00	Sensor B Circuit High - No sub type information	**NOTE:** Circuit reference TPS ▪ Harness failure - Throttle position sensor circuit ▪ Throttle position sensor failure	diagrams and check the throttl position sensor circuits for shor circuit to ground, open circuit. Repair harness as required, clea the DTC and retest system ▪ Check and install a new throttle position sensor as required
	P0409-13	Exhaust Gas Recirculation Sensor A Circuit - Circuit open	**NOTE:** Circuit reference EVP_A ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Exhaust gas recirculation actuator circuit ▪ Exhaust gas recirculation actuator failure	▪ Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation senso signal circuit for open circuit. Check the power and ground circuits which supply the senso Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation sensor as required
	P0409-16	Exhaust Gas Recirculation Sensor A Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference EVP_A ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas recirculation actuator circuit short circuit to ground, high	▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator sensor circuit for short circuit to groun high resistance, open circuit, disconnected. Repair harness a required. Using the manufactu approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required

			resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	
	P0409-17	Exhaust Gas Recirculation Sensor A Circuit - Circuit voltage above threshold	**NOTE:** Circuit reference EVP_A ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas recirculation actuator circuit short circuit to power ▪ Exhaust gas recirculation actuator failure	▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator sensor circuit for short circuit to power Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation actuator as required
	P0409-92	Exhaust Gas Recirculation Sensor A Circuit - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Exhaust gas recirculation actuator circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator sensor circuit for short circuit to groun high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation actuator as required
	P0426-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank 1, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit

			temperature sensor post turbocharger circuit to ground, high resistance, open circuit, disconnected	diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground, high resistanc open circuit, disconnected ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required
	P0427-00	Catalyst Temperature Sensor Circuit Low (Bank 1, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger circuit to ground, high resistance, open circuit, disconnected	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground, high resistanc open circuit, disconnected ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required
	P0428-00	Catalyst Temperature Sensor Circuit High (Bank 1, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger circuit to power	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to power

				▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required
	P042B-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank1, Sensor Circuit 2) - No sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post catalytic converter short circuit to ground, open circuit, high resistance ▪ Exhaust gas temperature sensor post catalytic converter failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post cataly converter circuit for short circui to ground, open circuit, high resistance ▪ Check and install a new exhaus gas temperature sensor post catalytic converter as required
	P042C-00	Catalyst Temperature Sensor Circuit Low (Bank 1, Sensor Circuit 2) - No sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post catalytic converter short circuit to ground, open circuit, high resistance ▪ Exhaust gas temperature sensor post catalytic converter failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post cataly converter circuit for short circui to ground, open circuit, high resistance ▪ Check and install a new exhaus gas temperature sensor post catalytic converter as required
	P042D-00	Catalyst Temperature Sensor Circuit High (Bank 2, Sensor Circuit 2) - No sub type information		▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 2 Sensor 2

			NOTE: Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 circuit short circuit to power ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 failure	(0x03F8), Exhaust Gas Temperature Sensor Bank 2 Sensor 2 Voltage (0x03E9) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 2 circuit for short circuit to power Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 2 as required
	P042E-77	Exhaust Gas Recirculation A Control Stuck Open - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Check for other related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, operate the exhaust recirculation valve through the full operating rang whilst monitoring the position sensor signal value. If the position signal does not change smoothly in proportion to the commands check the operation of the valve ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation actuator as required

P042F-77	Exhaust Gas Recirculation A Control Stuck Closed - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation actuator circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation actuator failure	▪ Check for other related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, operate the exhaust recirculation valve through the full operating rang whilst monitoring the position sensor signal value. If the position signal does not change smoothly in proportion to the commands check the operation of the valve ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation actuator circuit for short circuit to ground, short circuit to power, high resistance open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation actuator as required
P0435-00	Catalyst Temperature Sensor Circuit (Bank 2, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 1 circuit for short circuit to groun short circuit to power, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas temperature sensor post

				turbocharger - bank 2, sensor 1 as required
	P0436-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank 2, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference STOT ▪ Exhaust gas temperature sensor post turbocharger overheating ▪ Exhaust gas temperature sensor post turbocharger short circuit short circuit to ground, open circuit, high resistance ▪ Exhaust gas temperature sensor post turbocharger failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground, open circuit, high resistance ▪ Clear the DTC and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger as required
	P0437-00	Catalyst Temperature Sensor Circuit Low (Bank 2, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 1 circuit for short circuit to groun high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required

P0438-00	Catalyst Temperature Sensor Circuit High (Bank 2, Sensor Circuit 1) - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 circuit short circuit to power ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7). ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 1 circuit for short circuit to power Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required
P043A-00	Catalyst Temperature Sensor Circuit (Bank 2, Sensor Circuit 2) - No sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 2 Sensor 2 (0x03F8), Exhaust Gas Temperature Sensor Bank 2 Sensor 2 Voltage (0x03E9) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 2 circuit for short circuit to groun short circuit to power, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 2 as required
P043B-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank 2, Sensor Circuit 2) - No		▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 2 Sensor 2

		sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 failure	(0x03F8), Exhaust Gas Temperature Sensor Bank 2 Sensor 2 Voltage (0x03E9) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 2 circuit for short circuit to groun short circuit to power, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 2 as required
	P043C-00	Catalyst Temperature Sensor Circuit Low (Bank 2, Sensor Circuit 2) - No sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 2 Sensor 2 (0x03F8), Exhaust Gas Temperature Sensor Bank 2 Sensor 2 Voltage (0x03E9) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 2 circuit for short circuit to groun high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 2 as required
	P043D-00	Catalyst Temperature Sensor Circuit High (Bank 2, Sensor Circuit 2) - No		▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga

		sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 circuit short circuit to power ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 2 failure	Temperature Bank 2 Sensor 2 (0x03F8), Exhaust Gas Temperature Sensor Bank 2 Sensor 2 Voltage (0x03E9) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 2 circuit for short circuit to powe Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 2 as required
	P044A-13	Exhaust Gas Recirculation Sensor C Circuit - Circuit open	**NOTE:** Circuit reference EVP_B ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Exhaust gas recirculation valve bank 2 circuit high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 2 (0x052F) ▪ Refer to the electrical circuit diagrams and check the bank 2 exhaust gas recirculation senso signal circuit for open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P044A-16	Exhaust Gas Recirculation Sensor C Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference EVP_B	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 2 (0x052F) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui

			▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, high resistance, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	for short circuit to ground, high resistance, disconnected. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P044A-17	Exhaust Gas Recirculation Sensor C Circuit - Circuit voltage above threshold	**NOTE:** Circuit reference EVP_B ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to power ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 2 (0x052F). ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P044A-92	Exhaust Gas Recirculation Sensor C Circuit - Performance or incorrect operation	**NOTE:** Circuit reference EVP_B ▪ The engine control module has detected that the component performance is outside	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 2 (0x052F) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, shor circuit to power, open circuit, high resistance, disconnected. Repair harness as required. Usi

			its expected range or operating in an incorrect way ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, short circuit to power, open circuit, high resistance, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P044C-00	Exhaust Gas Recirculation Sensor C Circuit Low - No sub type information	**NOTE:** Circuit reference EVP_B ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 2 (0x052F) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P044D-00	Exhaust Gas Recirculation Sensor C Circuit High - No sub type information	**NOTE:** Circuit reference EVP_B ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to power ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Valve Position Bank 2 (0x052F) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required

	P045A-13	Exhaust Gas Recirculation B Control Circuit - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Exhaust gas recirculation valve bank 2 circuit open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check the bank 2 exhaust gas recirculation contro circuit for open circuit, disconnected. Check both wire Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation control actuat as required
	P045A-16	Exhaust Gas Recirculation B Control Circuit - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, high resistance, open circuit, disconnected. Check both circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045A-19	Exhaust Gas Recirculation B Control Circuit - Circuit current above threshold	▪ The engine control module has measured current flow above a specified range ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, shor circuit to power, high resistance open circuit, disconnected. Check both circuits. Repair harness as required. Using the

		valve bank 2 failure	manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
P045A-1D	Exhaust Gas Recirculation B Control Circuit - Circuit current out of range	▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, shor circuit to power, high resistance open circuit, disconnected. Check both circuits. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
P045A-4B	Exhaust Gas Recirculation B Control Circuit - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, short circuit to power, high resistance ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, shor circuit to power, high resistance Check both circuits. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
P045A-	Exhaust Gas Recirculation	▪ The engine control	▪ Using the manufacturer approv

	71	B Control Circuit - Actuator stuck	module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas recirculation valve bank 2 failure	diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, shor circuit to power, open circuit, high resistance. Check both circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045B-19	Exhaust Gas Recirculation B Control Circuit Range/Performance - Circuit current above threshold	▪ The engine control module has measured current flow above a specified range ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, shor circuit to power, high resistance open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045C-00	Exhaust Gas Recirculation B Control Circuit Low - No sub type information	▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525) ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to ground, high resistance, open circuit, disconnected. Check both circuits. Repair harness as required. Using the manufactur

				approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045C-11	Exhaust Gas Recirculation B Control Circuit Low - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525). Refer to the electrical circuit diagrams a check exhaust gas recirculation valve bank 2 circuit for short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045C-77	Exhaust Gas Recirculation B Control Circuit Low - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525). Refer to the electrical circuit diagrams a check exhaust gas recirculation valve bank 2 circuit for short circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045D-00	Exhaust Gas Recirculation B Control Circuit High - No sub type information	▪ Exhaust gas recirculation valve bank 2 circuit short circuit to power ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525). Refer to the electrical circuit diagrams a check exhaust gas recirculation valve bank 2 circuit for short circuit to power. Repair harness

				as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045D-12	Exhaust Gas Recirculation B Control Circuit High - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to power ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525). Refer to the electrical circuit diagrams a check exhaust gas recirculation valve bank 2 circuit for short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045D-77	Exhaust Gas Recirculation B Control Circuit High - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to power ▪ Exhaust gas recirculation valve bank 2 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande EGR Bank 2 (0x0525). Refer to the electrical circuit diagrams a check exhaust gas recirculation valve bank 2 circuit for short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P045E-77	Exhaust Gas Recirculation B Control Stuck Open - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the	▪ Check for other related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, operate the

			commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation valve B circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve B failure	Exhaust Recirculation Valve through the full operating rang whilst monitoring the position sensor signal value. If the position signal does not change smoothly in proportion to the commands check the operation of the valve ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve B circuit for short circuit to ground, short circuit to power, high resistance open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve B as required
	P045F-77	Exhaust Gas Recirculation B Control Stuck Closed - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment ▪ Exhaust gas recirculation valve B circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve B failure	▪ Check for other related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, operate the Exhaust Recirculation Valve through the full operating rang whilst monitoring the position sensor signal value. If the position signal does not change smoothly in proportion to the commands check the operation of the valve ▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve B circuit for short circuit to ground, short circuit to power, high resistance open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve B as required

P046E-00	Exhaust Gas Recirculation Sensor "B" Circuit Range/Performance - No sub type information	**NOTE:** Circuit reference TPS ▪ Throttle position sensor short circuit to ground, short circuit to power, open circuit, high resistance ▪ Throttle position sensor failure	▪ Refer to the electrical circuit diagrams and check throttle position sensor for short circuit ground, short circuit to power, open circuit, high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new throttle position sensor as required
P0480-11	Fan 1 Control Circuit - Circuit short to ground	**NOTE:** Circuit reference ECFC ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Fan 1 control circuit - Short circuit to ground ▪ Cooling fan 1 component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03F9). Refer to the electrical circuit diagrams and check the cooling fan 1 control circuit for short circuit to ground. Check t power and ground supplies to the cooling fan 1 control modu Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 1 control module as require
P0480-12	Fan 1 Control Circuit - Circuit short to battery	**NOTE:** Circuit reference ECFC ▪ The engine control module has detected a vehicle power	▪ Using the manufacturer approv diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03F9). Refer to the electrical circuit diagrams and check the cooling fan 1 control circuit for short circuit to power. Check th power and ground supplies to the cooling fan 1 control modu Repair harness as required. Usi

			measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Fan 1 control circuit - Short circuit to power ▪ Cooling fan 1 component failure	the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 1 control module as require
	P0480-13	Fan 1 Control Circuit - Circuit open	**NOTE:** Circuit reference ECFC ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Fan 1 control circuit - Open circuit ▪ Cooling fan 1 component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03F9). Refer to the electrical circuit diagrams and check the cooling fan 1 control circuit for open circuit. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 1 control module as require
	P0480-16	Fan 1 Control Circuit - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ The engine control module has determined that the fan 1 supply voltage is below the calibrated threshold ▪ Fan 1 control circuit - Open circuit, short circuit to ground, high resistance ▪ Cooling fan 1 control	▪ Refer to the electrical circuit diagrams and check the fan 1 supply circuit for short circuit to ground, open circuit, high resistance. Check the power an ground supplies to the cooling fan 1 control module. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 1 control module as require

		module failure	
P0480-17	Fan 1 Control Circuit - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ The engine control module has determined that the fan 1 supply voltage is above the calibrated threshold ▪ Fan 1 control circuit - Short circuit to power ▪ Cooling fan 1 control module failure	▪ Refer to the electrical circuit diagrams and check the fan 1 supply circuit for short circuit to power. Check the power and ground supplies to the cooling fan 1 control module. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 1 control module as require
P0480-71	Fan 1 Control Circuit - Actuator stuck	▪ The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Engine cooling fan **partially** stalled ▪ Engine cooling fan stalling caused by deep water wading ▪ Engine cooling fan stalling caused by obstruction in fan cowling	▪ Confirm if customer has been deep water wading ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Using the manufacturer approv diagnostic system check datalogger signal - Electric Fan PWM Control - Commanded - (0x03F9) ▪ Using the manufacturer approv diagnostic system, operate cooling fan 1 through the full operating range and check that the DTC does not reset
P0480-97	Fan 1 Control Circuit - Component or system operation obstructed or blocked	▪ The engine control module has detected that the operation of a component is prevented by an obstruction ▪ Engine cooling fan stalled ▪ Engine cooling fan stalling caused by deep water wading ▪ Engine cooling fan stalling caused by	▪ Confirm if customer has been deep water wading ▪ Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required ▪ Using the manufacturer approv diagnostic system check datalogger signal - Electric Fan PWM Control - Commanded - (0x03F9) ▪ Using the manufacturer approv diagnostic system, operate

			obstruction in fan cowling	cooling fan 1 through the full operating range and check that the DTC does not reset
	P0481-11	Fan 2 Control Circuit - Circuit short to ground	**NOTE:** Circuit reference ECFC_2 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Fan 2 control circuit - Short circuit to ground ▪ Cooling fan 2 component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03FA). Refer to the electrica circuit diagrams and check the cooling fan 2 control circuit for short circuit to ground. Check t power and ground supplies to the cooling fan 2 control modu Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 2 control module as require
	P0481-12	Fan 2 Control Circuit - Circuit short to battery	**NOTE:** Circuit reference ECFC_2 ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Fan 2 control circuit - Short circuit to power ▪ Cooling fan 2 component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03FA). Refer to the electrica circuit diagrams and check the cooling fan 2 control circuit for short circuit to power. Check th power and ground supplies to the cooling fan 2 control modu Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 2 control module as require

	P0481-13	Fan 2 Control Circuit - Circuit open	**NOTE:** Circuit reference ECFC_2 ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Fan 2 control circuit - Open circuit ▪ Cooling fan 2 component failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Electric Fan PWM Control - Commanded (0x03FA). Refer to the electrica circuit diagrams and check the cooling fan 2 control circuit for open circuit. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 2 control module as require
	P0481-16	Fan 2 Control Circuit - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ The engine control module has determined that the fan 2 supply voltage is below the calibrated threshold ▪ Fan 2 control circuit - Open circuit, short circuit to ground, high resistance ▪ Cooling fan 2 control module failure	▪ Refer to the electrical circuit diagrams and check the fan 2 supply circuit for short circuit to ground, open circuit, high resistance. Check the power an ground supplies to the cooling fan 2 control module. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan 2 control module as require
	P0481-17	Fan 2 Control Circuit - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ The engine control module has determined that the fan 2 supply voltage is above the calibrated threshold	▪ Refer to the electrical circuit diagrams and check the fan 2 supply circuit for short circuit to power. Check the power and ground supplies to the cooling fan 2 control module. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			- Fan 2 control circuit - Short circuit to power - Cooling fan 2 control module failure	Check and install a new cooling fan 2 control module as require
	P0481-71	Fan 2 Control Circuit - Actuator stuck	- The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay - Engine cooling fan **partially** stalled - Engine cooling fan stalling caused by deep water wading - Engine cooling fan stalling caused by obstruction in fan cowling	- Confirm if customer has been deep water wading - Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required - Using the manufacturer approv diagnostic system check datalogger signal - Viscous Fan PWM Control - Commanded - (0x03FA) - Using the manufacturer approv diagnostic system, operate cooling fan 2 through the full operating range and check that the DTC does not reset
	P0481-97	Fan 2 Control Circuit - Component or system operation obstructed or blocked	- The engine control module has detected that the operation of a component is prevented by an obstruction - Engine cooling fan stalled - Engine cooling fan stalling caused by deep water wading - Engine cooling fan stalling caused by obstruction in fan cowling	- Confirm if customer has been deep water wading - Check for damage to or blockages in fan and fouling of fan cowling. Rectify as required - Using the manufacturer approv diagnostic system check datalogger signal - Viscous Fan PWM Control - Commanded - (0x03FA) - Using the manufacturer approv diagnostic system, operate cooling fan 2 through the full operating range and check that the DTC does not reset
	P0483-27	Fan Performance - Signal rate of change above threshold	- The signal transitions more quickly than is reasonably allowed - Engine cooling fan acceleration plausibility defect - This DTC is set if the calculated engine	Refer to relevant section of workshop manual and check th viscous fan unit

			cooling fan acceleration value is greater than a calibrated value	
	P0483-36	Fan Performance - Signal frequency too low	- The engine control module detected excessive duration for one cycle of the output across a specified sample size - Engine cooling fan speed below maximum threshold - Viscous fan circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to electrical circuit diagrams and check the viscous fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance
	P0483-37	Fan Performance - Signal frequency too high	- The engine control module detected insufficient duration for one cycle of the output across a specified sample size - Engine cooling fan speed above maximum threshold - Viscous fan circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to electrical circuit diagrams and check the viscous fan circuit for short circuit to ground, short circuit to power, open circuit, high resistance
	P0486-13	Exhaust Gas Recirculation Sensor B Circuit - Circuit open	**NOTE:** Circuit reference TPS - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Harness failure - Throttle	This DTC is set when the engin control module detects a no loa error on the throttle motor position sensor circuit. This sensor is a potentiometer. Refe to the electrical circuit diagram and check the sensor 5 volt supply and ground circuits. Check the sensor circuits for open circuits, short circuit to ground, short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			motor position sensor circuit open circuit ▪ Throttle motor position sensor failure	▪ Check and install a new throttle motor position sensor failure
	P0486-16	Exhaust Gas Recirculation Sensor B Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference TPS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas recirculation valve bank 2 failure	▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve B circuit for short circuit to ground, high resistance, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation valve B as required
	P0486-17	Exhaust Gas Recirculation Sensor B Circuit - Circuit voltage above threshold	**NOTE:** Circuit reference TPS ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Exhaust gas recirculation valve bank 2 circuit short circuit to power ▪ Exhaust gas recirculation valve bank 2 failure	▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation valve bank 2 circui for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation valve bank 2 a required
	P0486-92	Exhaust Gas Recirculation Sensor B Circuit -		▪ Refer to the electrical circuit diagrams and check throttle

		Performance or incorrect operation	**NOTE:** Circuit reference TPS - The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Throttle position sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit, disconnected - Throttle position sensor failure	position sensor circuit for short circuit to ground, short circuit t power, high resistance, open circuit, disconnected. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new throttle position sensor as required
	P0487-00	Exhaust Gas Recirculation Throttle Control Circuit A / Open - No sub type information	- Harness failure - Throttle motor control circuit open circuit, short circuit to ground, short circuit to power - Throttle motor failure	- Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control (0xF44C). This DTC is set when the engine control module detects an open load error on the throttle motor control circu Refer to the electrical circuit diagrams and check the throttle control circuits for open circuit, short circuit to ground, short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new throttle motor control actuator as required
	P0487-19	Exhaust Gas Recirculation Throttle Control Circuit A / Open - Circuit current above threshold	- The engine control module has measured current flow above a specified range	- Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control

			- Harness failure - Throttle motor control circuit short circuit to ground, high resistance, open circuit - Throttle motor failure	(0xF44C). Refer to the electrica circuit diagrams and check the throttle control circuits for shor circuit to ground, high resistanc open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new throttle motor control actuator as required
	P0487-1D	Exhaust Gas Recirculation Throttle Control Circuit A / Open - Circuit current out of range	- The engine control module has detected a current outside of the expected range, but not identified as too high or too low - Harness failure - Throttle motor control circuit short circuit to ground, short circuit to power, short circuit to other circuit - Throttle motor failure	- Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control (0xF44C). Refer to the electrica circuit diagrams and check the throttle plate actuator control circuits for short circuit to ground, short circuit to power, short circuit to other circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new throttle motor control actuator as required
	P0488-16	Exhaust Gas Recirculation Throttle Control Circuit A Range/Performance - Circuit voltage below threshold	- The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Harness failure - Throttle motor control circuit open circuit, short circuit to ground, short circuit to power, high resistance, short circuit to another circuit - Throttle motor failure	- Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control (0xF44C). Refer to the electrica circuit diagrams and check the throttle plate actuator control circuits for open circuit, short circuit to ground, short circuit t power, high resistance, short circuit to other circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new throttle motor control actuator as

			required
P0488-19	Exhaust Gas Recirculation Throttle Control Circuit A Range/Performance - Circuit current above threshold	▪ The engine control module has measured current flow above a specified range ▪ Harness failure - Throttle motor control circuit short circuit to ground, high resistance ▪ Throttle motor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control (0xF44C). Refer to the electrica circuit diagrams and check the throttle plate actuator control circuit for short circuit to groun high resistance. Repair harness required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new throttle motor control actuator as required
P0488-1D	Exhaust Gas Recirculation Throttle Control Circuit A Range/Performance - Circuit current out of range	▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ▪ Harness failure - Throttle motor control circuit high resistance, open circuit, short circuit to ground, short circuit to power, short circuit to other circuit ▪ Throttle motor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control (0xF44C). Refer to the electrica circuit diagrams and check the throttle plate actuator control circuits for high resistance, ope circuit, short circuit to ground, short circuit to power, short circuit to other circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new throttle motor control actuator as required
P0488-4B	Exhaust Gas Recirculation Throttle Control Circuit A Range/Performance - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Throttle motor control circuit short circuit to ground, short circuit to power, high resistance	▪ Using the manufacturer approv diagnostic system check datalogger signals, Commande Throttle Actuator Control (0xF44C). Refer to the electrica circuit diagrams and check the throttle motor control circuit fo short circuit to ground, short circuit to power, high resistance Repair harness as required. Usi the manufacturer approved

		▪ Throttle motor failure	diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new throttle motor control actuator as required
P0488-72	Exhaust Gas Recirculation Throttle Control Circuit A Range/Performance - Actuator stuck open	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment ▪ Harness failure - Exhaust gas recirculation throttle inlet control circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Exhaust gas recirculation throttle inlet control actuator failure	▪ Refer to the electrical circuit diagrams and check the exhaus gas recirculation throttle inlet control circuit for short circuit t ground, short circuit to power, high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation throttle inlet control actuator as required
P0488-73	Exhaust Gas Recirculation Throttle Control Circuit A Range/Performance - Actuator stuck closed	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Harness failure - Exhaust gas recirculation throttle inlet control circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Exhaust gas recirculation throttle inlet control actuator failure	▪ Refer to the electrical circuit diagrams and check the exhaus gas recirculation throttle inlet control circuit for short circuit t ground, short circuit to power, high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation throttle inlet control actuator as required
P0489-77	Exhaust Gas Recirculation Control Circuit Low - Commanded position not reachable		▪ Check for related DTCs and ref to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the throttle

			NOTE: Circuit reference SENSGND_5 - The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - Throttle position sensor circuit open circuit	position sensor circuit for open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes
	P0490-77	Exhaust Gas Recirculation A Control Circuit High - Commanded position not reachable	- The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - Throttle position sensor circuit, short circuit to ground, short circuit to power, open circuit	- Check for related DTCs and ref to the relevant DTC index - Refer to the electrical circuit diagrams and check the throttle position sensor circuit for short circuit to ground, short circuit t power, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P049D-00	Exhaust Gas Recirculation A Control Position Exceeded Learning Limit - No sub type information	- Harness failure - Exhaust gas recirculation valve bank 1 control circuit short circuit to ground, short circuit to power, high resistance, open circuit - Exhaust gas recirculation valve bank 1 actuator failure	- Refer to the electrical circuit diagrams and check the bank 1 exhaust gas recirculation contro circuit for short circuit to groun short circuit to power, high resistance, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new exhaus gas recirculation control actuat as required

P049E-00	Exhaust Gas Recirculation B Control Position Exceeded Learning Limit - No sub type information	▪ Harness failure - Exhaust gas recirculation valve bank 2 control circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Exhaust gas recirculation valve bank 2 actuator failure	▪ Refer to the electrical circuit diagrams and check the bank 2 exhaust gas recirculation contro circuit for short circuit to groun short circuit to power, high resistance, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation control actuato as required
P0504-27	Brake Switch A / B Correlation - Signal rate of change above threshold	**NOTE:** Circuit reference BOO_1 BOO_2 ▪ The signal transitions more quickly than is reasonably allowed ▪ Brake pedal switch plunger failure ▪ Brake pedal switch not fitted correctly ▪ Brake switch A failure	▪ Check for related brake pressur DTCs within the anti-lock brake system control module ▪ Check brake pedal switch is fitted correctly. Clear DTC, star the engine and press the brake pedal, using maximum travel, fo greater than 1 second taking ca not to press the accelerator pedal ▪ Check and install a new brake switch as required
P0504-62	Brake Switch A / B Correlation - Signal compare failure	**NOTE:** Circuit reference BOO_1 BOO_2 ▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Error check for brake plausibility ▪ Brake pedal switch	▪ Refer to the electrical circuit diagrams and check brake peda switch circuit for short circuit to power, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			circuit short circuit to power, open circuit	
	P050E-00	Cold Start Engine Exhaust Temperature Too Low - No sub type information	▪ Engine coolant temperature sensor circuit short circuit to ground, short circuit to power, open circuit, disconnected ▪ Engine coolant temperature sensor failure	▪ Refer to the electrical circuit diagrams and check engine coolant temperature sensor circuit for short circuit to groun short circuit to power, open circuit, disconnected. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine coolant temperature sensor as required
	P050F-00	Brake Assist Vacuum Too Low - No sub type information	▪ Harness failure - Vacuum switch circuit short circuit to ground, short circuit to power, open circuit, disconnected ▪ Vacuum switch failure	▪ Refer to the electrical circuit diagrams and check vacuum switch circuit for short circuit to ground, short circuit to power, open circuit, disconnected. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new vacuum switch as required
	P0512-24	Starter Request Circuit - Signal stuck high	**NOTE:** Circuit reference CRNK ▪ The engine control module measures a signal that remains high when transitions are expected ▪ Harness failure - Starter request circuit failure ▪ Central junction box failure	▪ This DTC is set when the engin control module detects the engine crank signal from the central junction box is stuck hig If this DTC is logged on its own clear the DTC using the manufacturer approved diagnostic system. Select PARK position. Set ignition OFF, wait minutes for post drive to complete. Set ignition ON but NOT crank. Wait 25 seconds. Crank engine and check if DTC cleared. If this DTC is logged with lost communication DTCs refer to electrical circuit diagrar and check CAN circuit. If the engine cannot be cranked read

			- Post drive cycle not completed - Harness failure - CAN circuit	the DTCs stored in the central junction box and refer to the relevant DTC index. Check the engine crank signal input circuit for open circuit, short circuit to power, short circuit to ground, short circuit to other circuits. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0512-64	Starter Request Circuit - Signal plausibility failure	**NOTE:** Circuit reference CRNK - The engine control module detected plausibility failures - Harness failure - Starter request circuit failure - Central junction box failure - Postdrive cycle not completed - Harness failure - CAN circuit	- This DTC is set when the engin control module detects the engine crank signal from the passenger compartment fuse b is stuck high. If this DTC is logged on its own clear the DT using the manufacturer approv diagnostic system. Select PARK position. Set ignition OFF, wait minutes for post drive to complete. Set ignition ON but NOT crank. Wait 25 seconds. Crank engine and check if DTC cleared. If this DTC is logged with lost communication DTCs refer to electrical circuit diagra and check CAN circuit. If the engine cannot be cranked read the DTCs stored in the central junction box and refer to the relevant DTC index. Check the engine crank signal input circui for open circuits, short circuit to power, short circuit to ground, short to other circuits. Repair wiring harness as required, Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest system
	P0513-00	Incorrect Immobilizer Key - No sub type information	- Security key invalid	- Check for CAN network and engine control module related DTCs. Re-configure the engine control module using the manufacturer approved diagnostic system. Re-configure the instrument cluster using the manufacturer approved

				diagnostic system
	P0528-00	Fan Speed Sensor Circuit No Signal - No sub type information	▪ Harness failure - Cooling fan monitor circuit	▪ Using the manufacturer approv diagnostic system check datalogger signals, Fan Speed Indicated (0x0702). This DTC is set when the engine control module does not receive a sign on the cooling fan monitor circuit. Check the cooling fan monitor signal input circuit for open circuits, short circuit to power, short circuit to ground o other circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes
	P052F-00	Glow Plug Control Module System Voltage - No sub type information	▪ Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, disconnected ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check glow plug control module circuit for short circuit to ground, short circuit t power, open circuit, disconnected. Repair harness a required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new glow plug control module as require
	P0544-00	Exhaust Gas Temperature Sensor Circuit - Bank 1 Sensor 1 - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 circuit short circuit to ground ▪ Exhaust gas temperature sensor post turbocharger bank 2, sensor 1 failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger bank 2, sensor 1 circuit for short circuit to groun Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

				▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required
	P0545-00	Exhaust Gas Temperature Sensor Circuit Low - Bank 1 Sensor 1 - No sub type information	**NOTE:** Circuit reference STOT ▪ Exhaust gas temperature sensor, bank 1 sensor 1 circuit short circuit to ground, high resistance, open circuit, disconnected ▪ Exhaust gas temperature sensor, bank 1 sensor 1 failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor, bank 1 sensor 1 circuit for short circuit ground, high resistance, open circuit, disconnected. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor, bank 1 sensor 1 as required
	P0546-00	Exhaust Gas Temperature Sensor Circuit High - Bank 1 Sensor 1 - No sub type information	**NOTE:** Circuit reference STOT ▪ Exhaust gas temperature sensor, bank 1 sensor 1 circuit short circuit to power ▪ Exhaust gas temperature sensor, bank 1 sensor 1 failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor, bank 1 sensor 1 circuit for short circuit power. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas temperature sensor, bank1 sensor 1 as required
	P0547-00	Exhaust Gas Temperature Sensor Circuit - Bank 2 Sensor 1 - No sub type information	**NOTE:** Circuit reference STOT ▪ Exhaust gas temperature sensor post turbocharger short circuit to ground	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground ▪ Clear the DTC and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger as required

			Exhaust gas temperature sensor post turbocharger failure	
	P0548-00	Exhaust Gas Temperature Sensor Circuit Low - Bank 2 Sensor 1 - No sub type information	**NOTE:** Circuit reference STOT - Exhaust gas temperature sensor post turbocharger short circuit short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground, open circuit, high resistance - Clear the DTC and retest - Check and install a new exhaus gas temperature sensor post turbocharger as required
	P0549-00	Exhaust Gas Temperature Sensor Circuit High - Bank 2 Sensor 1 - No sub type information	**NOTE:** Circuit reference STOT - Exhaust gas temperature sensor post turbocharger short circuit short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to power, open circuit, high resistance - Clear the DTC and retest - Check and install a new exhaus gas temperature sensor post turbocharger as required
	P0571-62	Brake Switch A Circuit - Signal compare failure	- The engine control module detected failure when comparing two or more input parameters for plausibility - The brake switch signal received over CAN is defective	- Using the manufacturer approv diagnostic system carry out network integrity test - Check for related DTCs and ref to the relevant DTC index - Check and install a new brake switch as required

			- Brake switch (footbrake switch) failure	
	P0571-68	Brake Switch A Circuit - Event information	- The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module - Brake pedal pressed by driver at same time as accelerator pedal pressed - Brake pedal switch A circuit - Short circuit to power - Brake switch (footbrake switch) failure	- Ensure driver has not lightly pressed brake pedal and accelerator pedal at the same time - Refer to the electrical circuit diagrams and check brake peda switch A circuit for short circuit power. Repair wiring harness as required, Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new brake switch as required
	P0575-81	Cruise Control Input Circuit - Invalid serial data received	- The engine control module has indicated a signal was received with the corresponding validity bit equal to "invalid" or post processing of the signal determines it is invalid - The DTC sets if the cancel, set minus, set plus, resume, headway increase and headway decrease speed control buttons have been pressed for longer than a calibrated period of time. The system then assumes a stuck/damaged button and will cancel and/or disable cruise. The failure will be healed in the next driving cycle if the failure is removed	- Check speed control buttons ar not jammed/contaminated/damage Check speed control module fo related DTCs and refer to the relevant DTC index

P0602-00	Powertrain Control Module Programming Error - No sub type information	▪ Mismatch between car configuration file and engine control module calibration for expected engine power output ▪ Incorrect engine control module calibration for vehicle specification	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P0605-00	Internal Control Module Read Only Memory (ROM) Error - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circui Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
P0606-00	Control Module Processor - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circui Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
P0606-44	Control Module Processor - Data memory failure	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit

			ground supply failure ▪ Engine control module failure	diagrams and check the engine control module power and ground supplies for open circui Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P0606-49	Control Module Processor - Internal electronic failure	▪ The engine control module detected an internal circuit failure ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circui Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P0607-00	Control Module Performance - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new engine control module as required
	P060A-00	Internal Control Module Monitoring Processor Performance - No sub type information	▪ Engine control module has reset due to hardware failure	▪ Check engine control module power and ground supply circu for open circuits. Re-configure the engine control module. Usi the manufacturer approved diagnostic system clear all store

				DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P060A-16	Internal Control Module Monitoring Processor Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new engine control module as required
	P060A-17	Internal Control Module Monitoring Processor Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circui Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P060B-00	Internal Control Module A/D Processing Performance - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground supplies for open circui Repair harness as required. Usi the manufacturer approved diagnostic system clear all store

			DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
P060B-16	Internal Control Module A/D Processing Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
P060B-17	Internal Control Module A/D Processing Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store

				DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P060B-46	Internal Control Module A/D Processing Performance - Calibration / parameter memory failure	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P060B-49	Internal Control Module A/D Processing Performance - Internal electronic failure	▪ The engine control module detected an internal circuit failure ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store

				DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P060B-63	Internal Control Module A/D Processing Performance - Circuit / component protection time-out	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P060C-00	Internal Control Module Main Processor Performance - No sub type information	▪ Engine control module has reset due to hardware or software failure	▪ Clear DTC and road test vehicle including at least 5 ignition cycles. Re-check DTC and repe above procedure. If DTC resets Check and install a new engine control module as required
	P060D-00	Internal Control Module Accelerator Pedal Position Performance - No sub type information	▪ Accelerator pedal position sensor circuit, short circuit to ground, short circuit to power, high resistance, open circuit ▪ Accelerator pedal position sensor failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sensor. Check signal circuits for high resistanc open circuits, short circuit to power, short circuit to ground. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

				▪ Check and install a accelerator pedal position sensor as require ▪ Check and install a new engine control module as required
	P0610-00	Control Module Vehicle Options Error - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P0615-00	Starter Relay Circuit - No sub type information	**NOTE:** Circuit reference SM_COMMAND_H SM_COMMAND_L ▪ Starter relay circuit failure	▪ Refer to the electrical circuit diagrams and check starter rela circuit for short circuit to groun short circuit to power, open circuit, high resistance
	P061A-00	Internal Control Module Torque Performance - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply's

			▪ Engine control module failure	for open circuit. Repair wiring harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply' for open circuit. Repair wiring harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P061B-00	Internal Control Module Torque Calculation Performance - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply's for open circuit. Repair wiring harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply' for open circuit. Repair wiring harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P061B-21	Internal Control Module Torque Calculation Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short	▪ Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first ▪ Using the manufacturer approv

			circuit to ground, gain low ▪ Vehicle supervisory controller torque failure (software contained within the engine control module for control of the HEV system) ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P061B-22	Internal Control Module Torque Calculation Performance - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Vehicle supervisory controller torque failure (software contained within the engine control module for control of the HEV system) ▪ Other HEV system failures ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

	P061B-35	Internal Control Module Torque Calculation Performance - Signal high time > maximum	- This DTC is set when the engine control module detects that the signal high time is greater than the maximum value - Other transmission system failures - Corrupt transmission control module software - Transmission control module power supply failure - Transmission control module ground supply failure - Corrupt engine control module software - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check transmission control module for additional DTCs and refer to relevant DTC index. Rectify these first - Using the manufacturer approv diagnostic system, re-configure the transmission control modul with the latest level software - Refer to the electrical circuit diagrams and check the transmission control module power and ground circuits for open circuit, high resistance - Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software - Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance - Check and install a new engine control module as required - Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P061B-62	Internal Control Module Torque Calculation Performance - Signal compare failure	- The engine control module detected failure when comparing two or more input parameters for plausibility - Speed synchronisation failure, set by engine control module torque calculation error - Corrupt engine control module software - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first - Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software - Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance - Check and install a new engine control module as required - Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men

			tab and retest
P061B-64	Internal Control Module Torque Calculation Performance - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Speed synchronisation failure, set by engine control module torque calculation error ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P061B-65	Internal Control Module Torque Calculation Performance - Signal has too few transitions / events	▪ The engine control module monitored a parameter over time within specified limits and detected fewer than the expected number of transitions ▪ Engine control module torque failure ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P061B-66	Internal Control Module Torque Calculation Performance - Signal has too many transitions /	▪ The engine control module monitored a parameter over time within specified limits	▪ Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first

		events	and detected more than the expected number of transitions ▪ Engine control module torque failure ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P061B-78	Internal Control Module Torque Calculation Performance - Alignment or adjustment incorrect	▪ The engine control module has detected incorrectly adjusted or aligned components ▪ Speed synchronisation failure, set by engine control module torque calculation error ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power and ground circuits for open circuit, high resistance ▪ Check and install a new engine control module as required ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P061C-00	Internal Control Module Engine RPM Performance - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply f open circuit. Repair harness as required ▪ Refer to the electrical circuit

				diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P061C-11	Internal Control Module Engine RPM Performance - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Internal failure within engine control module - Engine speed output short to ground	▪ Refer to the electrical circuit diagrams and check the power and ground supplies to the engine control module
	P061C-12	Internal Control Module Engine RPM Performance - Circuit short to battery	▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Internal failure within engine control module - Engine speed output short to power	▪ Refer to the electrical circuit diagrams and check the power and ground supplies to the engine control module
	P061C-13	Internal Control Module Engine RPM Performance - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Internal failure within	▪ Refer to the electrical circuit diagrams and check the power and ground supplies to the engine control module

			engine control module - Engine speed output open circuit	
	P0620-00	Generator Control Circuit - No sub type information	- Electrical failure has been detected by the generator and reported to the engine control module by LIN bus - Generator failure	- Refer to the electrical circuit diagrams and check the generator circuit, for short circu to power, short circuit to groun open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new generator as required
	P0627-13	Fuel Pump A Control Circuit / Open - Circuit open	**NOTE:** Circuit reference LPPR - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Wiring harness failure between engine control module and central junction box - Lift pump relay failure	**NOTE:** When this DTC is set the engine control module limits engine torque to protect the engine and avoid pulling air into the fuel system under high load - Check the operation of the lift pump relay. Replace if required Refer to the electrical circuit diagrams and check the fuel lift pump relay control circuit between the engine control module and the central junctio box for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0628-11	Fuel Pump A Control Circuit Low - Circuit short to ground	**NOTE:** Circuit reference LPPR - The engine control	

			module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Wiring harness failure between engine control module and central junction box ▪ Lift pump relay failure	**NOTE:** When this DTC is set the fuel lift pump will run continuously when the ignition is on, even when the engine is stopped ▪ Check the operation of the lift pump relay. Replace if required Refer to the electrical circuit diagrams and check the fuel lift pump relay control circuit between the engine control module and the central junctior box for a short circuit to grounc Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0629-12	Fuel Pump A Control Circuit High - Circuit short to battery	**NOTE:** Circuit reference LPPR ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Wiring harness failure between engine control module and central junction box ▪ Lift pump relay failure	**NOTE:** When this DTC is set the engine control module limits engine torque to protect the engine and avoid pulling air into the fuel system under high load ▪ Check the operation of the lift pump relay. Replace if required Refer to the electrical circuit diagrams and check the fuel lift pump relay control circuit between the engine control module and the central junctior box for a short circuit to power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P062A-92	Fuel Pump A Control Circuit Range/Performance	▪ The engine control module has detected	▪ Check for related DTCs within the central junction box

		- Performance or incorrect operation	that the component performance is outside its expected range or operating in an incorrect way ▪ Lift pump relay has been driven by the central junction box, but has not been requested by the engine control module ▪ Lift pump relay has been requested by the engine control module, but has not been driven by the central junction box ▪ Wiring harness failure lift pump monitor circuit ▪ Lift pump relay	▪ Refer to the electrical circuit diagrams and check fuel lift pump relay circuit for short circ to ground, short circuit to powe open circuit. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check between th engine control module and the rear junction box for short circu to ground. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check the rear junction box to in tank fuel pum relay for short circuit to power. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel lift pump relay as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P062B-00	Internal Control Module Fuel Injector Control Performance - No sub type information	▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine

				control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-41	Internal Control Module Fuel Injector Control Performance - General checksum failure	▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-48	Internal Control Module Fuel Injector Control Performance - Supervision software failure	▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur

				approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-49	Internal Control Module Fuel Injector Control Performance - Internal electronic failure	▪ The engine control module detected an internal circuit failure ▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-61	Internal Control Module Fuel Injector Control Performance - Signal calculation failure	▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software

			ground supply failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-62	Internal Control Module Fuel Injector Control Performance - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-64	Internal Control Module Fuel Injector Control	▪ The engine control module detected	▪ Check for related injector DTCs and repair these first

		Performance - Signal plausibility failure	plausibility failures ▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-67	Internal Control Module Fuel Injector Control Performance - Signal incorrect after event	▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required

	P062B-91	Internal Control Module Fuel Injector Control Performance - Parametric	▪ The engine control module has detected that the component parameter e.g. capacitance or inductance is outside its expected range ▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-92	Internal Control Module Fuel Injector Control Performance - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store

				DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-94	Internal Control Module Fuel Injector Control Performance - Unexpected operation	▪ The engine control module has detected that the component is operating in a way or at a time that it has not been commanded to operate ▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062B-9A	Internal Control Module Fuel Injector Control Performance - Component or system operating conditions	▪ Injector failure(s) ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Check for related injector DTCs and repair these first ▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine

				control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P062D-00	Fuel Injector Driver Circuit Performance Bank 1 - No sub type information	▪ Bank 1 injector circuit(s) high resistance, short circuit to ground, short circuit to power, open circuit, disconnected ▪ Injector bank 1 failure	▪ Check for related injector DTCs Check injector connections to bank 1 cylinders. Refer to the electrical circuit diagrams and check bank 1 injector circuits fo short circuit to ground, short circuit to power, open circuit, high resistance. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new injector(s) as required
	P062D-11	Fuel Injector Driver Circuit Performance Bank 1 - Circuit short to ground	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Wiring harness failure - Short circuit to ground	▪ Refer to the electrical circuit diagrams and check the wiring harness between engine contro module and bank 1 fuel injecto for short circuit to ground
	P062D-16	Fuel Injector Driver Circuit Performance Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Bank 1 injector circuit(s) high resistance, short circuit to ground, high resistance, open circuit, disconnected	▪ Check for related injector DTCs Check injector connections to bank 1 cylinders. Refer to the electrical circuit diagrams and check bank 1 injector circuits fo short circuit to ground, open circuit, high resistance. Using th manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			• Injector bank 1 failure	• Check and install a new injector(s) as required
	P062D-17	Fuel Injector Driver Circuit Performance Bank 1 - Circuit voltage above threshold	• The engine control module measured a voltage above a specified range but not necessarily a short circuit to power • Bank 1 injector circuit(s) short circuit to power • Injector bank 1 failure	• Check for related injector DTCs Check injector connections to bank 1 cylinders. Refer to the electrical circuit diagrams and check bank 1 injector circuits fo short circuit to power. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest • Check and install a new injector(s) as required
	P062E-00	Fuel Injector Driver Circuit Performance Bank 2 - No sub type information	• Bank 2 injector circuit(s) high resistance, short circuit to ground, short circuit to power, high resistance, open circuit, disconnected • Injector bank 2 failure	• Check for related injector DTCs Check injector connections to bank 2 cylinders. Refer to the electrical circuit diagrams and check bank 2 injector circuits fo short circuit to ground, short circuit to power, open circuit, high resistance. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest • Check and install a new injector(s) as required
	P062E-11	Fuel Injector Driver Circuit Performance Bank 2 - Circuit short to ground	• The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected • Wiring harness failure - Short circuit to ground	• Refer to the electrical circuit diagrams and check the wiring harness between engine contro module and bank 2 fuel injecto for short circuit to ground
	P062E-17	Fuel Injector Driver Circuit Performance Bank 2 - Circuit voltage above threshold	• The engine control module measured a voltage above a specified range but not	• Check for related injector DTCs Check injector connections to bank 2 cylinders. Refer to the electrical circuit diagrams and

		▪ necessarily a short circuit to power ▪ Bank 2 injector circuit(s) short circuit to power ▪ Injector bank 2 failure	check bank 2 injector circuits fo short circuit to power. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new injector(s) as required
P0633-00	Immobilizer Key Not Programmed - ECM/PCM - No sub type information	▪ Security key not programmed	▪ Check for CAN network interference/engine control module related errors. Re-configure the engine control module using the manufacturer approved diagnostic system. Re configure the instrument cluste using the manufacturer approve diagnostic system
P0634-1B	Control Module Internal Temperature "A" Too High - Circuit resistance above threshold	▪ The engine control module has infered a circuit resistance above a specified range ▪ Engine control module internal temperature too high ▪ Engine control module failure	▪ Clear DTC, wait 10 minutes anc retest ▪ Check and install a new engine control module as required
P064C-00	Glow Plug Control Module - No sub type information	▪ Glow plug control module circuit short circuit to ground, short circuit to power, open circuit, disconnected ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check glow plug control module circuit for short circuit to ground, short circuit t power, open circuit, disconnected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module as require
P064D-00	Internal Control Module O2 Sensor Processor Performance - Bank 1 - No sub type information	▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit	▪ Check and repair related heate oxygen sensor DTCs. Refer to the electrical circuit diagrams a check heated oxygen sensor fo short circuit to ground, short

		▪ Heated oxygen sensor failure ▪ Engine control module failure	circuit to power, high resistance open circuit ▪ Check and install a new heated oxygen sensor as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
P064D-16	Internal Control Module O2 Sensor Processor Performance - Bank 1 - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Heated oxygen sensor circuit short circuit to ground, high resistance, open circuit ▪ Heated oxygen sensor failure ▪ Engine control module failure	▪ Check and repair related heate oxygen sensor DTCs. Refer to the electrical circuit diagrams a check heated oxygen sensor fo short circuit to ground, high resistance, open circuit ▪ Check and install a new heated oxygen sensor as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
P0658-00	Actuator Supply Voltage A Circuit Low - No sub type information	▪ Actuator supply circuit voltage below threshold	▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine ▪ Refer to the electrical circuit diagrams and check engine control module power and ground supplies for short circui to ground, open circuit
P0659-00	Actuator Supply Voltage A Circuit High - No sub type information	▪ Actuator supply circuit voltage above threshold	▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine ▪ Refer to the electrical circuit diagrams and check engine control module power and ground supplies for short circui to power

	P065A-00	Generator System Performance - No sub type information	▪ Generator circuit short circuit to ground, high resistance, open circuit ▪ Generator mechanical failure	▪ Refer to the electrical circuit diagrams and check generator circuit for short circuit to groun high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new generator as required
	P065B-16	Generator Control Circuit Range/Performance - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Charging circuit short circuit to ground, short circuit to power, open circuit ▪ Quiescent current high ▪ Battery failure/worn out ▪ Generator failure	▪ Refer to the electrical circuit diagrams and check charging circuit for short circuit to groun short circuit to power, open circuit. Clear DTC and repeat automated diagnostic procedu using the manufacturer approv diagnostic system ▪ If DTC remains check battery is fully charged and in a serviceab condition using the Midtronics battery tester and battery care requirements, in section 414-0 If ok suspect the generator ▪ Check and install a new generator as required
	P065B-17	Generator Control Circuit Range/Performance - Circuit voltage above threshold	▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Generator circuit short circuit to ground, high resistance, open circuit ▪ Generator mechanical failure	▪ Refer to the electrical circuit diagrams and check generator circuit for short circuit to groun high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new generator as required
	P0668-00	Control Module Internal Temperature Sensor "A" Circuit Low - No sub type information	▪ Engine control module internal temperature sensor failure	▪ Check and install a new engine control module as required
	P0669-00	Control Module Internal Temperature Sensor "A"	▪ Engine control module internal temperature	▪ Check and install a new engine control module as required

	Circuit High - No sub type information	sensor failure	
P066A-00	Cylinder 1 Glow Plug Control Circuit Low - No sub type information	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Cylinder 1 glow plug circuit short circuit to ground ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check the glow plug control circuit between th glow plug control module and the glow plug connector for short circuit to ground, check tl power supply and ground connections to the glow plug control module. Repair harness as required. Refer to relevant section of workshop manual, check cylinder 1 glow plug for internal short circuit to ground. Replace glow plug if required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module
P066C-00	Cylinder 2 Glow Plug Control Circuit Low - No sub type information	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Cylinder 2 glow plug circuit short circuit to ground ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check the glow plug control circuit between th glow plug control module and the glow plug connector for short circuit to ground, check tl power supply and ground connections to the glow plug control module. Repair harness as required. Refer to relevant section of workshop manual, check cylinder 2 glow plug for internal short circuit to ground. Replace glow plug if required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module
P066E-00	Cylinder 3 Glow Plug Control Circuit Low - No sub type information	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground	▪ Refer to the electrical circuit diagrams and check the glow plug control circuit between th glow plug control module and the glow plug connector for short circuit to ground, check tl

			▪ measurement when another value was expected ▪ Cylinder 3 glow plug circuit short circuit to ground ▪ Glow plug failure ▪ Glow plug control module failure	power supply and ground connections to the glow plug control module. Repair harness as required. Refer to relevant section of workshop manual, check cylinder 3 glow plug for internal short circuit to ground. Replace glow plug if required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module
	P0671-00	Cylinder 1 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check glow plug connector is correctly connected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug as required
	P0672-00	Cylinder 2 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check glow plug connector is correctly connected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

				▪ Check and install a new glow plug as required
	P0673-00	Cylinder 3 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check glow plug connector is correctly connected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug as required
	P0674-00	Cylinder 4 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check glow plug connector is correctly connected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug as required
	P0675-00	Cylinder 5 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

				▪ Check glow plug connector is correctly connected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug as required
	P0676-00	Cylinder 6 Glow Plug Circuit / Open - No sub type information	▪ Glow plug control circuit open circuit, high resistance ▪ Glow plug control connector disconnected ▪ Glow plug failure	▪ Refer to the electrical circuit diagrams and check glow plug circuit for open circuit, high resistance. Repair wiring harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check glow plug connector is correctly connected. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug as required
	P067A-00	Cylinder 4 Glow Plug Control Circuit Low - No sub type information	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Cylinder 4 glow plug circuit short circuit to ground ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check the glow plug control circuit between th glow plug control module and the glow plug connector for short circuit to ground, check t power supply and ground connections to the glow plug control module. Repair harness as required. Refer to relevant section of workshop manual, check cylinder 4 glow plug for internal short circuit to ground. Replace glow plug if required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module

	P067C-00	Cylinder 5 Glow Plug Control Circuit Low - No sub type information	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Cylinder 5 glow plug circuit short circuit to ground ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check the glow plug control circuit between th glow plug control module and the glow plug connector for short circuit to ground, check t power supply and ground connections to the glow plug control module. Repair harness as required. Refer to relevant section of workshop manual, check cylinder 5 glow plug for internal short circuit to ground. Replace glow plug if required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new glow plug control module
	P067E-00	Cylinder 6 Glow Plug Control Circuit Low - No sub type information	▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Cylinder 6 glow plug circuit short circuit to ground ▪ Glow plug failure ▪ Glow plug control module failure	▪ Refer to the electrical circuit diagrams and check the glow plug control circuit between th glow plug control module and the glow plug connector for short circuit to ground, check t power supply and ground connections to the glow plug control module, ensure that the harness is checked for intermittent failures. Repair harness as required. Refer to relevant section of workshop manual, check cylinder 6 glow plug for internal short circuit to ground. Replace glow plug if required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new glow plug control module
	P0683-00	Glow Plug Control Module to PCM Communication Circuit - No sub type information	**NOTE:** Circuit reference GPC	▪ This DTC is set when the data transmission on the control circ between the engine control module and the glow plug control module has an error. Check for other related DTCs and refer to the relevant DTC

			▪ Engine control module to glow plug control module circuit failure ▪ Glow plug control module failure ▪ Engine control module failure	index. Refer to the electrical circuit diagrams and check the glowplug control signal line fro the engine control module to t glow plug control module for failures such as intermittent op circuits, high resistance, short circuit to ground, short circuit t power. Check the glowplug monitor signal line from the glowplug control module to the engine control module for intermittent open circuits, high resistance, short circuit to ground, short circuit to power ▪ Refer to the electrical circuit diagrams and check power and ground supplies to the glowplu control module. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P068A-00	ECM/PCM Power Relay De-Energized - Too Early - No sub type information	▪ Vehicle battery disconnected before engine management system relay has powered down ▪ Engine management system high current relay failure ▪ Central junction box failure ▪ Harness failure - Relay control circuit	▪ This DTC is set when the engin management system high curre relay contacts open early - Indicating a power hold failure. Check the vehicle battery has n been disconnected before the engine management system rel has powered down. Check the operation of the engine management system high curre relay. Refer to the electrical circuit diagrams and check the engine management system hi current relay supply and contro circuits for open circuits, high resistance, short circuit to ground, short circuit to power, short circuit to other circuits. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new relay o

			central junction box as requirec
P06AF-00	Torque Management System - Forced Engine Shutdown - No sub type information	▪ Injection cut off demand for shut off coordinator ▪ The DTC sets when the engine speed has exceeded the hard engine speed limiter. When this DTC sets, a request to the shut-off coordinator is made for a reversible shut-off of the injection power stages, ensuring no injections take place. It is reset when engine speed falls less than or equal to the hard limiter ▪ Engine run away ▪ Excessive injected fuel ▪ Over boost from a turbocharger	▪ Check for related engine contro module DTCs and repair these first. Clear the DTC and retest
P0700-00	Transmission Control System (MIL Request) - No sub type information	▪ MIL request by automatic gearbox ▪ HS CAN network, short circuit to power, short circuit to ground, open circuit	▪ Check transmission control module for DTCs and refer to t relevant DTC index. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams a check HS CAN network for sho circuit to power, short circuit to ground, open circuit
P0850-64	Park / Neutral Switch Input Circuit - Signal plausibility failure	**NOTE:** Circuit reference PNS ▪ The engine control module detected plausibility failures	▪ Refer to the electrical circuit diagrams and check the park/neutral switch circuit, for short circuit to power, short circuit to ground, open circuit ▪ If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams a check HS CAN network for sho circuit to power, short circuit to

			▪ Hardwired park/neutral switch does not match the information within the CAN signal ▪ Park/neutral switch circuit, short circuit to power, short circuit to ground, open circuit ▪ HS CAN network failure	ground, open circuit
	P0A09-16	DC/DC Converter Status Circuit Low - Circuit voltage below threshold	▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P0A0F-00	Engine Failed to Start - No sub type information	▪ Engine control module has logged an extended crank with no engine start ▪ Harness failure - Starter motor circuit and relays ▪ Vehicle battery failure ▪ Starter motor mechanical failure	▪ Check for related engine contro module DTCs and repair these first. Using the manufacturer approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the starter motor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the battery care requirements section and verify that the vehicle battery is fully charged and serviceable before continuing with further diagnostic tests ▪ Check Starter motor for

			mechanical failure. Install a new starter motor as required
P0A0F-71	Engine Failed to Start - Actuator stuck	- The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay - Vehicle battery failure - Jammed starter motor	- Refer to the battery care requirements section and verify that the vehicle battery is fully charged and serviceable before continuing with further diagnostic tests - Check for a jammed or damage starter motor. Install a new star motor as required
P0A10-17	DC/DC Converter Status Circuit High - Circuit voltage above threshold	- The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Corrupt engine control module software - Engine control module power supply failure - Engine control module ground supply failure - Engine control module failure	- Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software - Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new engine control module as required
P0A14-13	Engine Mount Control A Circuit / Open - Circuit open	**NOTE:** Circuit reference AEM_1 - The engine control module has determined an open circuit via lack of bias voltage, low	- Refer to the electrical circuit diagrams and check the active engine mount solenoid control circuit for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

		current flow, no change in the state of an input in response to an output ▪ Wiring harness failure - Active engine mount solenoid circuit	
P0A15-11	Engine Mount Control A Circuit Low - Circuit short to ground	**NOTE:** Circuit reference AEM_1 ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Wiring harness failure - Active engine mount solenoid circuit	▪ Refer to the electrical circuit diagrams and check the active engine mount solenoid control circuit for short circuit to groun Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P0A16-12	Engine Mount Control A Circuit High - Circuit short to battery	**NOTE:** Circuit reference AEM_1 ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Wiring harness failure - Active engine mount solenoid circuit	▪ Refer to the electrical circuit diagrams and check the active engine mount solenoid control circuit for short circuit to power Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

P0A16-4B	Engine Mount Control A Circuit High - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Active engine mount solenoid circuit short circuit to ground, short circuit to power ▪ Active engine mount solenoid failure	▪ Refer to the electrical circuit diagrams and check active engine mount solenoid circuit f short circuit to ground, power. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine mount control vacuum solenoi as required
P0A1A-00	Generator Control Module - No sub type information	▪ Generator to engine control module LIN circuit, open circuit ▪ Generator/engine control module failure	▪ Check for good/clean contact a generator and engine control module LIN circuit connectors/pins. Refer to the electrical circuit diagrams and check generator to engine control module LIN circuit for open circuit. Check for engine control module hardware DTCs and refer to relevant DTC index Clear DTCs and repeat automated diagnostic procedu using the manufacturer approv diagnostic system ▪ Check and install a new generator / engine control module as required
P0A1A-87	Generator Control Module - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module communication with generator) ▪ Generator to engine control module LIN circuit open circuit	▪ Check for good/clean contact a generator and engine control module LIN circuit connectors/pins. Refer to the electrical circuit diagrams and check generator to engine control module LIN circuit for open circuit. Check for engine control module hardware DTCs and refer to relevant DTC index Clear DTCs and repeat automated diagnostic procedu using the manufacturer approv diagnostic system

	P0A3B-00	Generator Over Temperature - No sub type information	▪ Generator wiring/connectors heat damaged ▪ Generator circuit short circuit to ground, high resistance ▪ Generator failure	▪ Check the generator wiring anc connectors for heat damage ▪ Refer to the electrical circuit diagrams and check generator wiring for short circuit to groun high resistance. Repair harness required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new generator as required
	P0A7B-00	Battery Energy Control Module Requested MIL Illumination - No sub type information	▪ MIL request by battery energy control module ▪ HS CAN network, short circuit to power, short circuit to ground, open circuit	▪ Check battery energy control module for DTCs and refer to relevant DTC index. Rectify the first ▪ If this DTC is logged with other HS CAN related DTCs refer to the electrical circuit diagrams a check HS CAN network for sho circuit to power, short circuit to ground, open circuit ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0A94-00	DC/DC Converter Performance - No sub type information	**NOTE:** Circuit reference V_V_5V10 V_V_5V9 V_V_5V8 V_V_5V7 V_V_5V6 V_V_5V5 V_V_5V4 V_V_5V3 V_V_5V2 V_V_5V1 ▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check the engine control module power supply fo open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the engine control module ground supply for open circuit. Repair harness as required. Using the manufacturer approved

			- Engine control module failure	diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new engine control module as required
	P0A94-53	DC/DC Converter Performance - Deactivated	- The engine control module has indicated that some portion of the software program has been disabled - This DTC has set because the hybrid DC to DC converter has been disabled by the service routine	- Using the manufacturer approv diagnostic system carry out 'Hybrid DC to DC converter mode' routine (0x4089) using tl 'Service Functions' tab - Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P0AB9-68	Hybrid System Performance - Event information	- The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module - The DTC is set when the vehicle supervisory controller sends the engine control module a flag to indicate that an inertia start has been attempted. An inertia start is where vehicle supervisory controller analyses vehicle speed and gear position when driving in EV mode and decides if it is possible to bump start the engine, by closing the K0 clutch and connecting the engine to the spinning drivetrain. The only time inertia start is used is when the electric power	- Check engine control module f additional DTCs and refer to relevant DTC index. Rectify the first - Refer to the battery care requirements, section 414-00, inspect the vehicle battery and ensure it is fully charged and serviceable - Check the vehicle charging system performance to ensure the voltage regulation is correc - Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

			inverter converter reports that it is no longer capable of supporting 12 volt loads. Therefore to use the starter motor to initiate a restart from EV mode in these conditions could risk the 12 volt supply dropping too low to keep all modules alive. Therefore if vehicle is moving fast enough an inertia start is performed and engine control module is told to inhibit the starter motor. If vehicle is not moving fast enough or the inertia start fails then the starter motor will be used when the vehicle comes to rest	
	P115A-00	Low Fuel Level - Forced Limited Power - No sub type information	- Low level fuel condition - Fuel level sensor signal circuit - Short circuit to ground, short circuit to power, open circuit - Fuel level sensor failure	- Check the fuel level, add fuel if required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - If there is sufficient fuel, refer to the electrical circuit diagrams a check the fuel level sensor circuits for short, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new fuel lev sensor as required
	P115A-68	Low Fuel Level - Forced Limited Power - Event information	The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from	- Check the fuel level, add fuel if required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - If there is sufficient fuel, refer to the electrical circuit diagrams a check the fuel level sensor circuits for short, open circuit.

			another system or control module ▪ Low level fuel condition ▪ Fuel level sensor signal circuit - Short circuit to ground, short circuit to power, open circuit ▪ Fuel level sensor failure	Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel lev sensor as required
	P115B-00	Low Fuel Level - Forced Engine Shutdown - No sub type information	▪ Low level fuel condition enabling run dry strategy ▪ Fuel level sensor signal circuit - Short circuit to ground, short circuit to power, open circuit ▪ Fuel level sensor failure	▪ Check the fuel level, add fuel if required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ If there is sufficient fuel, refer to the electrical circuit diagrams a check the fuel level sensor circuits for short, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new fuel lev sensor as required
	P115F-11	Electronic Control Module Cooling Fan Circuit - Circuit short to ground	**NOTE:** Circuit reference EBF ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Engine control module cooling fan circuits ▪ Cooling fan failure	▪ Refer to the electrical circuit diagrams and check the engine control module cooling fan circ for short circuit to ground. Rep harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan as required

P115F-12	Electronic Control Module Cooling Fan Circuit - Circuit short to battery	**NOTE:** Circuit reference EBF ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure - Engine control module cooling fan circuits	▪ Refer to the electrical circuit diagrams and check the engine control module cooling fan circ for short circuit to power. Repa harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P115F-13	Electronic Control Module Cooling Fan Circuit - Circuit open	**NOTE:** Circuit reference EBF ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Engine control module cooling fan circuits ▪ Cooling fan failure	▪ Refer to the electrical circuit diagrams and check the engine control module cooling fan circ for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new cooling fan as required
P115F-19	Electronic Control Module Cooling Fan Circuit - Circuit current above threshold	▪ The engine control module has measured current flow above a specified range ▪ Engine control module cooling fan circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the engine control module cooling fan circ for short circuit to ground. Rep harness as required. Using the manufacturer approved diagnostic system clear all store

			▪ Engine control module cooling fan failure or obstructed	DTCs using the 'Diagnosis men tab and retest ▪ Check engine control module cooling fan for obstruction. Check and install a new engine control module cooling fan as required
	P120F-00	Fuel Pressure Regulator Excessive Variation - No sub type information	▪ Fuel pressure regulator signal circuit short circuit to ground, short circuit to power, open circuit ▪ Fuel pressure regulator VREF circuit high resistance ▪ Fuel pressure regulator failure	▪ Using the manufacturer approv diagnostic system check for fue pump related DTCs ▪ Refer to the electrical circuit diagrams and check the fuel pressure regulator signal circuit for short circuit to ground, shor circuit to power, open circuit. Repair wiring harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the electrical circuit diagrams and check the fuel pressure regulator VREF circuit for high resistance. Repair wirin harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new Fuel pressure regulator as required
	P1247-00	Turbocharger Boost Pressure Low - No sub type information	▪ Air intake system, low pressure intake blocked or restricted ▪ Exhaust system blocked or restricted ▪ Charge air recirculation solenoid stuck open in bi-turbo mode ▪ Charge air solenoid stuck closed in bi-turbo mode ▪ Turbine intake solenoid stuck open ▪ Air intake system,	▪ Test air intake system for air lea and check exhaust system for blockages or restriction ▪ If this DTC is logged with P00B 00, suspect charge air recirculation solenoid stuck ope in bi-turbo mode ▪ If this DTC is logged with P00B 00, suspect charge air solenoid stuck closed in bi-turbo mode ▪ If this DTC is logged with P023 94, P00BD-07, P22D2-77 & P22CF-71 suspect turbine intak solenoid stuck open

			charge air system high pressure charge air leak on bank 1 ▪ Air intake system, charge air system high pressure charge air leak in bi-turbo mode on bank 2 ▪ Variable geometry turbocharger compressor wheel seized	▪ If this DTC is logged with P006 00, P0402-00 & P00BF-07 suspect air intake system, charg air system high pressure charge air leak on bank 1 ▪ If this DTC is logged with P006 00 suspect air intake system, charge air system high pressure charge air leak in bi-turbo mod on bank 2 ▪ If this DTC is logged with P00B 07 & P00BD-07 suspect variabl geometry turbocharger compressor wheel seized ▪ If this DTC is logged with P023 94, P00BD-07, P22D2-77 & P22CF-71 suspect Air intake system, low pressure intake blocked or restricted
	P1334-00	EGR Throttle Position Sensor Minimum/Maximum Stop Performance - No sub type information	▪ Exhaust gas recirculation throttle position sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Exhaust gas recirculation throttle position sensor failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check exhaust ga recirculation throttle position sensor circuit for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas recirculation throttle positio sensor as required ▪ Check and install a new engine control module as required
	P151B-00	Idle Speed Control - RPM Lower Than Expected - No sub type information	▪ Indicated torque at idle < minimum	▪ Using the manufacturer approv diagnostic system check for additional DTCs and refer to th relevant DTC index. Check engine oil level. Carry out cylinder compression tests. Check for mechanical failure of engine
	P151C-00	Idle Speed Control - RPM Higher Than Expected - No	▪ Indicated torque at idle > maximum	▪ Using the manufacturer approv diagnostic system check for

		sub type information		additional DTCs and refer to th relevant DTC index. Check engine oil level. Carry out cylinder compression tests. Check for mechanical failure of engine
	P1551-32	Cylinder 1 Injector Circuit Range/Performance - Signal low time < minimum	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Injector control circuit failure ▪ Injector failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the injecto control circuits for intermittent failures, open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the workshop manual and check the injector is to specification and has been installed correctly. Check and install a new injector as require ▪ Check and install a new engine control module as required
	P1551-35	Cylinder 1 Injector Circuit Range/Performance - Signal high time > maximum	**NOTE:** Circuit reference O_P_PVL11 O_P_PVH11 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Injector control circuit failure ▪ Injector failure ▪ Engine control module failure	▪ Refer to the electrical circuit diagrams and check the injecto control circuits for intermittent failures, open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Refer to the workshop manual and check the injector is to specification and has been installed correctly. Check and install a new injector as require ▪ Check and install a new engine control module as required

	P1552-32	Cylinder 2 Injector Circuit Range/Performance - Signal low time < minimum	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t ground ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other
	P1552-35	Cylinder 2 Injector Circuit Range/Performance - Signal high time > maximum	**NOTE:** Circuit reference O_P_PVL12 O_P_PVH12 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t power ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other
	P1553-32	Cylinder 3 Injector Circuit Range/Performance - Signal low time < minimum		▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t ground ▪ Refer to the electrical circuit diagrams and check fuel injecto

			NOTE: Circuit reference O_P_PVL13 O_P_PVH13 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	control circuit for short to each other
	P1553-35	Cylinder 3 Injector Circuit Range/Performance - Signal high time > maximum	**NOTE:** Circuit reference O_P_PVL13 O_P_PVH13 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t power ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other
	P1554-32	Cylinder 4 Injector Circuit Range/Performance - Signal low time < minimum	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t ground ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each

		▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	other
P1554-35	Cylinder 4 Injector Circuit Range/Performance - Signal high time > maximum	**NOTE:** Circuit reference O_P_PVL21 O_P_PVH21 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t power ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other
P1555-32	Cylinder 5 Injector Circuit Range/Performance - Signal low time < minimum	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t ground ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other

			injector control circuit short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	
	P1555-35	Cylinder 5 Injector Circuit Range/Performance - Signal high time > maximum	**NOTE:** Circuit reference O_P_PVL22 O_P_PVH22 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t power ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other
	P1556-32	Cylinder 6 Injector Circuit Range/Performance - Signal low time < minimum	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ This DTC is set when the engine control module detects that the injector signal low time is less than the minimum value ▪ Harness failure - fuel injector control circuit	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t ground ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other

		short circuit to ground ▪ Harness failure - fuel injector control circuit short to each other	
P1556-35	Cylinder 6 Injector Circuit Range/Performance - Signal high time > maximum	**NOTE:** Circuit reference O_P_PVL23 O_P_PVH23 ▪ This DTC is set when the engine control module detects that the injector signal high time is greater than the maximum value ▪ Harness failure - fuel injector control circuit short circuit to power ▪ Harness failure - fuel injector control circuit short to each other	▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short circuit t power ▪ Refer to the electrical circuit diagrams and check fuel injecto control circuit for short to each other
P1575-62	Pedal Position Out Of Self Test Range - No sub type information	▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Brake pedal switch - Circuit Brake_SW_2 - Short circuit to power ▪ Brake pedal switch - Circuit Brake_SW_2 - Open circuit ▪ Brake switch failure	▪ Refer to the electrical circuit diagrams and check brake peda switch - Circuit Brake_SW_2 - F short circuit to power ▪ Refer to the electrical circuit diagrams and check brake peda switch - Circuit Brake_SW_2 - F open circuit ▪ Check and install a new brake switch as required
P1627-16	Module Supply Voltage Out Of Range - Circuit voltage below threshold	**NOTE:** Circuit reference VBR1 VBR2	▪ Refer to the battery care requirements, section 414-00, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check the vehicle charging

			▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Vehicle battery failure ▪ Charging system failure ▪ Engine control module power or ground supply ▪ Engine junction box failure ▪ Power distribution failure ▪ Engine control module failure	system performance to ensure the voltage regulation is correc ▪ Using the manufacturer approv diagnostic system check datalogger signals, Control module voltage (0xF442). This DTC is set when the engine control module supply voltage below the threshold value. Refe to the electrical circuit diagram and check the four engine control module power ground circuits for high resistance or open circuits ▪ Check the switched power supply feeds to the engine control module which come fro the engine junction box though the engine management systen relay. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P1627-17	Module Supply Voltage Out Of Range - Circuit voltage above threshold	**NOTE:** Circuit reference VBR1 VBR2 ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Vehicle battery failure ▪ Charging system failure ▪ Engine control module power or ground supply ▪ Engine junction box failure ▪ Power distribution failure	▪ Refer to the battery care requirements, section 414-00, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Check the vehicle charging system performance to ensure the voltage regulation is correc ▪ Using the manufacturer approv diagnostic system check datalogger signals, Control module voltage (0xF442). This DTC is set when the engine control module supply voltage greater than the threshold valu Refer to the electrical circuit diagrams and check the four engine control module power ground circuits for high resistance or open circuits ▪ Check the switched power

				supply feeds to the engine control module which come fro the engine junction box though the engine management systen relay. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P162F-00	Starter Motor Disabled - Engine Crank Time Too Long - No sub type information	▪ Vehicle battery failure ▪ Other starting related failures	▪ Refer to the battery care requirements, section 414-00, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Using the manufacturer approv diagnostic system, check the engine control module, for related DTCs and refer to the relevant DTC index
	P1631-00	Main Relay (power hold) - No sub type information	**NOTE:** Circuit reference PWRSTN ▪ Vehicle battery disconnected before engine management system relay has powered down ▪ Relay control circuit failure ▪ Engine management system high current relay failure ▪ Central junction box failure	▪ This DTC is set when the engin management system high curre relay contacts open early - Indicating a power hold failure. Check the vehicle battery has n been disconnected before the engine management system rel has powered down. Check the operation of the engine management system high curre relay ▪ Refer to the electrical circuit diagrams and check the engine management system high curre relay supply and control circuits for open circuits, high resistanc short circuit to ground, short circuit to power, short circuit to other circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new relay as required ▪ Check and install a new central junction box as required

P1631-73	Main Relay (power hold) - Actuator stuck closed	**NOTE:** Circuit reference PWRSTN ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Engine management system high current relay failure ▪ Relay control circuit failure	▪ This DTC is set when the engin management system high curre relay contacts are detected stu closed by the engine control module. Check the operation o the engine management systen high current relay. Refer to the electrical circuit diagrams and check the engine management system high current relay suppl and control circuits for open circuits, high resistance, short circuit to ground, short circuit t power, short circuit to other circuits. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new relay as required
P165D-01	Grill Shutter "A" Control Circuit - General electrical failure	▪ Grill shutter actuator supply voltage falls outside the specified range ▪ Grill shutter actuator internal circuit failure	▪ Refer to the electrical circuit diagrams and check the grill shutter actuator circuits for sho circuit to ground, short circuit t power, open circuit, high resistance. Repair harness as required ▪ Using the manufacturer approv diagnostic system, operate the grill shutter through the full operating range and check that the DTC does not reset ▪ Check and install a new grill shutter actuator as required
P165D-07	Grill Shutter "A" Control Circuit - Mechanical failures	▪ Grill shutter "A" actuator has become disconnected from the grill shutter ▪ Linkage between the grill shutter "A" actuator and grill shutter has failed	▪ Check all vanes within the grill shutter are connected to the gr shutter actuator. Repair as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Using the manufacturer approv diagnostic system, operate the grill shutter through the full

				operating range and check that the DTC does not reset ▪ Check and install a new grill shutter as required
	P165D-13	Grill Shutter "A" Control Circuit - Circuit open	▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Grill shutter "A" control circuit - Open circuit ▪ Grill shutter actuator internal circuit failure	▪ Refer to the electrical circuit diagrams and check the grill shutter actuator control circuit f open circuit. Repair harness as required ▪ Check and install a new grill shutter actuator as required
	P165D-4B	Grill Shutter "A" Control Circuit - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ The grill shutter actuator internal circuits are overheated	▪ Check the grill shutter for damage, obstructions and blockages. Repair as required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Using the manufacturer approv diagnostic system, operate the grill shutter through the full operating range and check that the DTC does not reset ▪ Check and install a new grill shutter actuator as required
	P165D-87	Grill Shutter "A" Control Circuit - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module communication with grill shutter)	▪ Refer to the electrical circuit diagrams and check the grill shutter actuator control circuit f open circuit, high resistance. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new grill shutter actuator as required

			▪ The engine control module has not received the expected signal from the grill shutter "A" actuator ▪ Grill shutter "A" control circuit - Open circuit ▪ Grill shutter actuator internal circuit failure	
	P165D-97	Grill Shutter "A" Control Circuit - Component or system operation obstructed or blocked	▪ The engine control module has detected that the operation of a component is prevented by an obstruction ▪ Grill shutter damaged, obstructed or blocked ▪ Grill shutter actuator has failed all calibration attempts	▪ Check the grill shutter for damage, obstructions and blockages. Repair as required. Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Using the manufacturer approv diagnostic system, operate the grill shutter through the full operating range and check that the DTC does not reset ▪ Check and install a new grill shutter actuator as required
	P1695-00	CAN Link Injection Pump Control Module/Engine Control Module - No sub type information	▪ The engine control module internal powerstage is overheated	▪ Check engine control module f DTCs and refer to the relevant DTC index. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check the engine control modu is sufficiently able to be cooled
	P1703-00	Brake Switch Out Of Self Test Range - No sub type information	▪ Brake switch circuit, short circuit to ground, short circuit to power, high resistance, open circuit ▪ Brake switch incorrect adjustment ▪ Brake switch internal failure	▪ Refer to the electrical circuit diagrams and check the brake switch for short to ground, shor circuit to power, high resistance open circuit. Repair wiring harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and adjust brake switch

				required ▪ Check and install a new brake switch as required
	P1712-00	Transmission Torque Reduction Request Signal - No sub type information	▪ Unintended torque request signal sent over CAN from transmission control module to engine control module. The engine control module recognizes it is unintended and applies a torque limit and sets the DTC	▪ Check transmission control module for related DTCs and refer to relevant DTC index, Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P1719-68	Engine Torque Signal - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Working limitation information ▪ The DTC sets as a result of an engine torque limitation caused by an engine overheat situation	▪ Check engine control module f related DTCs and refer to relevant DTC index ▪ Refer to the workshop manual and check the cooling system is functioning correctly
	P2002-68	Diesel Particulate Filter Efficiency Below Threshold (Bank 1) - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Diesel particulate filter	▪ Using the manufacturer approv diagnostic system, check for related DTCs and refer to the relevant DTC index. Carry out a diesel particulate filter regeneration

			regeneration disabled by other DTCs logged	
	P2031-00	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 2 - No sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post catalytic converter short circuit to ground ▪ Exhaust gas temperature sensor post catalytic converter failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post cataly converter circuit for short circui to ground ▪ Check and install a new exhaus gas temperature sensor post catalytic converter as required
	P2032-00	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 2 - No sub type information	▪ Harness failure - Particulate filter inlet exhaust gas temperature sensor ▪ Particulate filter inlet exhaust gas temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 2 Voltage (0x03C4), Exhaust Gas Temperature Bank 1 Sensor 2 (0x03F5). This DTC is set if the particulate filter inlet exhaust g temperature sensor fails a diagnostic check due to the circuit voltage being less than t expected value. Refer to the electrical circuit diagrams and check the signal circuit for oper circuit, short circuit to power, short circuit to ground, high resistance. Check the ground circuit for open circuit, high resistance, short circuit to powe Repair harness as required ▪ Check and install a new exhaus gas temperature sensor as required
	P2033-00	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 2 - No sub type information	▪ Harness failure - Particulate filter inlet exhaust gas temperature sensor	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 2 Voltage (0x03C4), Exhaust Gas

			▪ Particulate filter inlet exhaust gas temperature sensor failure	Temperature Bank 1 Sensor 2 (0x03F5). This DTC is set if the particulate filter inlet exhaust g temperature sensor fails a diagnostic check due to the circuit voltage being greater th the expected value. Refer to th electrical circuit diagrams and check the signal circuit for oper circuit, short circuit to power, short circuit to ground, high resistance. Check the ground circuit for open circuit, high resistance, short circuit to powe Repair harness as required ▪ Check and install a new exhaus gas temperature sensor as required
	P2080-00	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1 - No sub type information	**NOTE:** Circuit reference CCCIT_B ▪ Exhaust gas temperature sensor post turbocharger overheating ▪ Exhaust gas temperature sensor post turbocharger circuit short circuit to ground	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 1 Voltage (0x03BF), Exhaust Gas Temperature Bank 2 Sensor 1 (0x03F7) ▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new exhaus gas temperature sensor post turbocharger - bank 2, sensor 1 as required
	P2082-00	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 2 Sensor 1 - No sub type information	**NOTE:** Circuit reference STOT ▪ Exhaust gas	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post turbocharger circuit for short circuit to ground, open circuit, high resistance ▪ Clear the DTC and retest ▪ Check and install a new exhaus

			temperature sensor post turbocharger short circuit short circuit to ground, open circuit, high resistance ▪ Exhaust gas temperature sensor post turbocharger failure	gas temperature sensor post turbocharger as required
	P2084-00	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 2 - No sub type information	**NOTE:** Circuit reference CCCOT_A ▪ Exhaust gas temperature sensor post catalytic converter overheating ▪ Exhaust gas temperature sensor post catalytic converter circuit, short circuit to ground ▪ Exhaust gas temperature sensor post catalytic converter failure	▪ Refer to the electrical circuit diagrams and check exhaust ga temperature sensor post cataly converter circuit for short circui to ground ▪ Clear the DTC and retest ▪ Check and install a new exhaus gas temperature sensor post catalytic converter as required
	P2121-1F	Throttle/Pedal Position Sensor/Switch D Circuit Range/Performance - Circuit intermittent	**NOTE:** Circuit reference APP_1 ▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Accelerator Pedal Position D (0xF449), Peda Position Sensor Voltage - Senso 1 (0x0914). The accelerator pec position sensor consists of two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when t engine control module detects noise on the pedal demand 1 signal circuit. Refer to the electrical circuit diagrams and check the reference voltage an ground connections to the accelerator pedal position sens Check signal circuits for high

				resistance, open circuit, short circuit to power, short circuit to ground. Check all accelerator pedal position sensor circuits fo intermittent failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new accelerator pedal position sens as required
	P2122-00	Throttle/Pedal Position Sensor/Switch D Circuit Low - No sub type information	**NOTE:** Circuit reference APP_1 ▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Accelerator Pedal Position D (0xF449), Peda Position Sensor Voltage - Senso 1 (0x0914). The Accelerator Pedal Position sensor consists o two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control modul detects the pedal demand 1 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage and ground connections to the accelerator pedal position sens Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new accelerator pedal position sens as required
	P2123-00	Throttle/Pedal Position Sensor/Switch D Circuit High - No sub type information	**NOTE:** Circuit reference APP_1	▪ Using the manufacturer approv diagnostic system check datalogger signals, Accelerator Pedal Position D (0xF449), Peda Position Sensor Voltage - Senso 1 (0x0914). The Accelerator

			▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	Pedal Position sensor consists c two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control modul detects the pedal demand 1 signal range is high. Refer to th electrical circuit diagrams and check the reference voltage an ground connections to the accelerator pedal position sens Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new accelerator pedal position sens as required
	P2126-1F	Throttle/Pedal Position Sensor/Switch E Circuit Range/Performance - Circuit intermittent	**NOTE:** Circuit reference APP_2 ▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Accelerator Pedal Position E (0xF44A), Peda Position Sensor Voltage - Senso 2 (0x0915). The Accelerator Pedal Position sensor consists c two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control modul detects noise on the pedal demand 2 signal circuit. Refer t the electrical circuit diagrams a check the reference voltage an ground connections to the accelerator pedal position sens Check signal circuits for high resistance, open circuit, short circuit to power, short circuit to ground. Check all accelerator pedal position sensor circuits fo intermittent failures. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men

			tab and retest ▪ Check and install a new accelerator pedal position sens as required
P2127-00	Throttle/Pedal Position Sensor/Switch E Circuit Low - No sub type information	**NOTE:** Circuit reference APP_2 ▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Accelerator Pedal Position E (0xF44A), Peda Position Sensor Voltage - Senso 2 (0x0915). The Accelerator Pedal Position sensor consists c two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control modul detects the pedal demand 2 signal range is low. Refer to the electrical circuit diagrams and check the reference voltage an ground connections to the accelerator pedal position sens Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new accelerator pedal position sens as required
P2128-00	Throttle/Pedal Position Sensor/Switch E Circuit High - No sub type information	**NOTE:** Circuit reference APP_2 ▪ Harness failure - Accelerator pedal position sensor circuit ▪ Accelerator pedal position sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Accelerator Pedal Position E (0xF44A), Peda Position Sensor Voltage - Senso 2 (0x0915). The Accelerator Pedal Position sensor consists c two potentiometer circuits feeding independent pedal demand signals to the engine control module. This DTC is set when the engine control modul detects the pedal demand 2 signal range is high. Refer to th electrical circuit diagrams and check the reference voltage an

				ground connections to the accelerator pedal position sens Check signal circuits for high resistance, open circuits, short circuit to power, short circuit to ground. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new accelerator pedal position sens as required
	P2138-00	Throttle/Pedal Position Sensor/Switch D / E Voltage Correlation - No sub type information	▪ Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit ▪ Accelerator pedal position sensor failure	▪ Refer to the electrical circuit diagrams and check accelerato pedal position sensor 1 circuit f short circuit to ground, short circuit to power, open circuit. Refer to the electrical circuit diagrams and check accelerato pedal position sensor 2 circuit f short circuit to ground, short circuit to power, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new accelerator pedal position sens as required
	P2138-64	Throttle/Pedal Position Sensor/Switch D / E Voltage Correlation - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit ▪ Accelerator pedal position sensor failure	▪ Refer to the electrical circuit diagrams and check accelerato pedal position sensor 1 circuit f short circuit to ground, short circuit to power, open circuit. Refer to the electrical circuit diagrams and check accelerato pedal position sensor 2 circuit f short circuit to ground, short circuit to power, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new accelerator pedal position sens

			as required
P2138-67	Throttle/Pedal Position Sensor/Switch D / E Voltage Correlation - Signal incorrect after event	▪ Accelerator pedal position sensor circuit short circuit to ground, short circuit to power, open circuit ▪ Accelerator pedal position sensor failure	▪ Refer to the electrical circuit diagrams and check accelerato pedal position sensor 1 circuit f short circuit to ground, short circuit to power, open circuit. Refer to the electrical circuit diagrams and check accelerato pedal position sensor 2 circuit f short circuit to ground, short circuit to power, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new accelerator pedal position sens as required
P213E-01	Fuel Injection System Fault - Forced Engine Shutdown - General electrical failure	▪ Quantity shut-off of the dual-mass flywheel	▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P2141-00	Exhaust Gas Recirculation Throttle Control Circuit A Low - No sub type information	▪ Harness failure - Throttle motor control circuit ▪ Throttle motor failure	▪ This DTC is set when the engin control module detects a short circuit to ground on the throttle motor control circuit. Refer to t electrical circuit diagrams and check both the throttle plate actuator control circuits for sho circuit to ground. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new throttle motor control actuator as required
P2142-00	Exhaust Gas Recirculation Throttle Control Circuit A High - No sub type	▪ Harness failure - Throttle motor control circuit	▪ This DTC is set when the engin control module detects a short circuit to power on the throttle

		information	▪ Throttle motor failure	motor control circuit. Refer to t electrical circuit diagrams and check both the Throttle Plate Actuator control circuits for sho circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new throttle motor control actuator as required
	P2177-00	System Too Lean Off Idle - Bank 1 - No sub type information	▪ Oxygen concentration implausibly high ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for excess fuel at exhaus manifold, downpipe, heated oxygen sensor ▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit t ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P2178-00	System Too Rich Off Idle - Bank 1 - No sub type information	▪ Oxygen concentration implausibly low ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor ▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit t ground, high resistance, open

				circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P2191-00	System Too Lean at Higher Load - Bank 1 - No sub type information	▪ Oxygen concentration implausibly high ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for excess fuel at exhaus manifold, downpipe, heated oxygen sensor ▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit t ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P2192-00	System Too Rich at Higher Load - Bank 1 - No sub type information	▪ Oxygen concentration implausibly low ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor ▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit t ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated

				oxygen sensor as required
	P2195-00	O2 Sensor Signal biased/Stuck Lean - Bank 1, Sensor 1 - No sub type information	▪ Air leak at exhaust manifold, downpipe, heated oxygen sensor bank 1 ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P2196-00	O2 Sensor Signal biased/Stuck Rich - Bank 1, Sensor 1 - No sub type information	▪ Fuel injection system failure ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P2226-62	Barometric Pressure Sensor A Circuit - Signal compare failure	▪ The engine control module detected failure when comparing two or more input parameters for plausibility ▪ Differential pressure	▪ Refer to the electrical circuit diagrams and check differential pressure sensor circuit for short circuit to ground, short circuit t power, open circuit, high resistance

			sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Differential pressure sensor failure ▪ The engine control module has been submersed in water or mud ▪ The engine control module has been sealed in a non approved covering ▪ Differential pressure sensor failure ▪ Engine control module failure	▪ Check the engine control modu is clean and dry ▪ Check the engine control modu is not sealed by any non approved covering ▪ Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new differential pressure sensor as required ▪ Check and install a new engine control module as required
	P2228-00	Barometric Pressure Sensor A Circuit Low - No sub type information	▪ Corrupt engine control module software ▪ Engine control module power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with the latest level software ▪ Refer to the electrical circuit diagrams and check engine control module power supply circuits for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check engine control module ground supply circuits for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P2229-00	Barometric Pressure Sensor A Circuit High - No sub type information	▪ Corrupt engine control module software ▪ Engine control module	▪ Using the manufacturer approv diagnostic system, re-configure the engine control module with

			power supply failure ▪ Engine control module ground supply failure ▪ Engine control module failure	the latest level software ▪ Refer to the electrical circuit diagrams and check engine control module power supply circuits for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check engine control module ground supply circuits for open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new engine control module as required
	P2238-00	O2 Sensor Positive Current Control Circuit Low - Bank 1, Sensor 1 - No sub type information	▪ Heated oxygen sensor positive current control circuit short circuit to ground, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit t ground, high resistance, open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new heated oxygen sensor as required
	P2245-00	O2 Sensor Reference Voltage Circuit Low - Bank 1, Sensor 1 - No sub type information	**NOTE:** Circuit reference LPV_A ▪ Harness failure - Heated oxygen sensor circuit ▪ Oxygen sensor failure	▪ This DTC is set if the engine control module detects the bar 1, heated oxygen sensor 1 reference voltage is lower than expected. Refer to the electrica circuit diagrams and check the heated oxygen sensor circuits f open circuits, high resistance, short circuit to ground. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new heated

			oxygen sensor as required
P2246-00	O2 Sensor Reference Voltage Circuit High - Bank 1, Sensor 1 - No sub type information	**NOTE:** Circuit reference LPV_A ▪ Harness failure - Heated oxygen sensor circuit ▪ Oxygen sensor failure	▪ This DTC is set if the engine control module detects the bar 1, heated oxygen sensor 1 reference voltage is greater tha expected. Refer to the electrica circuit diagrams and check the heated oxygen sensor circuits f open circuits, high resistance, short circuit to power. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new heated oxygen sensor as required
P2261-73	Turbocharger/Supercharger Bypass Valve "A" - Mechanical - Actuator stuck closed	▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Charge air recirculation solenoid stuck shut during transition from mono turbo to bi-turbo mode	▪ This DTC is set when the engin control module detects that the turbocharger bypass valve is nc operating. Refer to the worksh manual and inspect the charge air recirculation solenoid for sticking shut during transition from mono turbo to bi-turbo mode
P2263-21	Turbocharger/Supercharger Boost System Performance - Signal amplitude < minimum	▪ The engine control module measured a signal voltage below a specified range but not necessarily a short circuit to ground, gain low ▪ Induction system air leak or blockage ▪ Charge air system leak or blockage ▪ Manifold absolute pressure sensor circuit short circuit to power, ground, open circuit	▪ Check induction system for leal blockages ▪ Check charge air system for leaks, blockages. Check for related DTCs ▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to power, ground, open circuit. Repair harness as required ▪ Check and install a new manifo absolute pressure sensor as required

			▪ Manifold absolute pressure sensor failure ▪ Variable geometry turbocharger actuator A sticking, failure ▪ Turbocharger A failure	▪ Check turbocharger rod connection and oil seals ▪ Check and install a new variabl geometry turbocharger actuato as required. Check install new turbocharger as required
	P2263-22	Turbocharger/Supercharger Boost System Performance - Signal amplitude > maximum	▪ The engine control module measured a signal voltage above a specified range but not necessarily a short circuit to power, gain too high ▪ Induction system air leak or blockage ▪ Charge air system leak or blockage ▪ Manifold absolute pressure sensor circuit short circuit to power, ground, open circuit ▪ Manifold absolute pressure sensor failure ▪ Variable geometry turbocharger actuator A sticking, failure ▪ Turbocharger A failure	▪ Check induction system for leal blockages ▪ Check charge air system for leaks, blockages. Check for related DTCs ▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine ▪ Refer to the electrical circuit diagrams and check manifold absolute pressure sensor circuit for short circuit to power, ground, open circuit. Repair harness as required ▪ Check and install a new manifo absolute pressure sensor as required ▪ Check turbocharger rod connection and oil seals ▪ Check and install a new variabl geometry turbocharger actuato as required. Check install new turbocharger as required
	P2264-00	Water in Fuel Sensor Circuit - No sub type information	**NOTE:** Circuit reference WIF ▪ Fuel supply incorrect/contaminated ▪ Harness failure - Water in fuel sensor circuit ▪ Water in fuel sensor	▪ This DTC is set if the engine control module receives information indicating an implausible fuel level. Refer to the workshop manual and chec the fuel system to ensure there an adequate fuel level in the ta and the system is not leaking fu or suffering from air ingress. Check the fuel for contaminatic by other fluids (petrol or water etc) ▪ Refer to the electrical wiring

			failure ▪ Central junction box power distribution failure	diagrams and check the water i fuel sensor signal circuit betwe the sensor and the engine control module for high resistance, open circuit, short circuit to ground, short circuit t power, intermittent failures. Repair harness as required ▪ Check and install a new water i fuel sensor ▪ Refer to the electrical wiring diagrams and check the water i fuel sensor power supply from the central junction box for hig resistance, open circuit, short circuit to ground. Check the water in fuel sensor ground supply circuit for high resistanc open circuit, short circuit to power. Check the voltage supp from the central junction box, check and replace fuses as required. Replace central junction box if there is no volta supply to the water in fuel sens
	P2269-68	Water in Fuel Condition - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Water in fuel ▪ Water in fuel sensor circuit short circuit to power, open circuit ▪ Water in fuel sensor failure	▪ Drain water from fuel filter housing. Clear DTC and wait 3(seconds and re-check DTC has cleared ▪ Refer to the electrical circuit diagrams and check water in fu sensor circuit for short circuit to power, open circuit. Repair harness as required ▪ Check and install a new water i fuel sensor as required
	P226B-00	Turbocharger/Supercharger Boost Pressure Too High - Mechanical - No sub type	▪ Error path indicating whether over boost shut down is active or not	▪ Check for turbocharging relate DTCs and repair these first. Usi the manufacturer approved

		information	▪ This DTC is set when overboost shut down is active. This means that the vehicle has lost control of boost. The engine is shut down as a safety precaution. The engine control module checks for an overboost by system monitoring and also monitors vehicle acceleration, if in gear and engine acceleration at idle, also if in neutral	diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P228C-00	Fuel Pressure Regulator 1 Exceeded Control Limits - Pressure Too Low - No sub type information	▪ Fuel pressure control valve, fuel leak from the high pressure side ▪ Fuel injector stuck open / leaking ▪ Blocked fuel filter ▪ Low pressure fuel pipe leaking from the fuel tank fuel pump ▪ Fuel supply system failure ▪ Harness failure - Fuel rail pressure sensor A circuit high resistance ▪ Fuel rail pressure sensor A failure ▪ Fuel lift pump failure	▪ Check fuel pressure control valv for fuel leakage from the high pressure side ▪ Check for fuel injector stuck open / leaking ▪ Check for blocked fuel filter ▪ Check for low pressure fuel pip leakage from the fuel tank fuel pump ▪ Using the manufacturer approv diagnostic system check datalogger signals, Fuel Rail Pressure Sensor (0x0324), Fuel Rail Pressure (0xF423). Check fo related fuel system DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check fuel pump circuit for high resistance. Chec fuel rail pressure sensor A circu for high resistance ▪ Check and install a new fuel rai pressure sensor A as required ▪ Check and install a new fuel lift pump as required
	P2297-00	O2 Sensor Out of Range During Deceleration Bank 1, Sensor 1 - No sub type information	▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit	▪ Check for fuel / injector related DTCs and repair these first. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men

			Heated oxygen sensor failure	tab and retest. Refer to the electrical circuit diagrams and check heated oxygen sensor fo short circuit to ground, high resistance, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest Check and install a new heated oxygen sensor as required
	P2299-68	Brake Pedal Position/Accelerator Pedal Position Incompatible - Event information	- The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module - Brake pedal switch - Circuit Brake_SW_1 - Short circuit to ground, open circuit - Brake pedal switch - Circuit Brake_SW_2 - Short circuit to ground - Brake pedal switch incorrect adjustment - Brake pedal switch failure - Brake pedal pressed by driver at same time as accelerator pedal pressed	- Refer to the electrical circuit diagrams and check brake peda switch - Circuit Brake_SW_1 - F short circuit to ground, open circuit. Repair harness as required. - Refer to the electrical circuit diagrams and check brake peda switch - Circuit Brake_SW_2 - F short circuit to ground - Repair wiring harness as required, clear DTC. Start the engine press the brake pedal, using maximum travel for great than 1 second taking care not t press the accelerator pedal. Check the system is operating correctly and the DTC does not return - Check and adjust brake switch required - Check and install a new brake pedal switch as required
	P22C5-11	Turbocharger Compressor Outlet Valve Control Circuit / Open - Circuit short to ground	**NOTE:** Circuit reference CSOV	Refer to the electrical circuit diagrams and check the turbocharger charge air solenoi circuit between the engine control module and the control valve for a short circuit to ground. Check power supply tc

			- The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Harness failure - Turbocharger charge air solenoid circuit short circuit to ground	the control valve. Repair harnes as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P22C5-12	Turbocharger Compressor Outlet Valve Control Circuit / Open - Circuit short to battery	**NOTE:** Circuit reference CSOV - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Harness failure - Turbocharger charge air solenoid circuit short circuit to power - Turbocharger charge air solenoid failure	- Refer to the electrical circuit diagrams and check the turbocharger charge air solenoi circuit between the engine control module and the control valve for a short circuit to powe Check power supply to the control valve. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - Check and install a new turbocharger charge air solenoi as required
	P22C5-13	Turbocharger Compressor Outlet Valve Control Circuit / Open - Circuit open	**NOTE:** Circuit reference CSOV - The engine control module has determined	- Refer to the electrical circuit diagrams and check the turbocharger charge air solenoi circuit between the engine control module and the control valve for open circuit. Check power supply to the control valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store

			an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Turbocharger charge air solenoid circuit open circuit ▪ Turbocharger charge air solenoid failure	DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new turbocharger charge air solenoi as required
	P22C5-4B	Turbocharger Compressor Outlet Valve Control Circuit / Open - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Turbocharger charge air solenoid circuit open circuit ▪ Turbocharger charge air solenoid failure	▪ Refer to the electrical circuit diagrams and check the turbocharger charge air solenoi circuit between the engine control module and the control valve for open circuit. Check power supply to the control valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new turbocharger charge air solenoi as required. Check and install a new turbocharger charge air solenoid as required
	P22CF-00	Turbocharger Turbine Inlet Valve Control Circuit / Open - No sub type information	**NOTE:** Circuit reference TSOV ▪ Harness failure - Turbine intake solenoid circuit open circuit ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer to the electrical circuit diagrams a check the turbine intake soleno circuit between the engine control module and the control valve for open circuit. Check power supply to the control valve. Repair harness as require Using the manufacturer approv diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest

				▪ Check and install a new turbine intake solenoid as required
	P22CF-16	Turbocharger Turbine Inlet Valve Control Circuit / Open - Circuit voltage below threshold	**NOTE:** Circuit reference TSOV ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Turbine intake solenoid circuit failure ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer to the electrical circuit diagrams a check the turbine intake soleno circuit between the engine control module and the control valve for a high resistance, intermittent open circuit, short circuit to ground. Check power supply to the control valve. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new turbine intake solenoid as required
	P22CF-19	Turbocharger Turbine Inlet Valve Control Circuit / Open - Circuit current above threshold	▪ The engine control module has measured current flow above a specified range ▪ Harness failure - Turbine intake solenoid circuit failure ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer to the electrical circuit diagrams a check the turbine intake valve control circuit for short circuit to ground. Check power supply to the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new turbine intake solenoid as required
	P22CF-1D	Turbocharger Turbine Inlet Valve Control Circuit / Open - Circuit current out of range	**NOTE:** Circuit reference TSOV	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer to the electrical circuit diagrams a check the turbine intake soleno control circuit for short circuit to

			▪ The engine control module has detected a current outside of the expected range, but not identified as too high or too low ▪ Harness failure - Turbine intake solenoid circuit failure ▪ Turbine intake solenoid failure	ground. Check power supply tc the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new turbine intake solenoid as required
	P22CF-4B	Turbocharger Turbine Inlet Valve Control Circuit / Open - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Turbine intake solenoid circuit failure ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer tc the electrical circuit diagrams a check the turbine intake valve control circuit for short circuit tc ground. Check power supply tc the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new turbine intake solenoid as required
	P22CF-71	Turbocharger Turbine Inlet Valve Control Circuit / Open - Actuator stuck	▪ The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay ▪ Turbine intake solenoid sticking closed ▪ Turbine intake solenoid sticking open ▪ Intake air system, blocked low pressure air intake. This failure mode can be caused by snow packing in the intake system. Symptoms often disappear after the vehicle has been warmed and heat	▪ If this DTC is logged with P22D 77 & P00BC-00 suspect, turbine intake solenoid sticking closed ▪ If this DTC is logged with P023 94, P00BD-07, P22D2-77 & P1247-00 suspect, turbine intak solenoid sticking open ▪ If this DTC is logged with P00B 07, P22D2-77, P1247-00 & P0235-94 suspect, intake air system, blocked low pressure a intake ▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine

		soaked. Similar symptoms to seized primary turbo	
P22D0-00	Turbocharger Turbine Inlet Valve Control Circuit Low - No sub type information	**NOTE:** Circuit reference TSOV ▪ Harness failure - Turbine intake solenoid circuit failure ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer tc the electrical circuit diagrams a check the turbine intake solenc control circuit for short circuit t ground. Check power supply tc the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new turbine intake solenoid as required
P22D0-11	Turbocharger Turbine Inlet Valve Control Circuit Low - Circuit short to ground	**NOTE:** Circuit reference TSOV ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Turbine intake solenoid bank 2 circuit short circuit to ground ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer tc the electrical circuit diagrams a check the turbine intake solenc control circuit on bank 2 for shc circuit to ground. Check power supply to the solenoid. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new turbine intake solenoid as required
P22D0-12	Turbocharger Turbine Inlet Valve Control Circuit Low - Circuit short to battery		▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position -

			NOTE: Circuit reference TSOV ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Harness failure - Turbine intake solenoid bank 2 circuit short circuit to power ▪ Turbine intake solenoid failure	Commanded (0x03F0). Refer to the electrical circuit diagrams a check the turbine intake soleno control circuit on bank 2 for sho circuit to power. Check power supply to the solenoid. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new turbine intake solenoid as required
	P22D1-00	Turbocharger Turbine Inlet Valve Control Circuit High - No sub type information	**NOTE:** Circuit reference TSOV ▪ Harness failure - Turbine intake solenoid circuit short circuit to power ▪ Turbine intake solenoid failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0). Refer to the electrical circuit diagrams a check the turbine intake soleno control circuit for short circuit t power. Check power supply to the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new turbine intake solenoid as required
	P22D2-77	Turbocharger Turbine Inlet Valve Stuck Open - Commanded position not reachable	▪ The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in	▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine ▪ Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position -

			- the actuator or its mechanical environment - Turbine intake solenoid sticking open - Blocked low pressure air intake	Commanded (0x03F0) and Turk Shut Off Valve Opening Positio Measured (0x03F1). Refer to th electrical circuit diagrams and check the turbine intake soleno control circuit for short circuit t power. Check power supply to the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - If this DTC is logged with P023 94, P00BD-07, P1247-00 & P22CF-71, suspect turbine intal solenoid sticking open - If this DTC is logged with P124 00, P0235-94, P00BD-07, & P22CF-71, suspect blocked low pressure air intake - Check and install a new turbine intake solenoid as required
	P22D3-77	Turbocharger Turbine Inlet Valve Stuck Closed - Commanded position not reachable	- The engine control module is unable to command a motor, solenoid or relay, to move a piece of equipment to the commanded position either due to a failure in the actuator or its mechanical environment - Turbine intake solenoid sticking closed	- Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine - Using the manufacturer approv diagnostic system check datalogger signals, Turbo Shut Off Valve Opening Position - Commanded (0x03F0) and Turk Shut Off Valve Opening Positio Measured (0x03F1). Refer to th electrical circuit diagrams and check the turbine intake soleno control circuit for short circuit t power. Check power supply to the solenoid. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes - If this DTC is logged with P00B 00, & P22CF-71, suspect turbin intake solenoid sticking closed - Check and install a new turbine intake solenoid as required

	P22D4-13	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Circuit open	**NOTE:** Circuit reference TSVP ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Turbine intake valve position sensor circuit open circuit, high resistance	▪ Using the manufacturer approv diagnostic system perform the (Turbo, EGR and air path dynan test) routine ▪ Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit for open circuit, high resistance. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes
	P22D4-16	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Circuit voltage below threshold	**NOTE:** Circuit reference TSVP ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Turbine intake valve position sensor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit short circuit to ground. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P22D4-17	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Circuit voltage above threshold - Actuator stuck	**NOTE:** Circuit reference TSVP ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power	▪ Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit for short circuit to powe Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new turbine intake valve position sensor as required

			- This is the long term adaption limit diagnostic. It diagnoses that the adapted values for the actuator end stops is outside of tolerance. This could be caused by sensor drift over time - Turbine intake valve position sensor circuit short circuit to power - Turbine intake valve position sensor stuck	
	P22D4-71	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Actuator stuck	- The engine control module has not detected any motion, in response to energizing a motor, solenoid or relay - This is the long term adaption limit diagnostic. It diagnoses that the adapted values for the actuator end stops is outside of tolerance. This could be caused by sensor drift over time - Turbine intake valve position sensor stuck	Check and install a new turbine intake valve position sensor as required
	P22D4-92	Turbocharger Turbine Inlet Valve Position Sensor Circuit - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - This is the long term adaption limit diagnostic. It diagnoses that the adapted values for the actuator end stops is outside of tolerance. This could be caused by sensor drift over time	Check and install a new turbine intake valve position sensor as required

			- Turbine intake valve position sensor stuck	
	P22D5-92	Turbocharger Turbine Inlet Valve Position Sensor Circuit Range/Performance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - This is the long term adaption limit diagnostic. It diagnoses that the adapted values for the actuator end stops is outside of tolerance. This could be caused by sensor drift over time - Turbine intake valve position sensor stuck	- Check and install a new turbine intake valve position sensor as required
	P22D6-11	Turbocharger Turbine Inlet Valve Position Sensor Circuit Low - Circuit short to ground	**NOTE:** Circuit reference TSVP - The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected - Turbine intake valve position sensor circuit short circuit to ground	- Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit for short circuit to groun Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P22D7-12	Turbocharger Turbine Inlet Valve Position Sensor Circuit High - Circuit short to battery		- Refer to the electrical circuit diagrams and check turbine intake valve position sensor circuit for short circuit to power Repair harness as required. Usi the manufacturer approved

			NOTE: Circuit reference TSVP - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Turbine intake valve position sensor circuit short circuit to power	diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
	P242A-00	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 3 - No sub type information	**NOTE:** Circuit reference PFOT - Harness failure - Exhaust gas temperature sensor post DPF - Exhaust gas temperature sensor post DPF failure	- Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is s if the exhaust gas temperature sensor post DPF fails a cold sta diagnostic check by the engine control module. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check t sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness required - Check and install a new exhaus gas temperature sensor post D as required
	P242B-00	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 3 - No sub type		- Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 3

		information	**NOTE:** Circuit reference PFOT ▪ Exhaust gas temperature sensor post DPF overheating ▪ Harness failure - Exhaust gas temperature sensor post DPF ▪ Exhaust gas temperature sensor post DPF failure	(0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is s if the exhaust gas temperature sensor post DPF fails a cold sta diagnostic check by the engine control module. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check t sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness required ▪ Clear the DTC and retest ▪ Check and install a new exhaus gas temperature sensor post D as required
	P242C-00	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 3 - No sub type information	**NOTE:** Circuit reference PFOT ▪ Harness failure - Exhaust gas temperature sensor post DPF ▪ Exhaust gas temperature sensor post DPF failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is s if the exhaust gas temperature sensor post DPF signal voltage less than the engine control module was expecting. Refer t the electrical circuit diagrams a check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check t sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness required ▪ Check and install a new exhaus gas temperature sensor post D as required
	P242D-00	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 3 - No sub type		▪ Using the manufacturer approv diagnostic system check datalogger signals, Exhaust Ga

		information	**NOTE:** Circuit reference PFOT ▪ Exhaust gas temperature sensor post DPF circuit, short circuit to power, open circuit ▪ Exhaust gas temperature sensor post DPF failure	Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6) ▪ Refer to the electrical circuit diagrams and check the exhaus gas temperature sensor post D circuit, for short circuit to powe open circuit ▪ Check and install a new exhaus gas temperature sensor post D as required
	P242F-00	Diesel Particulate Filter Restriction - Ash Accumulation (Bank 1) - No sub type information	▪ Maximum ash load	**NOTE:** The setting value of this DTC is inhibited ▪ Contact dealer technical suppo
	P244A-00	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - No sub type information	▪ Diagnostic failure check for minimum pressure differential characteristics ▪ Diesel particulate filter internal components are missing or destroyed	**NOTE:** If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) ▪ Using the manufacturer approv diagnostic system, check for related DTCs and refer to the relevant DTC index ▪ Clear DTC and re-test ▪ Check and install a new diesel particulate filter as required
	P244A-96	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - Component internal failure	▪ Destroyed particulate filter	

				NOTE: If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) ▪ Refer to the relevant pinpoint test in section 309-00
	P244B-68	Diesel Particulate Filter Differential Pressure Too High (Bank 1) - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Engine protection back pressure high ▪ Sudden increases in differential pressure across the diesel particulate filter	**NOTE:** This DTC when logged on its own is advisory only and no further action should be taken ▪ Using the manufacturer approv diagnostic system, check for related DTCs. If this DTC exists with any other diesel particulat filter differential pressure senso DTCs, follow the advise given f that DTC
	P244E-00	Exhaust Temperature Too Low For Particulate Filter Regeneration, Bank 2 - No sub type information	▪ Error path for not reaching the setpoint of the inner loop with maximal control variable	▪ Using the manufacturer approv diagnostic system check for related DTCs and refer to the relevant DTC index. Clear DTC and re-test. This DTC has been calibrated not to flag
	P244F-00	Exhaust Temperature Too High For Particulate Filter Regeneration, Bank 2 - No sub type information	▪ Error path for not reaching the setpoint of the inner loop with minimal control variable	▪ Using the manufacturer approv diagnostic system check for related DTCs and refer to the relevant DTC index. Clear DTC and re-test. This DTC has been calibrated not to flag
	P2452-23	Diesel Particulate Filter Pressure Sensor A Circuit - Signal stuck low	▪ The engine control module measures a signal that remains low	▪ Refer to the electrical circuit diagrams and check the differential pressure sensor

			when transitions are expected ▪ Differential pressure sensor circuit, short circuit to ground ▪ Diesel differential pressure sensor A circuit, hose line error	circuit, for short circuit to grour ▪ Using the manufacturer approv diagnostic system, check for related DTCs and refer to the relevant DTC index
	P2452-29	Diesel Particulate Filter Pressure Sensor A Circuit - Signal invalid	▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ Diagnostic failure check for frozen differential pressure sensor	▪ Using the manufacturer approv diagnostic system check for related DTCs and refer to the relevant DTC index
	P2452-95	Diesel Particulate Filter Pressure Sensor A Circuit - Incorrect assembly	**NOTE:** In cold climates differential pressure sensor hose lines or metal pipes may be frozen ▪ The engine control module has detected that the component has been incorrectly installed e.g. hydraulic pipes crossed over, circuits cross wired or polarity errors ▪ Differential pressure sensor hoses connected incorrectly ▪ Differential pressure sensor hoses crushed, blocked, split	

NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor Circuit reference repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Using the manufacturer approv diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sens Voltage - Bank 1 (0x03DB). Refe to the workshop manual and check differential pressure sens

				hoses are installed correctly ▪ Check differential pressure sensor hoses for crushed, blockage, split
	P2453-00	Diesel Particulate Filter Pressure Sensor A Circuit Range/Performance - No sub type information	**NOTE:** Circuit reference DPS ▪ Harness failure - Differential pressure sensor ▪ Differential pressure sensor failure	**NOTE:** If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor Circuit reference repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa ▪ Using the manufacturer approv

				diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sens Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor fails a plausibili check. Refer to the electrical circuit diagrams and check the differential pressure sensor sigr circuit for open circuit, short circuit to ground, short circuit t other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness required ▪ Check and install a new differential pressure sensor as required
	P2453-16	Diesel Particulate Filter Pressure Sensor A Circuit Range/Performance - Circuit voltage below threshold	**NOTE:** Circuit reference DPS ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Diagnostic failure check for signal range check low in flow resistance ▪ Harness failure - Differential pressure sensor ▪ Differential pressure sensor failure	▪ Using the manufacturer approv diagnostic system check for related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sens Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor fails a plausibili check. Refer to the electrical circuit diagrams and check the differential pressure sensor sigr circuit for open circuit, short circuit to ground, short circuit t other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness required ▪ Check and install a new differential pressure sensor as required

	P2453-17	Diesel Particulate Filter Pressure Sensor A Circuit Range/Performance - Circuit voltage above threshold	**NOTE:** Circuit reference DPS - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Diagnostic failure check for signal range check high in flow resistance - Harness failure - Differential pressure sensor - Differential pressure sensor failure	- Using the manufacturer approv diagnostic system check for related DTCs and refer to the relevant DTC index - Using the manufacturer approv diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sens Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor fails a plausibili check. Refer to the electrical circuit diagrams and check the differential pressure sensor sigr circuit for open circuit, short circuit to ground, short circuit t other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness required - Check and install a new differential pressure sensor as required
	P2454-00	Diesel Particulate Filter Pressure Sensor A Circuit Low - No sub type information	**NOTE:** Circuit reference DPS - Harness failure - Differential pressure sensor - Differential pressure sensor crossed hose lines - Differential pressure sensor failure	

NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor Circuit reference repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Using the manufacturer approv diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sens Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor voltage is less

				than the threshold set in the engine control module diagnos check. Refer to the electrical circuit diagrams and check the differential pressure sensor sigr circuit for open circuit, short circuit to ground, short circuit t other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness required ▪ Check differential pressure sensor hose lines are installed correctly ▪ Check and install a new differential pressure sensor as required
	P2455-00	Diesel Particulate Filter Pressure Sensor A Circuit High - No sub type information	**NOTE:** Circuit reference DPS ▪ Harness failure - Differential pressure sensor ▪ Differential pressure sensor failure ▪ Exhaust back pressure is too high	

NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor Circuit reference repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Using the manufacturer approv diagnostic system, check engin control module, for related DT(and refer to the relevant DTC index
- Using the manufacturer approv diagnostic system check

				datalogger signals, Particulate Filter Differential Pressure Sens Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor voltage is greater than the threshold set i the engine control module diagnostic check. Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to power, short circuit to other circuits. Check the sensor ground circui for open circuit, short circuit to power, high resistance. Check t sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required ▪ Check and install a new differential pressure sensor as required
	P2456-00	Diesel Particulate Filter Pressure Sensor A Circuit Intermittent/Erratic - No sub type information	▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ In cold climate or off road driving differential pressure sensor hoses crushed, blocked, split ▪ Differential pressure sensor failure	

NOTE:

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor Circuit reference repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

- Refer to the electrical circuit diagrams and check differential pressure sensor circuit for short circuit to ground, short circuit t power, open circuit, high resistance
- Inspect connectors for signs of

				water ingress, and pins for damage and/or corrosion ▪ Check differential pressure sensor hoses for crushed, blocked, split ▪ Check and replace diesel particulate filter differential pressure sensor as required
	P2458-66	Diesel Particulate Filter Regeneration Duration (Bank 1) - Signal has too many transitions / events	▪ The engine control module monitored a parameter over time within specified limits and detected more than the expected number of transitions ▪ Permanent regeneration	**NOTE:** This code is enabled for JLR engineering detailed diagnostics only. No further action should be taken ▪ Using the manufacturer approv diagnostic system, check for related DTCs and refer to the relevant DTC index
	P2459-65	Diesel Particulate Filter Regeneration Frequency (Bank 1) - Signal has too few transitions / events	▪ The engine control module monitored a parameter over time within specified limits and detected fewer than the expected number of transitions ▪ Blocked regeneration ▪ Customer driving routine does not allow the system to clean the particulate filter	**NOTE:** If DTC is P2459-65 or AMBER DPF FULL REFER TO HANDBOOK message is displayed with no other reported messages. No repair is required, if the vehicle is driven on a highway AS DIRECTED IN THE HANDBOOK then the light will be extinguished and the system self healed, nothing more than this is required ▪ Refer to the diesel particulate filter regeneration procedure a carry out a diesel particulate filt regeneration ▪ Advise customer of driving routine required to regenerate diesel particulate filter as state in the vehicle handbook

P2459-66	Diesel Particulate Filter Regeneration Frequency (Bank 1) - Signal has too many transitions / events	▪ The engine control module monitored a parameter over time within specified limits and detected more than the expected number of transitions ▪ Regeneration frequency	**NOTE:** This code is enabled for JLR engineering detailed diagnostics only. No further action should be taken ▪ Using the manufacturer approv diagnostic system, check for related DTCs and refer to the relevant DTC index
P245A-11	Exhaust Gas Recirculation Cooler Bypass Control Circuit - Circuit short to ground	**NOTE:** Circuit reference EGRCBV ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Exhaust gas recirculation cooler bypass valve solenoid circuit short circuit to ground	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrica circuit diagrams and check the exhaust gas recirculation cooler bypass valve solenoid circuit fo short circuit to ground. Check power supply to the solenoid. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
P245A-12	Exhaust Gas Recirculation Cooler Bypass Control Circuit - Circuit short to battery	**NOTE:** Circuit reference EGRCBV ▪ The engine control module has detected a vehicle power measurement for a period longer than	▪ Using the manufacturer approv diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrica circuit diagrams and check the exhaust gas recirculation cooler bypass valve solenoid circuit fo short circuit to power. Check power supply to the solenoid. Repair harness as required. Usi the manufacturer approved

			- expected or has detected a vehicle power measurement when another value was expected - Harness failure - Exhaust gas recirculation cooler bypass valve solenoid circuit short circuit to power - Exhaust gas recirculation cooler bypass valve solenoid failure	diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new exhaus gas recirculation cooler bypass valve solenoid as required
	P245A-13	Exhaust Gas Recirculation Cooler Bypass Control Circuit - Circuit open	**NOTE:** Circuit reference EGRCBV - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Harness failure - Exhaust gas recirculation cooler bypass valve solenoid circuit open circuit - Exhaust gas recirculation cooler bypass valve solenoid failure	- Using the manufacturer approv diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrica circuit diagrams and check the exhaust gas recirculation cooler bypass valve solenoid circuit fo open circuit. Check power supp to the solenoid. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new exhaus gas recirculation cooler bypass valve solenoid as required
	P245B-19	Exhaust Gas Recirculation Cooler Bypass Control Circuit Range/Performance - Circuit current above threshold	- The engine control module has measured current flow above a specified range - Harness failure - Exhaust gas recirculation cooler bypass valve solenoid circuit short circuit to ground, high resistance - Exhaust gas recirculation cooler bypass valve solenoid failure	- Using the manufacturer approv diagnostic system check datalogger signals, EGR Cooler Bypass Valve Duty Cycle (0x03C5). Refer to the electrica circuit diagrams and check the exhaust gas recirculation cooler bypass valve solenoid circuit fo short circuit to ground, high resistance. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the

				'Diagnosis menu' tab and retes ▪ Check and install a new exhaus gas recirculation cooler bypass valve solenoid as required
	P2463-00	Diesel Particulate Filter Restriction - Soot Accumulation (Bank 1) - No sub type information	▪ Maximum soot mass	▪ Refer to the relevant pinpoint test in section 309-00
	P246C-00	Diesel Particulate Filter Restriction - Forced Limited Power (Bank 1) - No sub type information	▪ Diagnostic failure check for maximum pressure differential characteristics	▪ Using the manufacturer approv diagnostic system check for related DTCs and refer to the relevant DTC index
	P250A-36	Engine Oil Level Sensor Circuit - Signal frequency too low	**NOTE:** Circuit reference OTL ▪ The engine control module detected excessive duration for one cycle of the output across a specified sample size ▪ Oil level and temperature sensor circuit short circuit to ground, high resistance, open circuit ▪ Oil level and temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperatur - Measured (0x03F3). Refer to t electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, high resistance, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new oil leve and temperature sensor as required
	P250A-37	Engine Oil Level Sensor Circuit - Signal frequency too high	**NOTE:** Circuit reference OTL ▪ The engine control module detected insufficient duration for one cycle of the output	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperatur - Measured (0x03F3). Refer to t electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to power. Repair

			▪ across a specified sample size ▪ Oil level and temperature sensor circuit short circuit to power ▪ Oil level and temperature sensor failure	harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new oil leve and temperature sensor as required
	P250A-38	Engine Oil Level Sensor Circuit - Signal frequency incorrect	**NOTE:** Circuit reference OTL ▪ The engine control module measured an incorrect number of cycles in a given time period ▪ Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Oil level and temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperatur - Measured (0x03F3). Refer to t electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, short circuit to power, high resistance open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new oil leve and temperature sensor as required
	P250A-47	Engine Oil Level Sensor Circuit - Watchdog / safety micro controller failure	▪ Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Oil level and temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperatur - Measured (0x03F3). Refer to t electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, short circuit to power, high resistance open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes

			▪ Check and install a new oil leve and temperature sensor as required
P250A-92	Engine Oil Level Sensor Circuit - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Oil level and temperature sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Oil level and temperature sensor failure	▪ Using the manufacturer approv diagnostic system check datalogger signals, Engine Oil Level - Measured (0x03E6), Engine Oil Volume - Calculated (0x03F2), Sump Oil Temperatur - Measured (0x03F3). Refer to t electrical circuit diagrams and check the oil level and temperature sensor circuit for short circuit to ground, short circuit to power, high resistance open circuit. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new oil leve and temperature sensor as required
P2586-13	Turbocharger Boost Control Position Sensor B Circuit - Circuit open	**NOTE:** Circuit reference VGT_FB ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Harness failure - Variable geometry turbine vane actuator open circuit ▪ Variable geometry turbine vane actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator on bank for open circuit. This circuit consists of three wires between the engine control module and the variable geometry turbocharger control module. The three sensor wires are a 5 volt sensor supply, a sensor ground and a signal line. Check signal line for open circuit and power and ground supply to sensor. Repair harness as required. Using the manufactur approved diagnostic system cle all stored DTCs using the 'Diagnosis menu' tab and retes ▪ Check and install a new variabl geometry turbocharger positio

				sensor as required
	P2586-32	Turbocharger Boost Control Position Sensor B Circuit - Signal low time < minimum	**NOTE:** Circuit reference VGT_FB ▪ The engine control module detected the low pulse is too narrow with respect to time ▪ Harness failure - Variable geometry turbine vane actuator circuit short circuit to ground, high resistance, open circuit ▪ Variable geometry turbine vane actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator on bank for short circuit to ground, high resistance, open circuit. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new variabl geometry turbocharger positio sensor as required
	P2586-35	Turbocharger Boost Control Position Sensor B Circuit - Signal high time > maximum	**NOTE:** Circuit reference VGT_FB ▪ This DTC is set when the engine control module detects that the signal high time is greater than the maximum value ▪ Harness failure - Variable geometry turbine vane actuator ▪ Variable geometry turbine vane actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator on bank for open circuit, short circuit to power, short circuit to ground. Check the power and ground supply to the sensor. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new variabl geometry turbocharger positio sensor as required
	P2586-36	Turbocharger Boost Control Position Sensor B Circuit - Signal frequency too low		▪ Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured

			NOTE: Circuit reference VGT_FB - The engine control module detected excessive duration for one cycle of the output across a specified sample size - Harness failure - Variable geometry turbine vane actuator - Variable geometry turbine vane actuator failure	Position (0x0347). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator on bank for open circuit, short circuit to power, short circuit to ground. Check the power and ground supply to the sensor. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new variabl geometry turbocharger positio sensor as required
	P2586-37	Turbocharger Boost Control Position Sensor B Circuit - Signal frequency too high	**NOTE:** Circuit reference VGT_FB - The engine control module detected insufficient duration for one cycle of the output across a specified sample size - Harness failure - Variable geometry turbine vane actuator - Variable geometry turbine vane actuator failure	- Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator on bank for open circuit, short circuit to power, short circuit to ground. Check the power and ground supply to the sensor. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new variabl geometry turbocharger positio sensor as required
	P2587-92	Turbocharger Boost Control Position Sensor B Circuit Range/Performance - Performance or incorrect operation	- The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way - Harness failure - Variable	- Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). Refer to the electrical circuit diagrams and check the variable geometry turbine vane actuator on bank

			geometry turbine vane actuator ▪ Variable geometry turbine vane actuator failure	for open circuit, short circuit to power, short circuit to ground. Check the power and ground supply to the sensor. Repair harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new variabl geometry turbocharger positio sensor as required
	P2588-00	Turbocharger Boost Control Position Sensor B Circuit Low - No sub type information	**NOTE:** Circuit reference VGT_FB ▪ Harness failure - Variable geometry turbine vane actuator ▪ Variable geometry turbine vane actuator circuit short circuit to ground ▪ Variable geometry turbine vane actuator failure	▪ Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). This DTC is s when the engine control modul detects a low circuit voltage on the signal line from the variable geometry turbine vane actuato Check signal line for short circu to ground. Check power and ground supply to sensor. Repai harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check variable geometry turbine vane actuato circuit for short circuit to groun ▪ Check and install a new variabl geometry turbine vane actuato as required
	P2589-00	Turbocharger Boost Control Position Sensor B Circuit High - No sub type information	**NOTE:** Circuit reference VGT_FB ▪ Harness failure - Variable geometry turbine vane actuator ▪ Variable geometry	▪ Using the manufacturer approv diagnostic system check datalogger signal, Boost Pressu Actuator Bank 2 - Measured Position (0x0347). This DTC is s when the engine control modul detects a high circuit voltage o the signal line from the variable geometry turbine vane actuato Check signal line for short circu to power. Check power and ground supply to sensor. Repai

			turbine vane actuator circuit short circuit to power ▪ Variable geometry turbine vane actuator failure	harness as required. Using the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Refer to the electrical circuit diagrams and check variable geometry turbine vane actuato circuit for short circuit to powe ▪ Check and install a new variabl geometry turbine vane actuato as required
	P25A2-00	Brake System Control Module Requested MIL Illumination - No sub type information	**NOTE:** Monitor description. Anti-lock brake system control module is indicating a fault to the engine control module that could have emission failure conditions therefore requests the MIL ▪ Anti-lock brake system failure ▪ High speed CAN bus failure ▪ Fuse failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit ▪ Anti-lock brake system control module power circuit failure ▪ Anti-lock brake system control module ground circuit failure	▪ Check anti-lock brake system control module for additional DTCs and refer to relevant DTC index. Rectify these first ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit t ground, short circuit to power, open circuit ▪ Refer to the electrical circuit diagrams and check anti-lock brake system control module power and ground circuits for open circuit ▪ Clear the DTC and retest
	P268C-00	Cylinder 1 Injector Data Incompatible - No sub type	▪ Injector calibration data held in the engine	▪ Using the manufacturer approv diagnostic system reprogram th

		information	control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	injector codes
	P268C-51	Cylinder 1 Injector Data Incompatible - Not programmed	▪ The engine control module has indicated that programming is required ▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P268D-00	Cylinder 2 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P268D-51	Cylinder 2 Injector Data Incompatible - Not programmed	▪ The engine control module has indicated that programming is required ▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P268E-00	Cylinder 3 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read	▪ Using the manufacturer approv diagnostic system reprogram th injector codes

			from the injector ▪ Injector calibration data not stored / programmed	
	P268E-51	Cylinder 3 Injector Data Incompatible - Not programmed	▪ The engine control module has indicated that programming is required ▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P268F-00	Cylinder 4 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P268F-51	Cylinder 4 Injector Data Incompatible - Not programmed	▪ The engine control module has indicated that programming is required ▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P2690-00	Cylinder 5 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data	▪ Using the manufacturer approv diagnostic system reprogram th injector codes

			not stored / programmed	
	P2690-51	Cylinder 5 Injector Data Incompatible - Not programmed	▪ The engine control module has indicated that programming is required ▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P2691-00	Cylinder 6 Injector Data Incompatible - No sub type information	▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P2691-51	Cylinder 6 Injector Data Incompatible - Not programmed	▪ The engine control module has indicated that programming is required ▪ Injector calibration data held in the engine control module is different to that read from the injector ▪ Injector calibration data not stored / programmed	▪ Using the manufacturer approv diagnostic system reprogram th injector codes
	P2A00-16	O2 Circuit Range / Performance (Bank 1, Sensor 1) - Circuit voltage below threshold	**NOTE:** Circuit reference LPTR_A ▪ The engine control	▪ This DTC is set when the engin control module detects the voltage on the trim resistor circuit of the heated oxygen sensor is less than the voltage threshold. This may be caused b the heated oxygen sensor bein too hot to operate correctly. Refer to the workshop manual

			module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Heated oxygen sensor failure ▪ Heated oxygen sensor failure	and check the exhaust system and heated oxygen sensor harness for sign of mechanical damage. Refer to the electrical circuit diagrams and check all t heated oxygen sensor circuits f open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control modul power and ground supplies. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new heated oxygen sensor as required
	P2A00-17	O2 Circuit Range / Performance (Bank 1, Sensor 1) - Circuit voltage above threshold	**NOTE:** Circuit reference LPTR_A ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Heated oxygen sensor failure ▪ Heated oxygen sensor failure	▪ This DTC is set when the engin control module detects the voltage on the trim resistor circuit of the heated oxygen sensor is greater than the volta threshold. This may be caused the heated oxygen sensor bein too hot to operate correctly. Refer to the workshop manual and check the exhaust system and heated oxygen sensor harness for sign of mechanical damage. Refer to the electrical circuit diagrams and check all t heated oxygen sensor circuits f open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control modul power and ground supplies. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest ▪ Check and install a new heated oxygen sensor as required
	P2A00-26	O2 Circuit Range / Performance (Bank 1,		▪ This DTC is set when the engin control module detects the

		Sensor 1) - Signal rate of change below threshold	**NOTE:** Circuit reference LPTR_A - The signal transitions more slowly than is reasonably allowed - Harness failure - Heated oxygen sensor failure - Heated oxygen sensor failure	voltage on the trim resistor circuit of the heated oxygen sensor is greater than the volta threshold. This may be caused l the heated oxygen sensor bein too hot to operate correctly. Refer to the workshop manual and check the exhaust system and heated oxygen sensor harness for sign of mechanical damage. Refer to the electrical circuit diagrams and check all tl heated oxygen sensor circuits f open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control modul power and ground supplies. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest - Check and install a new heated oxygen sensor as required
	U0001-00	High Speed CAN Communication Bus - No sub type information	- High speed CAN bus failure - High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit	- Using the manufacturer approv diagnostic system carry out network integrity test - Refer to the electrical circuit diagrams and check high speec CAN network for short circuit t ground, short circuit to power, open circuit
	U0001-87	High Speed CAN Communication Bus - Missing message	- The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module	- Using the manufacturer approv diagnostic system, check the snapshot data to determine the missing message source contro module. Check the relevant control module for related DTC and refer to the relevant DTC index

		communication with another HS CAN control module) ▪ Missing message from another control module via the high speed CAN bus (powertrain)		
	U0001-88	High Speed CAN Communication Bus - Bus off	▪ The engine control module has determined failures where a data bus is not available ▪ High speed CAN bus failure ▪ High speed CAN bus circuit, short circuit to ground, short circuit to power, open circuit	▪ Using the manufacturer approv diagnostic system carry out network integrity test ▪ Refer to the electrical circuit diagrams and check high speed CAN network for short circuit t ground, short circuit to power, open circuit
	U0101-00	Lost Communication with TCM - No sub type information	▪ The engine control module has not received the expected CAN signal from the transmission control module within the specified time interval ▪ CAN harness link between engine control module and transmission control module network malfunction	▪ Using the manufacturer approv diagnostic system, check transmission control module fo DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check transmission control module power and ground circuits for open circuit. Check CAN harness between engine control module and transmissio control module, repair as necessary
	U0101-26	Lost Communication with TCM - Signal rate of change below threshold	▪ The signal transitions more slowly than is reasonably allowed ▪ The engine control module has not received the expected CAN signal from the transmission control module within the specified time interval ▪ CAN harness link	▪ Using the manufacturer approv diagnostic system, check transmission control module fo DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check transmission control module power and ground circuits for open circuit. Check

			between engine control module and transmission control module network malfunction	CAN harness between engine control module and transmissio control module, repair as necessary
	U0102-00	Lost Communication with Transfer Case Control Module - No sub type information	- The engine control module has not received the expected CAN signal from the transfer case control module within the specified time interval - CAN harness link between engine control module and transfer case control module network malfunction	- Using the manufacturer approv diagnostic system, check transf case control module for DTCs and refer to the relevant DTC index - Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check transfer case control module power and ground circuits for open circuit. Check CAN harness between engine control module and transfer ca control module, repair as necessary
	U0103-00	Lost Communication With Gear Shift Control Module A - No sub type information	- The engine control module has not received the expected CAN signal from the transmission control switch within the specified time interval - CAN harness link between engine control module and transmission control switch network malfunction	- Using the manufacturer approv diagnostic system, check transmission control switch for DTCs and refer to the relevant DTC index - Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check transmission control swit power and ground circuits for open circuit. Check CAN harne between engine control modul and transmission control switch repair as necessary
	U0103-87	Lost Communication With Gear Shift Control Module A - Missing message	- The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events	- Using the manufacturer approv diagnostic system, check transmission control switch for DTCs and refer to the relevant DTC index - Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check transmission control swit

			of the concerning component (e.g. engine control module communication with transmission control switch) ▪ The engine control module has not received the expected CAN signal from the transmission control switch within the specified time interval ▪ CAN harness link between engine control module and transmission control switch network malfunction	power and ground circuits for open circuit. Check CAN harne between engine control modul and transmission control switch repair as necessary
	U0104-00	Lost Communication With Cruise Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the speed control module within the specified time interval ▪ CAN harness link between engine control module and speed control module network malfunction	▪ Using the manufacturer approv diagnostic system, check speed control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check speed control module power and ground circuits for open circuit. Check CAN harne between engine control modul and speed control module, rep as necessary
	U0104-87	Lost Communication With Cruise Control Module - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module	▪ Using the manufacturer approv diagnostic system, check speed control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check speed control module power and ground circuits for open circuit. Check CAN harne between engine control modul and speed control module, rep as necessary

			communication with speed control module) ▪ The engine control module has not received the expected CAN signal from the speed control module within the specified time interval ▪ CAN harness link between engine control module and speed control module network malfunction	
	U010F-87	Lost Communication With Air Conditioning Control Module - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module communication with automatic temperature control module) ▪ The engine control module has not received the expected CAN signal from the automatic temperature control module within the specified time interval ▪ CAN harness link between engine control module and automatic temperature control module network malfunction	▪ Using the manufacturer approv diagnostic system, check automatic temperature control module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check automatic temperature control module power and ground circuits for open circuit. Check CAN harness between engine control module and automatic temperature control module, repair as necessary
	U0120-00	Lost Communication with Starter/ Generator Control Module - No sub type	▪ The engine control module has not received the expected CAN	▪ Using the manufacturer approv diagnostic system, check anti-lock brake system control

		information	signal from the anti-lock brake system control module within the specified time interval ▪ CAN harness link between engine control module and anti-lock brake system control module network malfunction	module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check generator system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti-lock brake system control module, repair as necessary
	U0121-00	Lost Communication With Anti-Lock Brake System (ABS) Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the anti-lock brake system control module within the specified time interval ▪ CAN harness link between engine control module and anti-lock brake system control module network malfunction	▪ Using the manufacturer approv diagnostic system, check anti-lock brake system control module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check anti-lock brake system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti lock brake system control module, repair as necessary
	U0126-00	Lost Communication With Steering Angle Sensor Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the steering angle sensor control module within the specified time interval ▪ CAN harness link between engine control module and steering angle sensor control module network malfunction	▪ Using the manufacturer approv diagnostic system, check steeri angle sensor control module fo DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check steering angle sensor control module power and ground circuits for open circuit. Check CAN harness between engine control module and steering angle sensor control module, repair as necessary

	U0128-00	Lost Communication With Park Brake Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the park brake control module within the specified time interval ▪ CAN harness link between engine control module and park brake control module network malfunction	▪ Using the manufacturer approv diagnostic system, check park brake control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check park brake control modu power and ground circuits for open circuit. Check CAN harne between engine control modul and park brake control module repair as necessary
	U012A-00	Lost Communication with Chassis Control Module "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the integrated suspension control module within the specified time interval ▪ CAN harness link between engine control module and integrated suspension control module network malfunction	▪ Using the manufacturer approv diagnostic system, check integrated suspension control module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and integrated suspension control module, repair as necessary
	U012A-87	Lost Communication with Chassis Control Module "A" - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module communication with integrated suspension	▪ Using the manufacturer approv diagnostic system, check integrated suspension control module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check integrated suspension control module power and ground circuits for open circuit ▪ Check CAN harness between

			control module) - The engine control module has not received the expected CAN signal from the integrated suspension control module within the specified time interval - CAN harness link between engine control module and integrated suspension control module network malfunction	engine control module and integrated suspension control module, repair as necessary
	U0131-00	Lost Communication With Power Steering Control Module - No sub type information	- The engine control module has not received the expected CAN signal from the power steering control module within the specified time interval - CAN harness link between engine control module and power steering control module network malfunction	- Using the manufacturer approv diagnostic system, check powe steering control module for DT and refer to the relevant DTC index - Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check power steering control module power and ground circuits for open circuit. Check CAN harness between engine control module and power steering control module, repair as necessary
	U0133-00	Lost Communication With Active Roll Control Module - No sub type information	- The engine control module has not received the expected CAN signal from the dynamic response control module within the specified time interval - CAN harness link between engine control module and dynamic response control module network malfunction	- Using the manufacturer approv diagnostic system, check dynamic response control module for DTCs and refer to t relevant DTC index - Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check dynamic response contro module power and ground circuits for open circuit. Check CAN harness between engine control module and dynamic response control module, repai as necessary

	U0138-00	Lost Communication with All Terrain Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the terrain response switchpack within the specified time interval ▪ CAN harness link between engine control module and terrain response switchpack network malfunction	▪ Using the manufacturer approv diagnostic system, check terrai response switchpack for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check terrain response switchpack power and ground circuits for open circuit. Check CAN harness between engine control module and terrain response switchpack, repair as necessary
	U0140-00	Lost Communication With Body Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the central junction box within the specified time interval ▪ CAN harness link between engine control module and central junction box network malfunction	▪ Using the manufacturer approv diagnostic system, check centra junction box for DTCs and refe to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check central junction box pow and ground circuits for open circuit. Check CAN harness between engine control modul and central junction box, repair as necessary
	U0146-00	Lost Communication With Gateway "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the gateway control module within the specified time interval ▪ CAN harness link between engine control module and gateway control module network malfunction	▪ Using the manufacturer approv diagnostic system, check gateway control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check gateway control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and gateway control module, repair

				as necessary
	U0151-00	Lost Communication With Restraints Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the restraints control module within the specified time interval ▪ CAN harness link between engine control module and restraints control module network malfunction	▪ Using the manufacturer approv diagnostic system, check restraints control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check restraints control module power and ground circuits for open circuit. Check CAN harne between engine control modul and restraints control module, repair as necessary
	U0151-08	Lost Communication With Restraints Control Module - Bus Signal / Message Failures	▪ The engine control module has not received the expected CAN signal from the restraints control module within the specified time interval ▪ CAN harness link between engine control module and restraints control module network malfunction	▪ Using the manufacturer approv diagnostic system, check restraints control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check restraints control module power and ground circuits for open circuit. Check CAN harne between engine control modul and restraints control module, repair as necessary
	U0155-00	Lost Communication With Instrument Panel Cluster Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the instrument cluster within the specified time interval ▪ CAN harness link between engine control module and instrument cluster network malfunction	▪ Using the manufacturer approv diagnostic system, check instrument cluster for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check instrument cluster power and ground circuits for open circuit. Check CAN harness between engine control modul and instrument cluster, repair a necessary

U0159-00	Lost Communication With Parking Assist Control Module "A" - No sub type information	▪ The engine control module has not received the expected CAN signal from the park distance control module within the specified time interval ▪ CAN harness link between engine control module and park distance control module network malfunction	▪ Using the manufacturer approv diagnostic system, check park distance control module for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check park distance control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and par distance control module, repair as necessary
U0164-00	Lost Communication With HVAC Control Module - No sub type information	▪ Power distribution system failure - Fuse failure ▪ CAN network failure ▪ Automatic temperature control module failure	▪ This DTC is set if the engine control module loses communication with the automatic temperature control module. Check the automatic temperature control module fo DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the automatic temperature control module ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Re to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require ▪ Check and install a new automatic temperature control module
U0167-00	Lost Communication With Vehicle Immobilizer Control Module - No sub type information	▪ Engine control module identity transfer failed	▪ Check the electric steering column lock for DTCs and refer to the relevant DTC index. Refe to the electrical circuit diagram and check the power supply an

				ground connections to the electric steering column lock. Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require ▪ Check and install a new electric steering column lock
	U0300-00	Internal Control Module Software Incompatibility - No sub type information	▪ CAN network failure ▪ Central junction box failure ▪ Car configuration file incorrect	▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required ▪ Car configuration signal not received. Check central junctio box for DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system check and up date the car configuration file a required
	U0402-00	Invalid Data Received from TCM - No sub type information	▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required
	U0402-83	Invalid Data Received from TCM - Value of signal protection calculation incorrect	▪ The engine control module has indicated that a message was processed with an incorrect protection (checksum) calculation ▪ CAN network failure ▪ Implausible CAN data received from transmission control module	▪ Check transmission control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required

U0404-68	Invalid Data Received from Gear Shift Control Module A - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ CAN network failure ▪ Implausible CAN data received from transmission control switch	▪ Check transmission control switch for related DTCs and ref to relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required
U0405-68	Invalid Data Received From Cruise Control Module - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Speed control system failure	▪ Check speed control module fo related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required ▪ Check and install a new speed control module
U0405-82	Invalid Data Received From Cruise Control Module - Alive/sequence counter incorrect/not updated	▪ The engine control module has indicated that a signal was received without the corresponding rolling count value being properly updated ▪ Speed control system failure ▪ CAN network failure ▪ Harness failure - Speed control module power	▪ Check speed control buttons ar not jammed/contaminated/damage Check speed control module fo related DTCs and refer to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required

			supply or ground failure	▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Check and install a new speed control module
	U0405-84	Invalid Data Received From Cruise Control Module - Signal below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Speed control system failure	▪ Check speed control module fo related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the speed control module ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required ▪ Check and install a new speed control module
	U0405-86	Invalid Data Received From Cruise Control Module - Signal invalid	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is not plausible given the operating conditions ▪ Speed control system failure	▪ Check speed control buttons ar not jammed/contaminated/damage Check speed control module fo related DTCs and refer to the relevant DTC index
	U0407-00	Invalid Data Received From Glow Plug Control Module - No sub type information	▪ This DTC is set when the glow plug control module does not match what is expected by the engine control module. Detection of the coding word is complete when 2 of 3 coding words match. This process is a one time operation and will be completed during assembly of the vehicle	▪ Check the correct glow plug control module is installed to th vehicle

U0407-81	Invalid Data Received From Glow Plug Control Module - Invalid serial data received	▪ The engine control module has indicated a signal was received with the corresponding validity bit equal to "invalid" or post processing of the signal determines it is invalid ▪ Glow plug control system failure	▪ Refer to electrical circuit diagrams and check the power supply and ground connections to the glow plug control modul Check the diagnostic circuit between the engine control module and the glow plug control module for short circuit to power, short circuit to groun open circuits. Clear the DTC an retest ▪ Check and install a new glow plug control module
U0415-68	Invalid Data Received From Anti-Lock Brake System (ABS) Control Module - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Signal error for vehicle speed over CAN	▪ Using the manufacturer approv diagnostic system, check anti-lock brake system control module for DTCs and refer to t relevant DTC index
U0416-46	Invalid Data Received From Vehicle Dynamics Control Module - Calibration / parameter memory failure	▪ Anti-lock brake system failure	▪ Using the manufacturer approv diagnostic system, check anti-lock brake system control module for DTCs and refer to t relevant DTC index
U0416-68	Invalid Data Received From Vehicle Dynamics Control Module - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ The engine control module has received the	▪ Using the manufacturer approv diagnostic system, check anti-lock brake system control module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system carry out network integrity test. Refer to the electrical circuit diagrams a check anti-lock brake system control module power and ground circuits for open circuit. Check CAN harness between engine control module and anti

			default brake pressure signal value over CAN from the anti-lock brake system control module for a specified time interval ▪ Anti-lock brake system failure ▪ CAN harness link between engine control module and anti-lock brake system control module network malfunction	lock brake system control module, repair as necessary
	U0416-92	Invalid Data Received From Vehicle Dynamics Control Module - Performance or incorrect operation	▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ Difference between anti-lock brake system speed signal value and instrument cluster speed value at low vehicle speeds	Check the anti-lock brake system control module for DTCs and refe to the relevant DTC index
	U0422-00	Invalid Data Received From Body Control Module - No sub type information	▪ The engine control module has not received the expected CAN signal from the central junction box within the specified time interval ▪ CAN harness link between engine control module and central junction box network malfunction	▪ Using the manufacturer approv diagnostic system, check centra junction box for DTCs and refe to the relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check central junction box power and ground circuits for open circuit ▪ Check CAN harness between engine control module and central junction box, repair as necessary
	U0424-00	Invalid Data Received From HVAC Control Module - No	▪ The engine control module has not received	▪ Using the manufacturer approv diagnostic system, check

		sub type information	the expected CAN signal from the automatic temperature control module within the specified time interval ▪ CAN harness link between engine control module and automatic temperature control module network malfunction	automatic temperature control module for DTCs and refer to t relevant DTC index ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test ▪ Refer to the electrical circuit diagrams and check automatic temperature control module power and ground circuits for open circuit ▪ Check CAN harness between engine control module and automatic temperature control module, repair as necessary
	U0424-68	Invalid Data Received From HVAC Control Module - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Power distribution system failure - Fuse failure ▪ CAN network failure ▪ Automatic temperature control module failure	▪ This DTC is set if the engine control module loses communication with the automatic temperature control module. Check the automatic temperature control module fo DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check the power supply and ground connections to the automatic temperature control module ▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require ▪ Check and install a new automatic temperature control module as required
	U0426-00	Invalid Data Received From Vehicle Immobilizer Control Module - No sub type information	▪ Electric steering column lock has received an invalid identity response ▪ Module substituted	▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair harness as required. Usi the manufacturer approved diagnostic system check and up date the car configuration file a

			required ▪ Ensure all modules installed in the vehicle which store vehicle identity are valid for this vehicle and are not substitutes from a donor vehicle
U0447-00	Invalid Data Received From Gateway "A" - No sub type information	▪ Missing/invalid data from the gateway module	▪ Using the manufacturer approv diagnostic system, check the gateway module for related DTCs and refer to the relevant DTC index
U0452-00	Invalid Data Received From Restraints Control Module - No sub type information	▪ Missing/invalid data from the restraints control module	▪ Using the manufacturer approv diagnostic system, check the restraints control module for related DTCs and refer to the relevant DTC index
U0A1A-87	LIN Bus "A" - Missing message	▪ The engine control module has determined failures where one (or more) expected message(s) is not received, e.g periodic transmission where the repetition time is too high, or message not received as a result of unforeseen reset events of the concerning component (e.g. engine control module communication with generator) ▪ Generator LIN bus communication circuit failure	▪ Refer to the electrical circuit diagrams and check the generator LIN bus circuit, for short circuit to power, short circuit to ground, open circuit. Repair harness as required. Usi the manufacturer approved diagnostic system clear all store DTCs using the 'Diagnosis men tab and retest
U1A14-00	CAN Initialization Failure - No sub type information	▪ Harness failure - CAN circuit failure	▪ Using the manufacturer approv diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2005-	Vehicle Speed - Signal	▪ The engine control	▪ This DTC is set when the engin

64	plausibility failure	module detected plausibility failures ▪ Anti-lock brake system failure	control module has recognized vehicle speed signal plausibility failure. Check the anti-lock brak system module for related DTC and refer to the relevant DTC index. Check the instrument cluster for related DTCs and re to the relevant DTC index. Usin the manufacturer approved diagnostic system, complete a CAN network integrity test. Re to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2005-84	Vehicle Speed - Below allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Anti-lock brake system failure	▪ This DTC is set when the engin control module has recognized vehicle speed signal which is below the allowable range. Check the anti-lock brake syste control module for related DTC and refer to the relevant DTC index. Check the instrument cluster for related DTCs and re to the relevant DTC index. Usin the manufacturer approved diagnostic system, complete a CAN network integrity test. Re to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2005-85	Vehicle Speed - Above allowable range	▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Anti-lock brake system failure	▪ This DTC is set when the engin control module has recognized vehicle speed signal which is above the allowable range. Check the anti-lock brake syste control module for related DTC and refer to the relevant DTC index. Check the instrument cluster for related DTCs and re to the relevant DTC index. Usin the manufacturer approved diagnostic system, complete a CAN network integrity test. Re to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2108-00	Adaptive Cruise Control - No sub type information	▪ Adaptive speed control system failure - Error	▪ Check adaptive speed control module for DTCs and refer to t

		indicating adaptive speed control failure flag set	relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2108-24	Adaptive Cruise Control - Signal stuck high	▪ The engine control module measures a signal that remains high when transitions are expected ▪ Adaptive speed control system failure - Adaptive speed control follow speed error	▪ Check adaptive speed control module for DTCs and refer to t relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2108-64	Adaptive Cruise Control - Signal plausibility failure	▪ The engine control module detected plausibility failures ▪ Adaptive speed control system failure - Adaptive speed control follow speed range error	▪ Check adaptive speed control module for DTCs and refer to t relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2108-68	Adaptive Cruise Control - Event information	▪ The engine control module indicated the detection of a system event that was not caused by the engine control module itself but forced the engine control module to store the DTC e.g. missing functionality from another system or control module ▪ Adaptive speed control system failure - Error indicating adaptive speed control follow speed check when stationary	▪ Check adaptive speed control module for DTCs and refer to t relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require
U2108-86	Adaptive Cruise Control - Signal invalid	▪ The engine control module has determined	▪ Check adaptive speed control module for DTCs and refer to t

			failures where some circuit quantity, reported via serial data, is not plausible given the operating conditions ▪ Adaptive speed control system failure - Error when invalid adaptive speed control resume requests are present	relevant DTC index. Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Ref to electrical circuit diagrams an check CAN circuits if required. Repair wiring harness as require